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I HEREBY RECOMMEND THAT THE THESIS PREPARED UNDER MY SUPERVISION BY

Colin Friedel
Zach Henderson
Cooper McCarthy
Elias Pedroza
Thatcher Steele-Maley

ENTITLED

IRIS - Integrated Reef Information System

BE ACCEPTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREES OF

BACHELOR OF SCIENCE IN MECHANICAL ENGINEERING
BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND ENGINEERING

Christopher Kitts	06/10/2026
Dr. Christopher Kitts, Thesis Advisor	Date
Michael Neumann Michael Neumann (Jun 10, 2026 09:06:01 PDT)	06/10/2026
Dr. Michael Neumann, Thesis Advisor	Date
M. Gil	
Dr. Michael Schimpf, Thesis Advisor	Date
Michael J. Taylor	06/11/2026
Dr. Michael Taylor, Department Chair	Date
Sylvia Figuera	06/11/2026
Dr. Sylvia Figuera, Department Chair	Date

IRIS - Integrated Reef Information System

by

Colin Friedel
Zach Henderson
Cooper McCarthy
Elias Pedroza
Thatcher Steele-Maley

Submitted in Partial Fulfillment of the Requirements
for the degrees of
Bachelor of Science in Mechanical Engineering
Bachelor of Science in Computer Science and Engineering
School of Engineering
Santa Clara University

Santa Clara, California
June 5, 2026

IRIS - Integrated Reef Information System

Colin Friedel
Zach Henderson
Cooper McCarthy
Elias Pedroza
Thatcher Steele-Maley

Department of Mechanical Engineering
Department of Computer Science and Engineering
Santa Clara University
June 5, 2026

ABSTRACT

Our project provides an independent addition to the existing REEF Sensor Network, which has been created by the Santa Clara University Robotics Systems Lab and is planned to be deployed in the Palos Verdes Reef. Our system enables wireless transmission of data from the submerged sensor network to the scientist on land by designing and implementing a submersible winch mechanism to automatically raise and lower a buoy from the ocean floor to the surface, and cellular connectivity to wirelessly transmit collected data from the buoy to shore. After the data is transmitted, the buoy is lowered back to the sea floor to prevent any collisions with boats and undesirable surface condition exposure, as well as reduce the risk of marine life entanglement. Data will be transmitted on a consistent interval, allowing data to be received frequently without the costs of sending a dive team to the sensor network. Our system is able to seamlessly integrate with REEF, operating independently apart from the input of data and power, which will open possibilities of integration with other networks and systems.

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We are also grateful to Kortney Opshaug of Blue Ocean Gear for her generous donation and thoughtful advice. Her guidance on buoy design and material selection shaped important aspects of our system.

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Table of Contents

Abstract iii

Table of Contents v

List of Figures ix

List of Tables xi

1 Introduction 1
- 1.1 Motivation . . . 1
- 1.2 Background or Related Work . . . 3
- 1.3 Objectives . . . 4
- 1.4 Approach . . . 5

2 Research 6
- 2.1 Background Research . . . 6
- 2.2 Customer Needs . . . 8
- 2.3 User stories . . . 10

3 System Overview and Rationale 11
- 3.1 System Definition . . . 11
- 3.2 System Requirements . . . 12
  - 3.2.1 List of Requirements . . . 13
  - 3.2.2 Rationale . . . 13
- 3.3 Concept of Operations . . . 14
- 3.4 System Components . . . 16
- 3.5 Wiring . . . 18
- 3.6 Software Overview . . . 19
  - 3.6.1 Subsea Electronics State Diagram . . . 19
  - 3.6.2 Buoy State Diagram . . . 20
  - 3.6.3 Use Cases . . . 21
  - 3.6.4 Class Diagram . . . 24
  - 3.6.5 Sequence Diagram . . . 24

4 Detailed Design 26
- 4.1 Fundamental System Architecture . . . 26
  - 4.1.1 Design Drivers . . . 26
  - 4.1.2 Proposed Solutions . . . 26
  - 4.1.3 Motion Trade-Off Drivers . . . 27
  - 4.1.4 Trade-Off Analysis . . . 29
  - 4.1.5 Tradeoff Conclusion . . . 29

4.1.6 Winch Structure . . . 30
4.1.7 Tethering . . . 32
4.2 Subsystem One - Communications . . . 33
4.2.1 Design Drivers . . . 33
4.2.2 Proposed Solutions . . . 34
4.2.3 Communication Trade-Off Drivers . . . 36
4.2.4 Trade-Off Analysis . . . 37
4.2.5 Trade-Off Conclusion - Buoy to Shore Communication . . . 37
4.2.6 Trade-Off Conclusion - Subsea Electronics to Buoy . . . 38
4.2.7 End to End Electrical . . . 40
4.2.8 Final Software Flow . . . 41
4.3 Subsystem Two - Winch . . . 41
4.3.1 Design Drivers . . . 42
4.3.2 Proposed Designs . . . 42
4.3.3 Trade-Off Analysis . . . 44
4.3.4 Trade-Off Conclusion . . . 45
4.3.5 Submersible Servo-Motor . . . 45
4.3.6 Acetal Gearbox Speed Reduction . . . 47
4.3.7 Ratchet & Pawl Braking . . . 47
4.3.8 Winch Design . . . 48
4.3.9 Drag Force Validation . . . 52
4.4 Subsystem Three - Buoy . . . 54
4.4.1 Design Drivers . . . 54
4.4.2 Buoy Exploration . . . 54
4.4.3 Final Buoy Design . . . 56
4.4.4 Subsea Electronics Enclosure . . . 59
4.4.5 Subsea Platform and Anchoring . . . 60

5 System Evaluation and Timeline **61**
5.1 Timeline . . . 61
5.2 Overview . . . 63
5.3 Bench-top (Unit) Testing . . . 63
5.3.1 Software . . . 63
5.3.2 Hardware . . . 63
5.4 Tank and Pool Testing . . . 65
5.4.1 Cellular Payload . . . 65
5.4.2 End to End IRIS Testing . . . 66
5.5 Bay Deployment . . . 66
5.6 Evaluation Results and Next Steps . . . 68
5.6.1 Validated Requirements . . . 68
5.6.2 Untested Requirements . . . 70
5.6.3 Incomplete Requirements . . . 71
5.7 Project Risks . . . 71

6 Constraints and Standards **73**
6.1 Constraints . . . 73
6.2 Standards . . . 74

7 Professional Considerations **75**
7.1 Ethics . . . 75
7.2 Science Technology and Society . . . 76
7.3 Civic Engagement . . . 77
7.4 Economic . . . 77
7.5 Manufacturability . . . 77
7.6 Environmental Impact . . . 78
7.7 Usability . . . 78
7.8 Lifelong learning . . . 78

8 Conclusion **79**
8.1 Summary and Discussion . . . 79
8.2 Future Work . . . 79
8.2.1 Near Term (1-3 Years) . . . 79
8.2.2 Long Term (5-10 Years) . . . 80
8.3 Reflection . . . 80

Bibliography **81**

A Expected Budget and Bill of Materials **84**

B Source Code **87**
B.1 Buoy Loop Function . . . 87
B.2 Subsea Electronics Loop Function . . . 92

C Buoy Tilt Equations **94**
C.1 Force Balance . . . 94
C.1.1 Net Buoyancy Force (Skyward - Z axis) . . . 94
C.1.2 Drag Force (Horizontally - X axis) . . . 94
C.1.3 Tether Force (X and Z components) . . . 95
C.1.4 Force Balance Equation . . . 95
C.2 Moment Balance . . . 95
C.2.1 Metacentric Righting Moment Analysis . . . 95
C.2.2 Drag Moment Analysis . . . 96
C.2.3 Moment Balance Equation . . . 96
C.3 Methodology . . . 96
C.4 Parametric Study . . . 96

D Drag Calculations **98**
D.1 Force Balance . . . 98
D.1.1 Drag Force . . . 98
D.2 Linear Wave Theory . . . 98
D.2.1 MATLAB Code . . . 98
D.3 Drag Values . . . 101

E Weight – Buoyancy Balance **103**
E.1 Method . . . 103
E.2 Inputs . . . 104
E.3 Results . . . 105
E.4 Ballast Position Sensitivity . . . 105

F Safety Form **107**

G Presentation Slides **129**

List of Figures

1.1 Topographical Map of the Palos Verdes Reef . . . 2
1.2 Vertical Profiling Robot . . . 3
1.3 REEF Sensor Node Deployed and On Land . . . 4

2.1 Wainman Retractable Autonomous Platform (WRAP) buoy . . . 7
2.2 NPS Ocean Sensing Buoy During Deployment . . . 9

3.1 REEF - Concept of Operations . . . 15
3.2 IRIS - Concept of Operations . . . 16
3.3 IRIS - Component Block Diagram . . . 17
3.4 IRIS - Wiring Diagram . . . 18
3.5 Subsea Electronics Enclosure State Diagram . . . 19
3.6 Buoy State Diagram . . . 20
3.7 Use Case 1: View Data from web endpoint . . . 21
3.8 Use Case 2: Initial deployment of system . . . 21
3.9 Use Case 3: Monitoring of system health and efficiency . . . 21
3.10 Use Case 4: Maintenance of system . . . 22
3.11 Use Case 5: Retrieval of system . . . 22
3.12 UML Use Case Diagram . . . 23
3.13 Class Diagram . . . 24
3.14 Sequence Diagram . . . 25

4.1 Diagram of variable buoyancy engine functionality . . . 27
4.2 CAD model of Spool . . . 31
4.3 Partial Exploded View of CAD Model . . . 31
4.4 CAD model of Buoy . . . 33
4.5 Particle Boron 404x Kit . . . 34
4.6 Iridium Certus 9704 . . . 35
4.7 Operating principle of acoustic communication . . . 35
4.8 Blue Robotics Fathom ROV Tether . . . 39
4.9 MOFLON Slip Ring Working Principle . . . 39
4.10 Succorfish Delphis Acoustic Modem . . . 40
4.11 Okeanus Subsea Winch . . . 42
4.12 Theoretical submerged air pocket with dry winch . . . 43
4.13 Magnetic coupling diagram . . . 43
4.14 Blue Robotics M200 submersible brushless DC motor . . . 44
4.15 Blue Trail SER-2130 Waterproof Servo . . . 45
4.16 Gearbox Design . . . 47
4.17 Ratchet & Pawl integration . . . 48
4.18 Winch Overview . . . 50

4.19 Winch Structure . . . 51
4.20 Torque Transfer through the Winch . . . 52
4.21 Winch Rope Guide . . . 52
4.22 Blue Ocean Gear Buoy . . . 55
4.23 Blue Robotics Watertight Enclosure . . . 55
4.24 Final IRIS buoy assembly . . . 56
4.25 Exploded view of the IRIS buoy assembly . . . 57
4.26 Metacentric stability of the IRIS buoy . . . 58
4.27 Blue Robotics Compression Gland Penetrator . . . 60

5.1 IRIS Project Timeline (June 2025 – June 2026) . . . 62
5.2 Repeatability Test - 100th Repetition . . . 64
5.3 Buoy transmitting cellular payload in tank . . . 66
5.4 IRIS ready for deployment in Foster City . . . 67

D.1 Water Speed for 1.5 m Wave Height and 15 m/s Wind Speed . . . 101

List of Tables

3.1 System Requirements 13

4.1 Vertical Motion Trade-Off 29
4.2 Communication Trade-Off 37

5.1 Risk Assessment Table 72

A.1 Bill of Materials 85

C.1 Current Speed Sweep 96

E.1 Constants and geometry used in the stability analysis. 104
E.2 Component inventory used in the stability analysis 104
E.3 Hydrostatic equilibrium and metacentric results. 105
E.4 Ballast position sensitivity 106

Chapter 1

Introduction

1.1 Motivation

The ocean is a critical foundation for life on Earth, regulating the global climate, supporting biodiversity, and producing more than half of the oxygen we breathe [1]. Because of its essential role in environmental and human well-being, it must be carefully monitored and protected. Effective stewardship depends on the ability to gather reliable, long-term ecological data that reflects the health and changes occurring within marine ecosystems.

The Palos Verdes Peninsula represents one of the most significant marine restoration efforts on the West Coast. After decades of ecological degradation, including sediment instability, kelp forest loss, and contamination, the region has become a focal point for scientific research and habitat recovery. The construction of the Palos Verdes Reef was led collaboratively by NOAA's Montrose Settlements Restoration Program, the Southern California Marine Institute (SCMI), and the Vantuna Research Group (VRG) at Occidental College [2]. Their combined work created more than 30 acres of new reef habitat, designed to enhance kelp forest recovery and restore marine biodiversity along the Southern California coastline. Shown below in Figure 1.1, the reef habitat has been created along a stretch of coastline by dumping quarry rock at specific distributed sites. Nearby thriving reefs were used as guidance for this work, and particular considerations were taken to maximize the functionality and integration of these new structures. The depth of the structures at 15-20 meters deep was chosen based on survey of other areas with the highest amount of rocky reef productivity [3]. These restored reef regions provide the basis on which the system described in this paper has been designed.

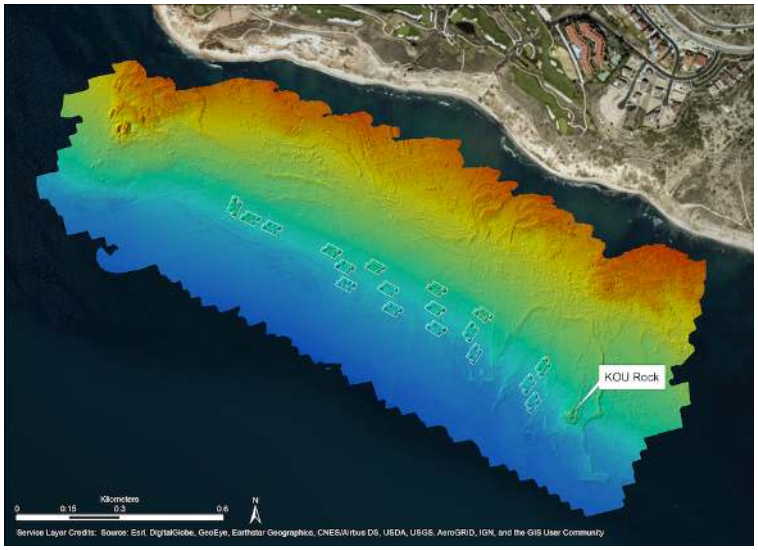


Figure 1.1: Topographical Map of the Palos Verdes Reef [4]

To support this restoration, long-term environmental monitoring is required. Santa Clara University's Robotics Systems Laboratory has been invested in the development of multiple robotic systems over multiple years to facilitate low cost, modular monitoring. Two notable systems developed over the past few years include REEF, a benthic reef health monitoring system (shown later in Figure 1.3), and an untethered Vertical Profiling robot shown in Figure 1.2. The Vantuna Research Group will deploy these systems to monitor the Palos Verdes artificial reef. The basis of this monitoring is to collect data on primary physicochemical parameters, including pH, dissolved oxygen, conductivity, and temperature. This suite of measurements is standard and used by scientists to gauge water health at various points in a water column. The vertical profiler described in [5] was designed with similar intentions and constraints as the project described in this paper, notably to aid in the ease of data collection for near shore oceanographic monitoring.

Long term installments in the area, like REEF, are intended for monitoring the evolution and changes of reef health without necessitating regular visits to the site, unlike the aforementioned vertical profiling robot. However, retrieving that data from underwater networks remains a challenge. Currently divers must still physically visit the deployment site based on the interval that they want to collect that data, creating redundant logistical complexity, increased cost, and potential safety risks simply to retrieve data for environmental monitoring.



Figure 1.2: Vertical Profiling Robot [5]

1.2 Background or Related Work

The REEF Sensor Network (Figure 1.3) is a Santa Clara University Robotic Systems Laboratory Master's Capstone project most recently worked on by Quinn Bates and advised by Dr. Michael Neumann and Dr. Christopher Kitts. The system is designed to take the necessary parameter measurements at the Palos Verdes Reef as specified by Dr. Jonathan Williams of Vantuna Research Group. The network is designed to be capable of gathering all of the data that was requested by Dr. Williams, which will aid in the restoration efforts of the group. According to Bates' Masters Capstone Report, "It was also determined that we should not design a system that needed physical interaction (such as batteries being replaced) more frequently than they already visit the restored reef, which is once a year" [6]. As stated, the planned interval of physical interaction with the network is once a year. This is due to the massive undertaking that is sending a diver down to retrieve data from the network or to make any physical changes to it. The intention of IRIS is to increase data accessibility for scientists, without increasing the frequency of diver deployment to the sensor network. The system outlined in this paper does not change any of the monitoring tactics or operating principle used by the existing REEF sensor network. Rather it operates independently, providing a critical additional level of communication functionality by passively integrating into the existing system.

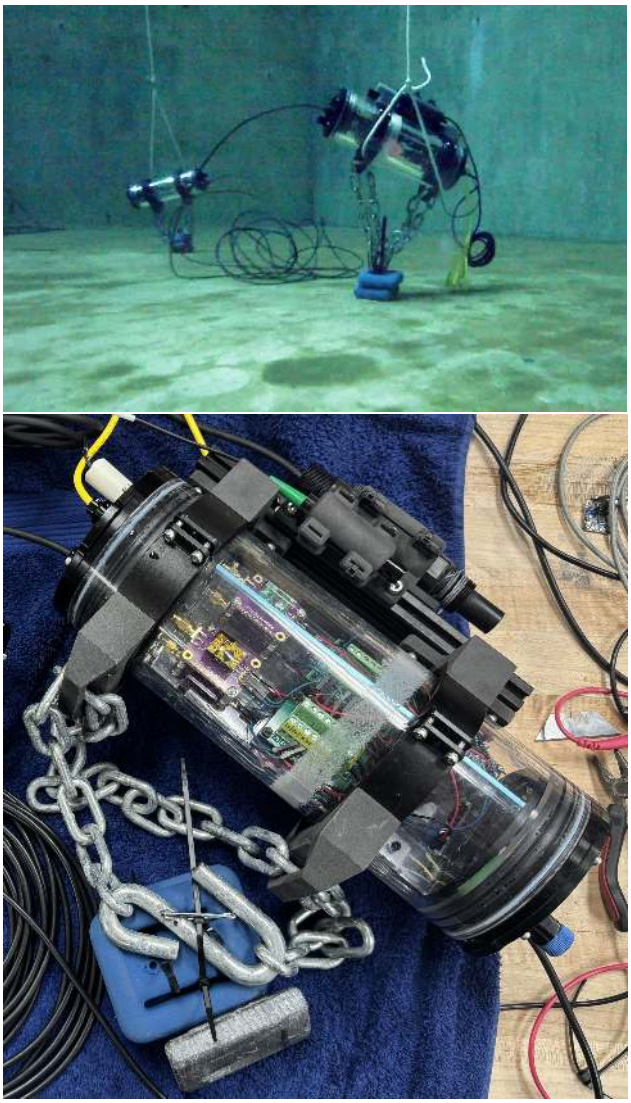


Figure 1.3: REEF Sensor Node Deployed and On Land [6]

1.3 Objectives

The goal of this project is to develop an autonomous, retractable communication system that enables reliable data retrieval from underwater environmental sensors while remaining protected from harsh surface conditions and human factors (e.g. vessel strikes, vandalism). This system, known as the Integrated Reef Information System (IRIS), is designed to support long-term ecological monitoring at the Palos Verdes Reef by surfacing only when needed to transmit data to shore. By reducing the need for diver-based data retrieval, IRIS aims to improve the safety, efficiency, and continuity of restoration monitoring efforts.

1.4 Approach

In this first iteration of the project, our team focused on developing a proof-of-concept system that demonstrates the core functionality required for a retractable data-relay buoy. This included designing and prototyping a winch-based actuation mechanism, developing a compact surface buoy equipped with communication hardware, and building a power and electronics module capable of supporting extended underwater operation. With IRIS still in its early stages, this proof-of-concept provided critical insights into winch performance, tether durability, buoyancy control, communication reliability, and long-term system stability. These findings will help to support the eventual deployment of a fully operational IRIS system at the Palos Verdes Reef and any other potential deployment sites.

Because this system represents a new direction in underwater ecological monitoring, the name IRIS—Integrated Reef Information System—has been adopted to reflect its role as a communication "gateway" between benthic sensors and surface research teams.

By improving data accessibility, reducing maintenance requirements, and enhancing the reliability of long-term monitoring, IRIS directly supports ongoing restoration work at Palos Verdes Reef. It provides marine scientists and resource managers with the continuous ecological information necessary to evaluate restoration progress, understand environmental change, and protect one of California's valuable reef ecosystems.

Chapter 2

Research

2.1 Background Research

In order to gather as much customer and stakeholder data as possible to inform our project, we used a combination of interviews with both experts in the industry and a representative scientist from the Vantuna Research Group that we expect to assist with this project, as well as benchmarking existing systems that could help inspire system development.

Modern reef monitoring faces several long-standing challenges. Traditional diver-based surveys provide valuable ecological data but are limited by diver safety, weather conditions, and the amount of time divers can spend underwater. Apprill et al. describes these limitations and emphasizes the need for a new generation of automated sensor systems capable of continuously measuring reef health indicators such as temperature, chemistry, and biological activity. Their work highlights a global shift toward integrated monitoring platforms that deliver higher-resolution data without relying on frequent human intervention [7]. This movement toward autonomous, persistent monitoring technologies forms the scientific foundation for our project.

Through our online research we found other examples of people trying to accomplish similar goals. Researchers have explored multiple methods of improving data retrieval from underwater sensors. One compelling solution is a retractable buoy system described by Wainman (Figure 2.1). This design stores the buoy underwater during normal conditions and uses compressed air to raise it to the surface when data transmission is needed. By retracting after each communication cycle, the buoy avoids wave action, weather exposure, and surface damage [8]. Wainman's prototype demonstrates that retractable systems can successfully operate for months at shallow depths and can house most electronics in a stationary docking base, reducing the payload on the surfacing buoy. This idea directly parallels our interest in a protected communication system that only surfaces when required, influencing several early concepts for IRIS.

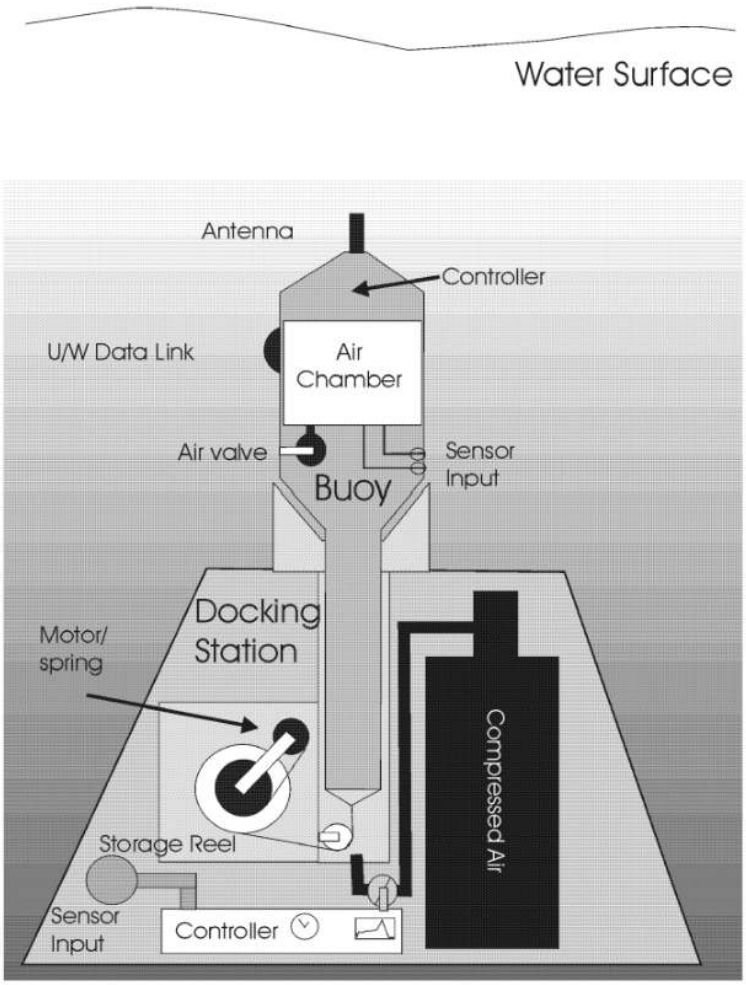


Figure 2.1: Wainman Retractable Autonomous Platform (WRAP) buoy [8]

Another relevant system is a self-elevating buoy platform utilizing a submersible winch, described by Weng et al. Unlike traditional fixed buoys, their design responds dynamically to ocean conditions: when water becomes rough, the buoy retracts below the surface for protection. When conditions improve, the winch raises it again so it can transmit environmental data via cellular networks [9]. This dual-buoy configuration proves that winch-driven actuation can reliably support repeated vertical cycling in real marine conditions.

Surface-standing buoys remain a dominant tool for ocean monitoring, and several real-world systems reinforce the reliability of satellite-based transmission. Martínez-Osuna et al. describe a coastal buoy deployed in the Gulf of Mexico, that uses solar panels, $LiFePO_4$ batteries, and the Iridium network to send hourly data to shore [10]. Their deployment highlights the strengths of satellite communications for marine telemetry: global coverage, proven reliability, and low maintenance. The main limitations of this system, being constant surface exposure and dependence

on solar charging, underscore exactly why IRIS must take a different approach. Because IRIS spends most of its time underwater without access to solar power, another power strategy is required.

The Palos Verdes Reef Restoration Project, constructed by NOAA, SCMI, and the Vantuna Research Group, provides the ecological and operational context for our work. Within 18 months of construction, researchers documented rapid increases in kelp canopy, fish biomass, and benthic cover, all of which point to significant ecological recovery [9]. This progress, however, depends on continuous data collection to track subtle, sometimes rapid, changes in reef conditions. Williams et al. report the use of a surface-deployed Spotter buoy [11] during restoration monitoring, which transmitted real-time data successfully but suffered from solar limitations and surface damage during storms [9]. This experience demonstrates that while surface buoys are effective communication tools, they are vulnerable in dynamic environments like Palos Verdes. Their challenges provide strong justification for a retractable, protected design like IRIS.

Together, these research efforts reveal several consistent themes: surface buoys are vulnerable, diver retrieval is costly and risky, and autonomous retractable systems offer promising advantages but remain underdeveloped for shallow coastal restoration sites. IRIS draws directly from these findings to create a system capable of supporting one of California's largest and most important reef restoration efforts.

2.2 Customer Needs

From interviews we found that there were five primary categories that customer and stakeholder needs fell under, Buoy Design, Winch System, Communication, Deployment, and Longevity. High priority objectives include buoy retractability, quality data, and buoy and system repeatability. Benchmarking also yielded interesting results as it shed light on existing systems. Other existing systems range from large, power-rich ocean platforms to compact, low-power prototypes. An NPS buoy (Figure 2.2) offers robust endurance but requires complex deployment and in the case of the NPS buoy, is enormous and designed for deeper marine environments [12]. The goal of the IRIS project is to develop a system that can complete many of the functional tasks that some of these other projects have demonstrated, yet have an easily deployable form factor, low cost, and efficient field serviceability.



Figure 2.2: NPS Ocean Sensing Buoy During Deployment [12]

We chose to designate specific needs centered around the system's core functionality and the operational environment as high priority. We chose to focus on these because addressing them mitigates the most immediate and critical risks identified by the experts we interviewed, which is crucial for a first prototype deployment.

First, the needs for the winch (tether management and depth of installment) were chosen as a high priority because the winch mechanism is the project's central technical challenge and potential failure point. As marine engineer Bill Kirkwood noted in discussion, entanglement is a high-risk failure mode so prioritizing tether management and stability is important. A single tether failure can result in a failed deployment, so repeatability is paramount. The depth needs must also be met to ensure the system can integrate well into the preexisting system and predetermined location (the reef at a depth of 60 feet).

Second, the needs for the buoy (depth and orientation for communication) were chosen because the physical survival of the system and effective data retrieval are critical. A minimum of 5 meters of retracting is vital for protecting the system from small boat traffic and hostile waves during storms, both of which can damage the equipment. Additionally, orientation within the water for communication is an important factor to ensure that when the buoy surfaces for the required data transfer, it remains stable enough to acquire a clean satellite or cellular signal despite potential wave interference.

Finally, data quality is prioritized over data frequency because the primary customer goal is scientifically useful data. If the data is frequent but imprecise, the entire project is unreliable and therefore invalidated.

Some needs were intrinsically connected and solving one would solve the other. For example, while biofouling is a major concern, it is partially mitigated by the high-priority need of limiting surface exposure time (as advised by Geoff Wheat). Others were found to be the opposite, offering conflicting paths forward particularly when consid-

ering manufacturability. An example of this is that while a low cost is desirable, the design cannot compromise the high-priority need for repeatability, and marine grade winches and specialized underwater components happen to be expensive. Despite this, the development focus will remain primarily on addressing the identified key needs while being aware of, and making plans for addressing the secondary needs.

2.3 User stories

The following user stories highlight specific user groups interaction with the system. These include some of the driving factors that guided the development of IRIS through a stakeholder focused lens to ensure the system is a thorough and seamless solution.

Vantuna Research Group Scientist

- **As a scientist** monitoring the Palos Verdes Reef, I want to have consistent data about the water, **so that** I can efficiently preserve and restore the health of the reef.

IRIS Engineer

- **As an IRIS engineer** and a part of the team responsible for the deployment of IRIS, I want the tethering system to be resilient, **so that** repeatable data transmission cycles are guaranteed.

IRIS Engineer

- **As an IRIS engineer** and a part of the team responsible for the reef sensor network, I want the buoy to be retracted to a distance of at least 5 meters below surface, **so that** the buoy will stay intact and last for its entire deployment period.

Vantuna Research Group Scientist

- **As a scientist** monitoring the Palos Verdes Reef, I want the data that I receive to be high quality and precise even if it comes at the cost of less frequent transmission, **so that** I can have accurate information about the Palos Verdes REEF.

IRIS Engineer

- **As an IRIS engineer** and a part of the team responsible for the reef sensor network, I want to ensure a resilient battery and electrical system, **so that** the system can operate properly for the entire duration of a deployment.

Chapter 3

System Overview and Rationale

3.1 System Definition

IRIS is a retractable, surface communication buoy designed to remain protected underwater during normal operation and surface only when data transfer is required based on a programmed schedule. When scheduled to transmit, the buoy rises to the surface, establishes a cellular link, and transmits environmental data back to researchers, eliminating the need for diver retrieval and dramatically increasing monitoring efficiency.

To inform the development of IRIS, prior work from the Santa Clara University Robotics Systems Lab (RSL) and our partner institutions was reviewed, including benthic monitoring platforms and surface communication buoys deployed along the Southern California coastline. Of particular relevance is previous work on the REEF sensor nodes in development for the Vantuna Research Group and NOAA's Montrose Settlements Restoration Program as part of the Palos Verdes Reef Restoration Project. These existing systems provide the foundation for IRIS and highlight the need for a protected, on-demand communication interface capable of surviving a dynamic coastal environment.

The first development effort of IRIS, as discussed in this paper, is centered on validating the feasibility of a retractable underwater data-relay platform through a convincing prototype system. The system integrates two primary physical subsystems: the winch and actuation mechanism, and the retractable cellular communications buoy, coordinated through an end-to-end software and communications architecture. Together, these subsystems allow IRIS to remain submerged and protected during dormant periods, surface on command to relay accumulated sensor data, and return below the surface to resume monitoring. While IRIS is still in its early stages, this first iteration will provide critical insights into winch performance, tether durability, buoyancy control, communication reliability, and long-term system stability. These findings will support the eventual deployment of a fully operational IRIS system at the Palos Verdes Reef.

The name IRIS, short for Integrated Reef Information System, reflects the system's function as a communication

gateway between benthic sensors and surface research teams. It gathers ecological data from instruments stationed on the reef, stores that data during submerged operation, and delivers it to researchers each time the buoy surfaces and establishes a communication link.

3.2 System Requirements

The requirements for the retractable-buoy communication system collectively prioritize long-term autonomous operation, reliable data delivery, and robust survivability in harsh marine environments. From discussion with customers and stakeholders, durability, environmental resilience, and consistent communication performance were found to be the dominant design drivers, reflecting the need for a system that can remain deployed for months with minimal intervention. Secondary priorities emerge around safe power management, ease of deployment and recovery, and responsible material selection, ensuring the buoy not only performs its sensing and transmission functions but also integrates cleanly into operational workflows and the surrounding marine ecosystem. These themes guided our design decisions when defining the following requirements.

3.2.1 List of Requirements

Table 3.1: System Requirements

1	The housing shall be waterproof and withstand submersion at 30 m depth without leakage, with a 6 month maintenance cycle.
2	The system shall resist corrosion and biofouling in seawater for a minimum deployment period of 6 months without loss of function.
3	The tether should be capable of retracting repeatedly to greater than 10 meters without tangling.
4	The buoy should survive wave heights up to 1.5 m and wind speeds up to 15 m/s without mechanical failure.
5	The system shall be constructed using materials rated for UV exposure $\geq$ 1 year in a marine environment.
6	The power system shall operate safely between 12°C – 22°C while including charge and short-circuit protection.
7	The system shall transmit sensor data at least once a week.
8	The system should transmit under normal oceanic conditions, and shall repeat the transmission until successful, if not received.
9	All sensor readings shall be time stamped with $\pm$1 second accuracy.
10	There should be less than 1% data loss as it transmits through the system.
11	There shall be an accessible interface by the end user for scientific purposes.
12	The system should be capable of 32 kb/week of data delivery.
13	The total system weight should not exceed 45 kg, allowing two-person deployment and recovery.
14	The system shall use non-toxic, reef-safe coatings and materials for any part in water contact.
15	The system shall be able to operate continuously for $\geq$ 180 days on DC battery.
16	The system should cost less to use per data collection event than the existing solution.
17	The buoy shall have high visibility coloring, as well as RSL identifying markings.
18	The system may be capable of wireless firmware updates over the cellular link.

3.2.2 Rationale

The primary objective of this project is to develop a reliable and practical ocean-deployed data communication system that prioritizes robustness, maintainability, and operational simplicity. Due to the limited development timeline of a single academic year, requirements were prioritized according to their necessity for core system operation. Functional requirements essential to minimum viable operation were given the highest priority, while secondary non-functional requirements were incorporated where feasible within schedule and resource constraints. The operational needs identified by both the Robotics Systems Lab at Santa Clara University and the Vantuna Research Group strongly influenced this prioritization process.

A key system requirement is the ability to transmit sensor data at least once per week, representing a substantial improvement over the previous recovery schedule of approximately once per year. To support long-term deployments, the system must also withstand harsh marine environmental conditions, including prolonged seawater exposure, wave loading, UV exposure, corrosion, and biofouling. These requirements are necessary to ensure continuous data collection without frequent maintenance or system recovery. Additionally, reliable tether deployment and retraction are critical for repeated operation throughout the deployment cycle without mechanical failure or tangling.

Non-critical requirements were included to improve usability, maintainability, and long-term system performance, but were not considered essential for baseline functionality. Examples include weight limitations to simplify deployment and recovery procedures, and minimizing power consumption to reduce battery requirements during extended deployments. Another desirable capability is wireless firmware updating through the cellular communication link. While not required for general functionality, remote updating would allow adjustment of transmission schedules and other operating parameters in response to changing environmental conditions or collection requirements. While many of these secondary requirements were able to be met regardless, they were deemed not mission-critical and were deprioritized.

3.3 Concept of Operations

The concept of operations shown below (Figure 3.1), from the most recent REEF Master's capstone by Quinn Bates [6] demonstrates the existing functionality of REEF, including the lack of easy data collection.

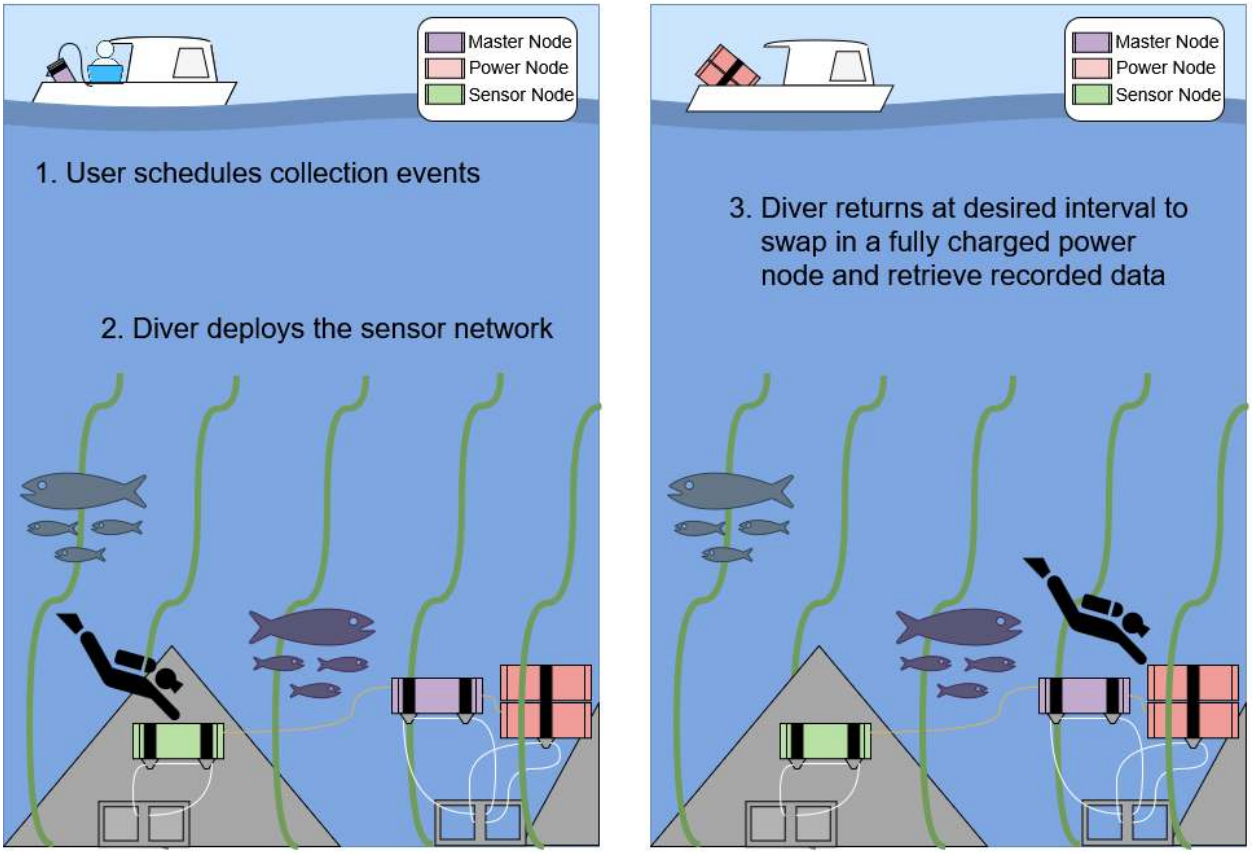


Figure 3.1: REEF - Concept of Operations [6]

The fundamental goal of IRIS is to replace the need for step three in Figure 3.1, particularly for the recorded data retrieval. An updated concept of operations for IRIS, including the deployment of REEF can be seen below (Figure 3.2). The entire system, both IRIS and the existing REEF network will be programmed with transmission and collection schedules respectively. Following their appropriate schedules, the data will be collected, and then transmitted through the IRIS system to the cloud, where the scientist can access it.

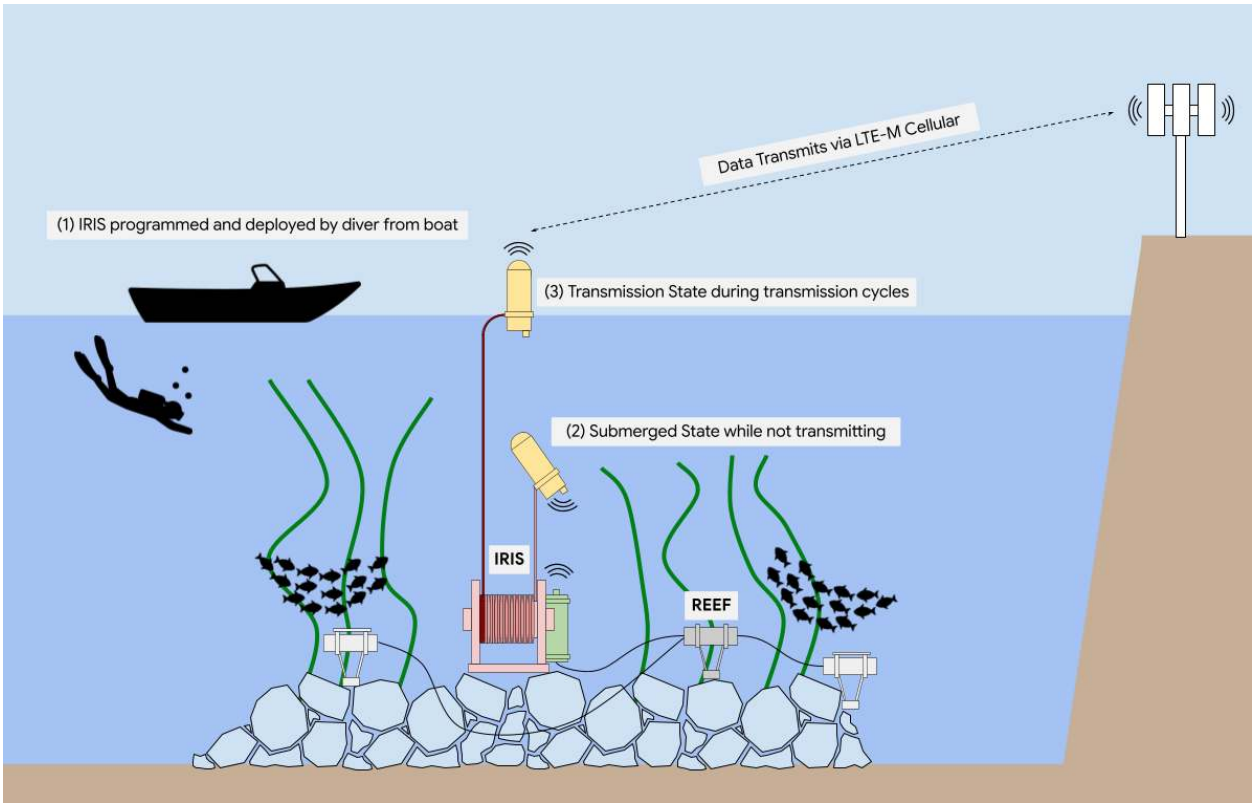


Figure 3.2: IRIS - Concept of Operations

3.4 System Components

Figure 3.3 illustrates the organizational architecture of the IRIS system, using color-coded blocks to distinguish between custom subsystems and existing infrastructure. Grayed out components represent pre-existing infrastructure, specifically the REEF Master Node. The green block delineates the Subsea Electronics, which serves as the data hub for incoming sensor network data, and actuation control for the winch. Powered by a Feather Adalogger, this enclosure pulls data from REEF through a WetMate connector and RS-485 serial communication. Using SPI microSD card storage, it provides underwater storage of the data before getting transmitted to our buoy. Connected to this subsea electronics enclosure, is one of our Succorfish Delphis acoustic modems which establishes connection between the enclosure and the buoy. The components within the blue block comprise the Winch sub-assembly, featuring two different models of Blue Trail Servos and a gear reduction system designed for open-water operation to control the buoy's depth. Finally, the yellow block represents the Buoy sub-assembly, which utilizes a Boron 404x Cell MCU and internal battery to relay subsea data from the acoustic modems to the shore-based cellular network. The Boron 404x receives data through its own attached acoustic modem which goes through RS-232 first, then it temporarily saves

this data to a microSD card in order to publish it to the web. The Boron 404x is set to a schedule, which allows it to activate the subsea electronics enclosure on a specified time interval.

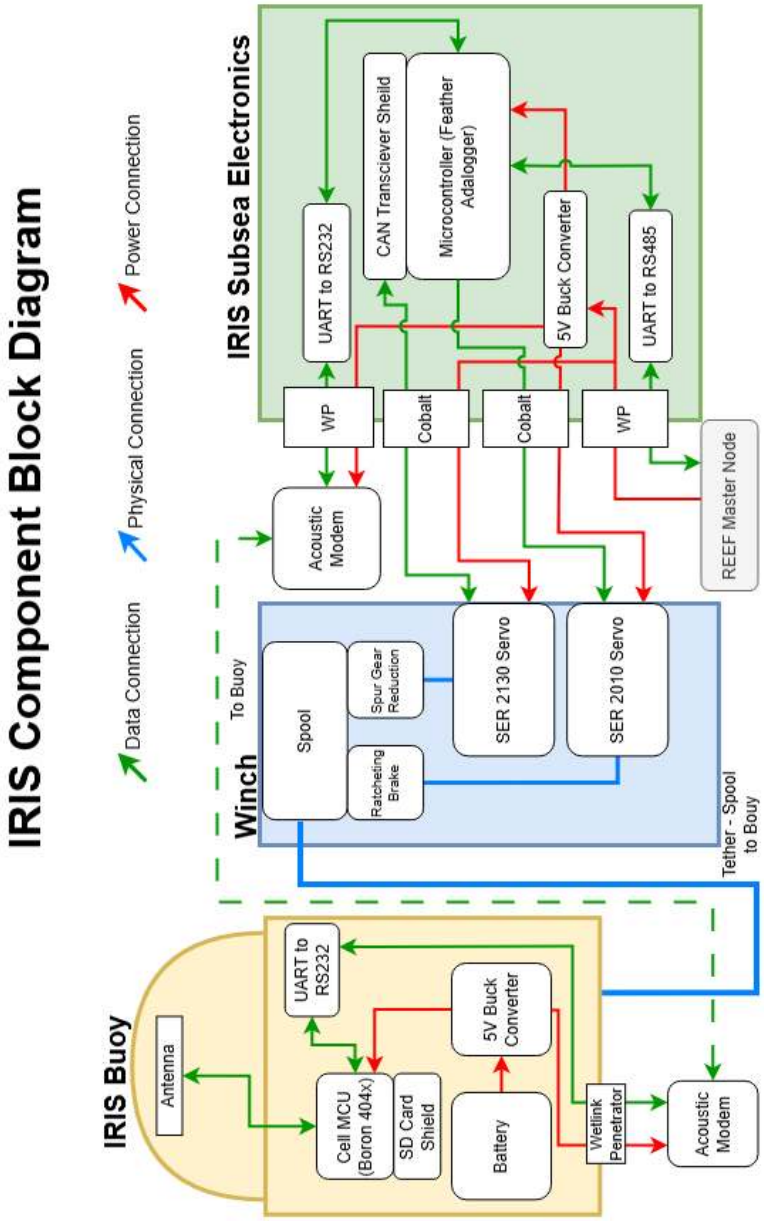


Figure 3.3: IRIS - Component Block Diagram

3.5 Wiring

The wiring diagram in Figure 3.4 below shows the exact wiring and pinouts for the system in its current state.

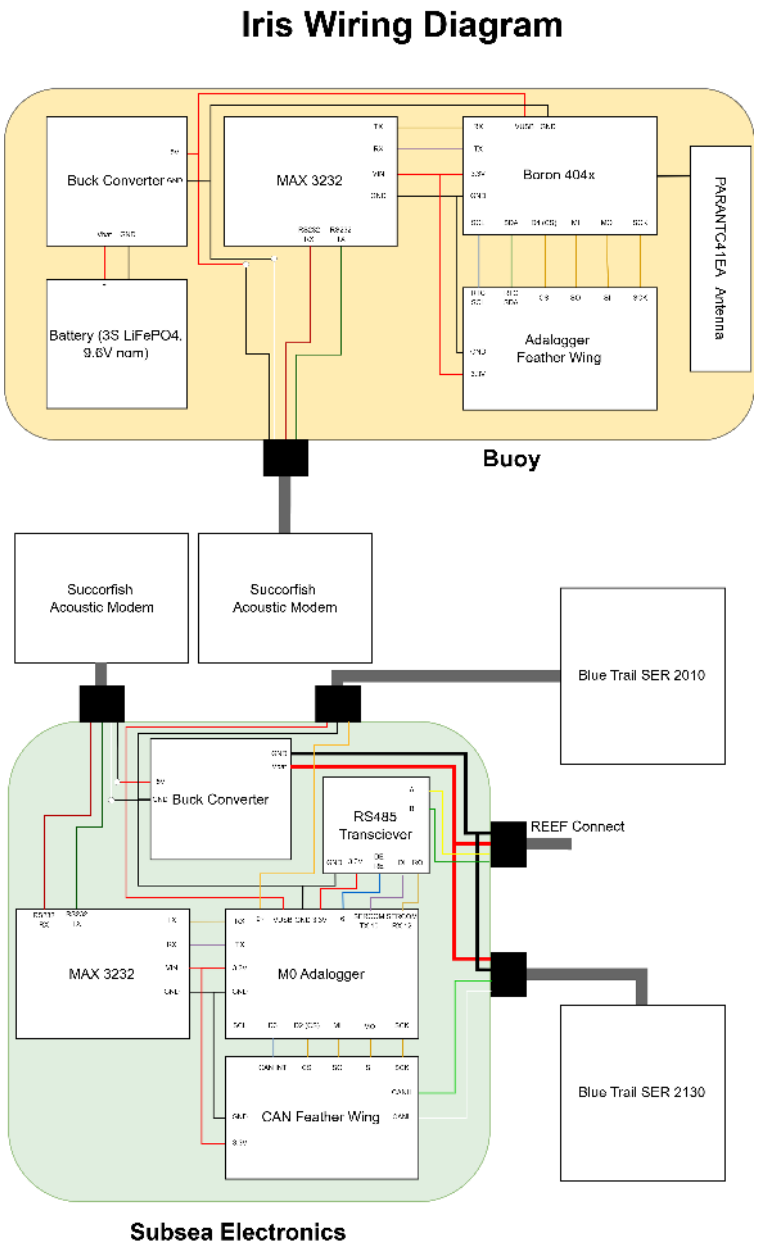


Figure 3.4: IRIS - Wiring Diagram

3.6 Software Overview

3.6.1 Subsea Electronics State Diagram

Figure 3.5 shows the software state diagram of the subsea electronics, run from a Feather Adalogger M0 microcontroller. It spends most of its time in a listening state, where it collects data coming in from REEF, and stores it on its own SD card. When it receives the command from the buoy, it completes a transmission cycle by dumping data to the buoy via acoustic, deploying the buoy, retracting after confirmed transmission, then returning to the listening state.

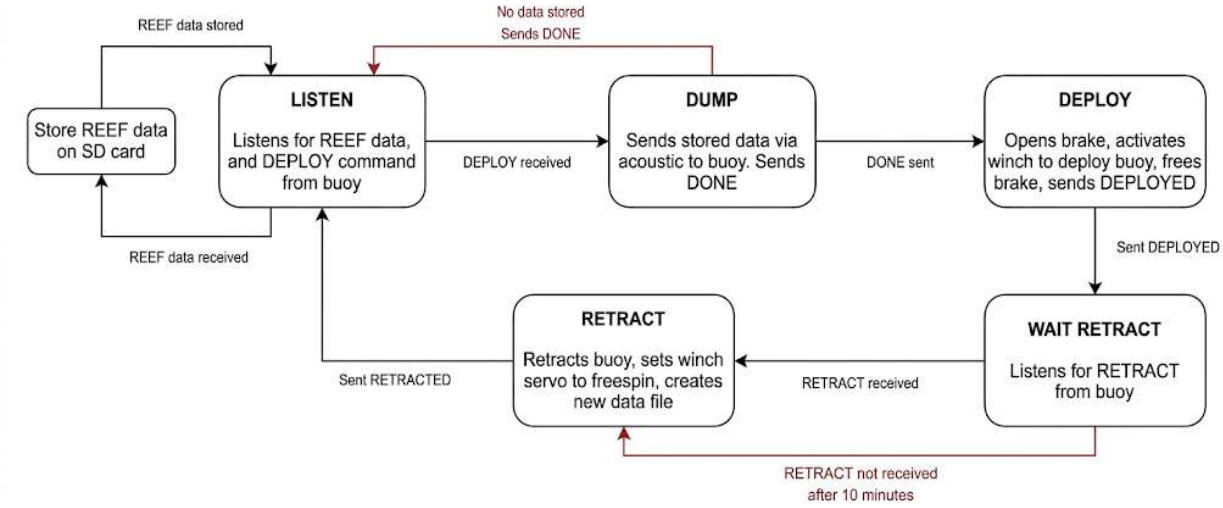


Figure 3.5: Subsea Electronics Enclosure State Diagram

3.6.2 Buoy State Diagram

Figure 3.6 shows the software state diagram of the buoy, run from a Particle Boron 404x microcontroller. Similar to the subsea electronics, most of the time is spent in an idle/waiting state. While in this state, it keeps an internal clock which keeps track of when a cycle should start. When the clock reaches the desired time interval, it sends the command to deploy to the subsea electronics, and goes through a deployment cycle which includes receiving the data, storing it temporarily, transmitting via cellular, sending the retract command to the subsea electronics, and returning to the idle state with the clock reset.

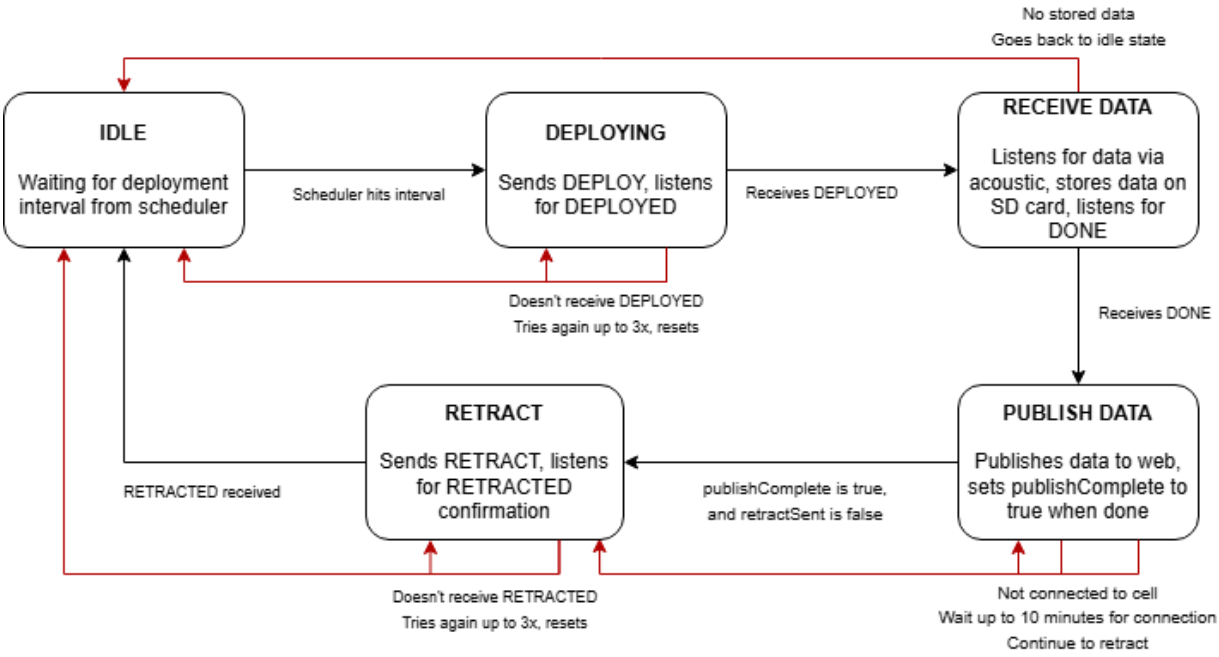


Figure 3.6: Buoy State Diagram

3.6.3 Use Cases

The following use cases were derived from the user stories, keeping in mind the real people who will be interacting with and benefiting from the IRIS/REEF design. The main actors are the scientist(s) who will be receiving the data, the operator who is responsible for the system as a whole, and the diver(s) who will be physically deploying and retrieving the system from the reef.

Use Case 1: View data from web endpoint

Actor: Vantuna Research Group Scientist **Preconditions:** Scientist has an established connection with REEF
Main Flow:

1. User opens the IRIS interface
2. User checks recent data updates and views past data

Alternate Flow: There are no new data updates, retrieval is still available for previous cycles **Postconditions:** Scientist has access to all current data

Figure 3.7: Use Case 1: View Data from web endpoint

Use Case 2: Initial deployment of system

Actor: IRIS Engineer and Divers **Preconditions:** Weather is cooperative of a deployment and location is chosen
Main Flow:

1. Group goes out on boat to the location of the sensor deployment
2. A dive team delivers the system to its proper locations

Alternate Flow: Weather is uncooperative with a deployment and it's postponed to a later date **Postconditions:** IRIS is successfully deployed and begins collecting data

Figure 3.8: Use Case 2: Initial deployment of system

Use Case 3: Monitoring of system health and efficiency

Actor: IRIS Engineer **Preconditions:** IRIS is deployed and actively gathering data
Main Flow:

1. User opens the web endpoint
2. User checks for any gaps in data transfer.

Alternate Flow: There are no errors, everything is running as planned **Postconditions:** Any issues with the system are made aware to the engineer

Figure 3.9: Use Case 3: Monitoring of system health and efficiency

Use Case 4: Maintenance of system

Actor: Operator and Divers **Preconditions:** An error has been identified which requires physical maintenance of the network.
Main Flow:

1. Dive team and engineer are sent to the location of IRIS.
2. Divers dive to the sensor, and adjust or replace necessary parts
3. Divers return to the surface and ensure proper functionality

Alternate Flow: If problems can't be resolved, the divers will extract IRIS from the ocean and it will be brought back to the SCU RSL. **Postconditions:** Problems are solved or actively being worked on.

Figure 3.10: Use Case 4: Maintenance of system

Use Case 5: Retrieval of system

Actor: Operator and Divers **Preconditions:** IRIS has completed its deployment.
Main Flow:

1. Operator and dive team take a boat to the location of the system
2. Divers dive down to the sensor network and bring it up to the surface
3. The sensor network and buoy are safely transported back to the SCU RSL

Alternate Flow: If weather conditions aren't cooperative, system will be left for longer, even if it isn't collecting data **Postconditions:** The entire system is taken out of the reef and returned safely to the RSL at SCU

Figure 3.11: Use Case 5: Retrieval of system

The UML use case diagram in Figure 3.12 highlights these previously mentioned use cases in a visual showing the three primary parties involved with IRIS.

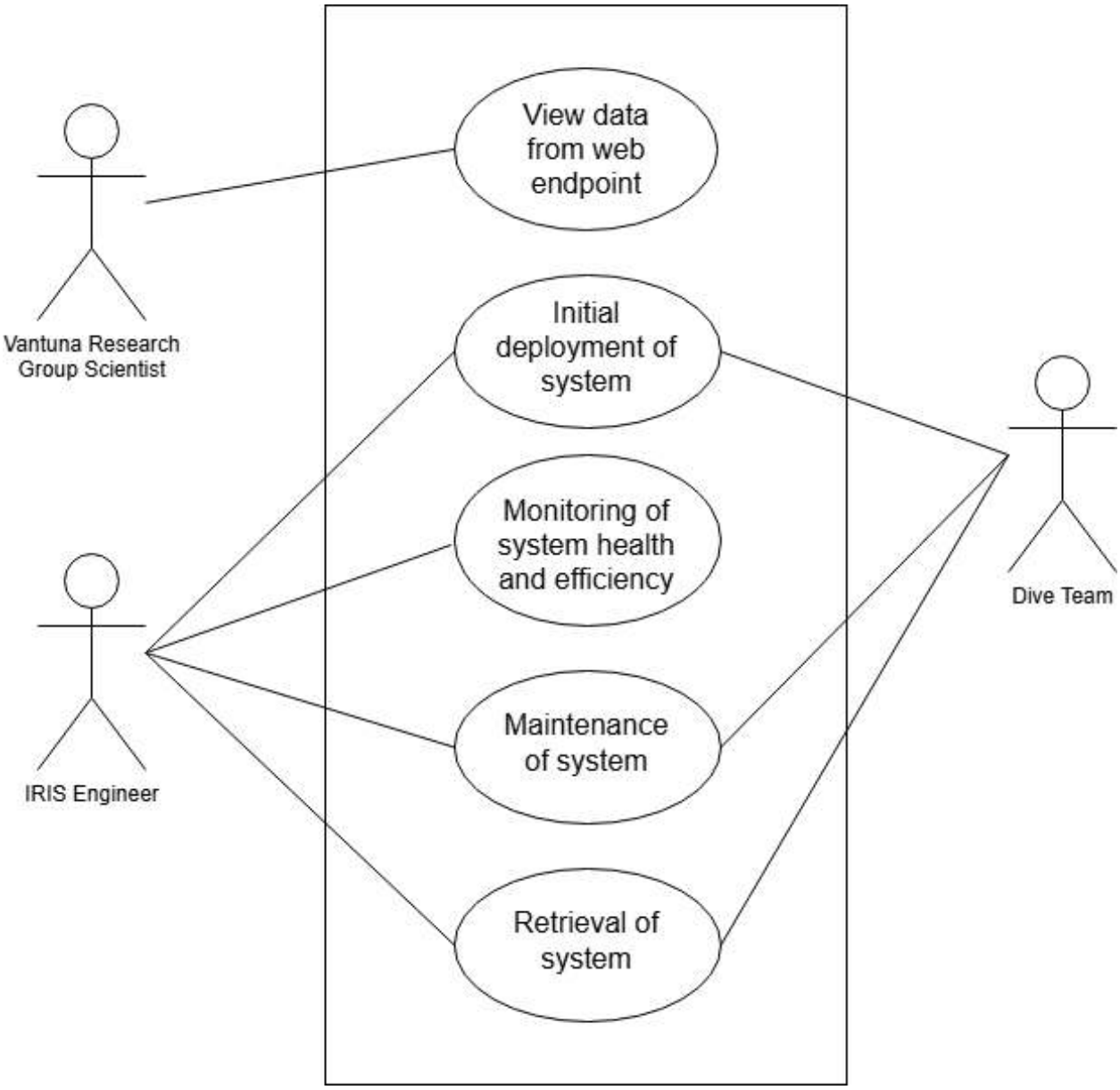


Figure 3.12: UML Use Case Diagram

3.6.4 Class Diagram

Figure 3.13 shows a class diagram for the IRIS system. The diagram shows the critical distinct aspects of IRIS and their unique functionalities, as well as their interdependencies.

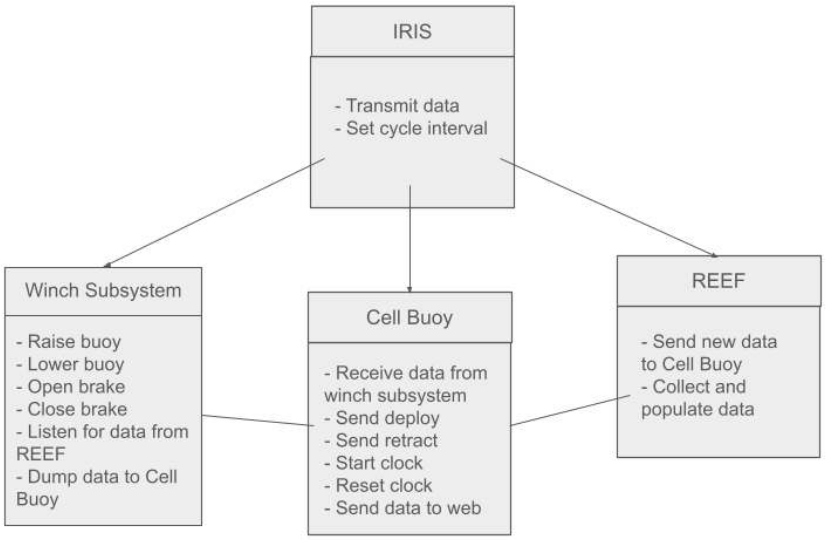


Figure 3.13: Class Diagram

3.6.5 Sequence Diagram

The sequence diagram in Figure 3.14 demonstrates the progression of actions through the system for the end to end functionality of IRIS. This starts with data being collected at sensor nodes, and ends with the data landing at the website endpoint via cellular.

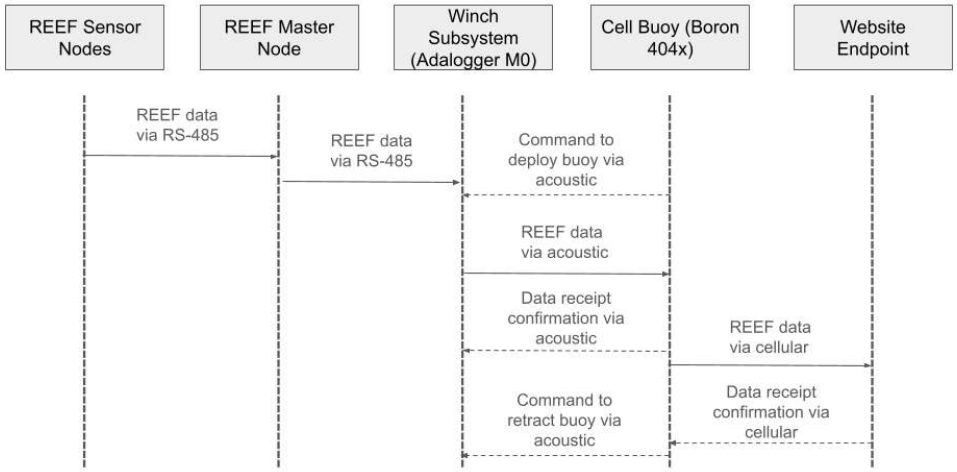


Figure 3.14: Sequence Diagram

Chapter 4

Detailed Design

4.1 Fundamental System Architecture

The contents of this section describe the method of motion that makes the IRIS buoy move up and down. This includes how force is generated, transmitted, and controlled to achieve reliable ascent, descent, and hold at position. The vertical motion choice drives mission performance, energy use, reliability in saltwater, maintainability, cost, and schedule. It also locks mechanical, electrical, and software interfaces for the rest of the system.

4.1.1 Design Drivers

- **Transmission Frequency:** Must be able to transmit weekly
- **Sea-keeping ability:** Withstand surface conditions of the Beaufort sea state of 7.
- **Mass and volume:** Total system limited to 45 kg and one cubic meter
- **Power:** Runs off of DC power from batteries with adequate protection circuitry.
- **Schedule:** Should be self-reliant on a 6-month service rotation
- **Environmental durability:** Durable in protecting against water ingress, UV-stable materials.

4.1.2 Proposed Solutions

- **Buoyancy engine (variable buoyancy system)**

  A mechanism that changes the buoy's buoyancy by adjusting internal volume or ballast mass, often through pumping fluid in or out of a chamber.

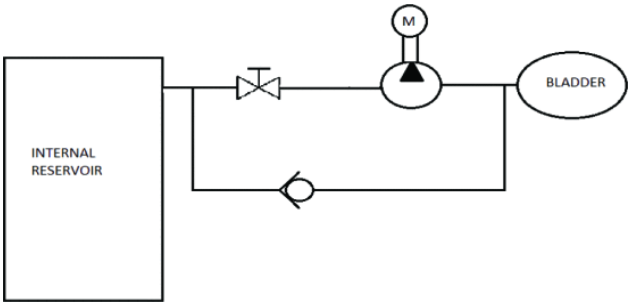


Figure 4.1: Diagram of variable buoyancy engine functionality [13]

- **Winch (Typical Solution)**

  A motorized spool connected to a tether that raises or lowers the buoy by reeling in or letting out the line. Depth control is achieved by controlling the motor's rotation direction and speed.

- **Disposable Buoy**

  A one-time-use buoyant device that detaches or inflates to bring a data capsule to the surface.

- **Spring/Thruster System**

  A system that uses a torsional spring or mechanical bias to move the buoy downward and a thruster to drive motion upward and keep the buoy at the surface during data transmission.

- **Linear arm**

  A system that moves the buoy through a rigid arm, lowering it through rotation about a point below the water surface.

4.1.3 Motion Trade-Off Drivers

A. **Long Term Relative Cost (20%)**

This category looks at the expected cost of the system over multiple deployment and maintenance cycles. A score of 0 means the system has a high cost per deployment, such as needing full replacements of major components. A score of 3 indicates that some parts must be replaced or extensively repaired each cycle, creating a moderate recurring expense. A score of 5 means the system experiences no part failures and requires only minimal replacement parts after each deployment, keeping long-term operational costs low. This driver is weighted heavily at 20% because our system must remain affordable across many years of repeated field deployments, making long-term operational cost one of the most critical factors.

B. **Environmental Impact (20%)**

Environmental impact measures how the system affects the surrounding natural environment, especially regarding pollutants or marine debris. A score of 0 means the system fouls the environment or creates harmful debris. A score of 3 reflects minimal pollutant leakage or limited disturbance. A score of 5 means the system does not rely on any pollutants or components that could enter the environment or generate debris during operation. This driver carries a 20% weight because minimizing negative ecological effects is essential for any long-term, in-water system and directly impacts regulatory acceptance and sustainability.

C. **Repeatability (15%)**

Repeatability describes how consistently the system can surface and submerge throughout the deployment. A score of 0 means the system is not repeatable and behaves inconsistently. A score of 3 indicates it is mostly repeatable but not perfect. A score of 5 represents full repeatability and consistent performance every cycle. This driver is weighted at 15% because the entire purpose of the system depends on reliable surfacing and submerging, and repeated failures would render the design unusable.

D. **Mechanical Complexity (15%)**

Mechanical complexity refers to the complexity and number of moving parts, seals, and required servicing. A score of 0 means the design uses very few actuators, moving parts, or seals. A score of 3 includes some complex moving systems and sealing mechanisms. A score of 5 is given to systems that are very mechanically complex, with multiple actuators, seals, and interconnected moving parts. This driver is weighted at 15% because overly complex systems increase failure modes, maintenance effort, and cost, while simpler systems are more robust in harsh marine environments.

E. **Biofouling/Corrosion Resistance (10%)**

This category evaluates how well the system withstands seawater exposure without losing function from corrosion or buildup of marine organisms. A score of 0 applies to systems heavily affected by corrosion and biofouling. A score of 3 reflects moderate resistance. A score of 5 means corrosion and biofouling have negligible effect even during extended deployments. This driver is weighted at 10% because long-term performance in seawater is essential, but the issue is partially mitigable through materials and coatings, giving it moderate, but not primary importance.

F. **Power Consumption (10%)**

Power consumption measures the total power needed for each transmission or submergence cycle. A score of 0 means the system draws excessive power or has significant idle draw. A score of 3 indicates manageable power usage with low static consumption. A score of 5 means the system uses minimal or no power over the deployment length. This driver is weighted at 10% because deployment longevity depends on power efficiency, but power systems can be optimized or scaled if needed, making it important but not the top priority.

G. **Communication Integration (10%)**

Communication integration assesses how easily sensor data from the network connects to the system for transmission. A score of 0 means the system has a complex or inconsistent data pathway. A score of 3 indicates adequate and consistent data transfer. A score of 5 means the system provides consistent, constant data communication without interruption. This driver is weighted at 10% because meaningful operation requires reliable data transfer, but communication issues are often solvable through software or interface improvements rather than large mechanical changes.

4.1.4 Trade-Off Analysis

Table 4.1: Vertical Motion Trade-Off

Driver Code	**A**	**B**	**C**	**D**	**E**	**F**	**G**	**Total**
Winched Tether	4	4	5	4	3	2	4	3.30
Buoyancy Engine	4	4	4	3	3	3	4	3.13
Disposable Buoys	1	1	1	3	4	5	3	1.89
Thruster Driven	3	4	3	3	3	1	4	2.66
Linear Pivot Arm	3	3	3	2	4	3	5	2.70

4.1.5 Tradeoff Conclusion

Based on the results of the tradeoff analysis, the disposable buoy, thruster-driven, and linear pivot arm options can be eliminated from further consideration. Their respective average weighted scores of 1.89, 2.66, and 2.70 indicate that they fall well below the threshold of viable performance compared to the other alternatives.

The winch and variable buoyancy engine are the two top contenders, with average weighted scores of 3.30 and 3.13, respectively. The winch performed especially well in repeatability, as it can reliably cycle between dormant and transmission states. However, its relatively low score in power draw introduces uncertainty of its feasibility. Additional analysis of power requirements, efficiency improvements, and potential optimization strategies is needed to determine whether it can be made energy-efficient enough to justify its use. The variable buoyancy engine did not exhibit any major weaknesses (no score below 3 out of 5 in any category), but it also lacked standout advantages, earning consistent 3's and 4's across the board. It remains a stable, well-rounded option; however, its benefits are

outweighed by the stronger repeatability and mechanical control offered by a winch-based system. Although both concepts performed well across most evaluation categories, the remaining uncertainty lies primarily in characterizing the power requirements of marine winches rather than comparing the two architectures further.

At this stage, the winch has been selected as the primary approach, and subsequent analysis efforts will focus on refining its design and validating its power and reliability characteristics in the marine environment.

4.1.6 Winch Structure

Figure 4.2 illustrates the CAD model of the winch spool, which serves as the structural foundation for the IRIS system's tether management. The assembly features a central drum mounted on reinforced vertical supports, designed to provide a stable base for the mechanical deployment and retrieval of the buoy. This assembly is built entirely out of high strength acetal and 316 stainless steel for good corrosion and moisture resistance. Acetal (Delrin): high dimensional stability and has nearly zero water absorption, preventing the drum from swelling or binding in subsea conditions. 316 Stainless Steel: This is marine grade standard due to its molybdenum content, which provides superior resistance to pitting and crevice corrosion compared to standard 304 stainless. There are some complexities in machining this assembly. The solutions to these issues are better seen in Figure 4.3. Due to the constraints of our machine shop, we cannot bolt through the acetal drum; therefore, we have settled on bolting directly into it, mating the motion of the shaft and drum together with a variety of machine bolts. Lastly this winch structure helps to manage the tether. There is an attachment point through the center of the drum, this will act as an anchor point for the tether, ensuring the line is secured before the first wrap begins. The flanges prevent tether bird-nesting and jumping off the spool during deployment and retrieval. Figure 4.2 illustrates the CAD model of the winch spool, which serves as the structural foundation for the IRIS system's tether management. The assembly features a central drum mounted on reinforced vertical supports, designed to provide a stable base for the mechanical deployment and retrieval of the buoy. This assembly is built entirely out of high strength acetal and 316 stainless steel for good corrosion and moisture resistance. Acetal (Delrin): high dimensional stability and has nearly zero water absorption, preventing the drum from swelling or binding in subsea conditions. 316 Stainless Steel: This is marine grade standard due to its molybdenum content, which provides superior resistance to pitting and crevice corrosion compared to standard 304 stainless. There are some complexities in machining this assembly. The solutions to these issues are better seen in Figure 4.3. Due to the constraints of our machine shop, we cannot bolt through the acetal drum; therefore, we have settled on bolting directly into it, mating the motion of the shaft and drum together with a variety of machine bolts. Lastly this winch structure helps to manage the tether. There is an attachment point through the center of the drum, this will act as an anchor point for the tether, ensuring the line is secured before the first wrap begins. The flanges prevent tether bird-nesting

and jumping off the spool during deployment and retrieval.

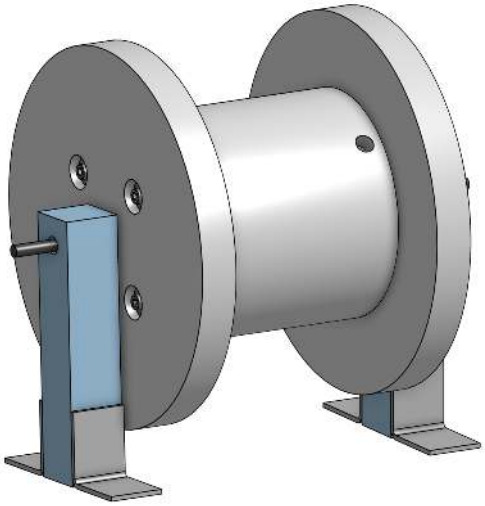


Figure 4.2: CAD model of Spool

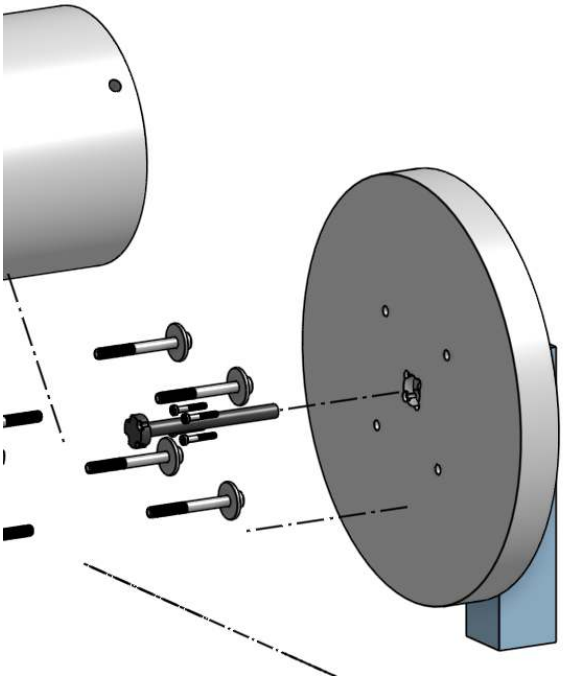


Figure 4.3: Partial Exploded View of CAD Model

4.1.7 Tethering

The IRIS system utilizes a single tether to manage both the structural and operational demands of the buoy's vertical motion. The single tether was selected over a dual cable configuration (one for structural, one for data) after it was determined that managing two cables simultaneously on a single spool introduced unacceptable complexity and entanglement risk. This decision also eliminated the need for a slip ring, which would otherwise be required to prevent a data carrying tether from twisting as the winch spool rotates. While waterproof slip rings exist, they are expensive, specialized, and introduce an additional potential failure point in a demanding marine environment. By moving the data transfer to an acoustic link between the subsea electronics enclosure and the buoy, the tether is relieved of any data carrying responsibility entirely. The acoustic communication protocol and its implementation are discussed in detail in Section 4.2.

The tether selected is a hollow braided polypropylene rope. Propylene was chosen for several reasons: it is naturally buoyant, resisting the tendency of a submerged tether to add downward load on the winch during the ascent, it is highly resistant to moisture absorption and saltwater degradation, and it exhibits low stretch which provides more predictable spool behavior and position control. The hollow braid construction offers the flexibility for clean spooling without kinking while maintaining sufficient tensile strength for the forces applied to the system during deployment. Given the scale of the IRIS buoy and the relatively shallow deployment depth of approximately 60 feet, the structural load on the tether is driven primarily by a net buoyant force and tether drag during retraction, both of which are well within the capabilities of the selected rope with safe margin. An image of the tether is shown in the next chapter in Figure 5.2.

The tether terminates at the buoy via a pair of designed attachment mounts fixed to a Blue Robotics 4" tube clamp on either side of the pressure vessel (buoy). These SLS nylon printed mounts each incorporate a teardrop-shaped lug with a shackle hole sized for a 1/4" 316 stainless steel anchor shackle. Both shackles connect to the tether, which merges to a single line running down the winch spool. The clamp position along the tube can be adjusted to shift the tether attachment point relative to the buoy's center of buoyancy and center of gravity, providing a simple mechanical tuning parameter for buoy stability without requiring redesign of any components. The CAD model of the buoy assembly including the clamp mounts is shown below in Figure 4.4.

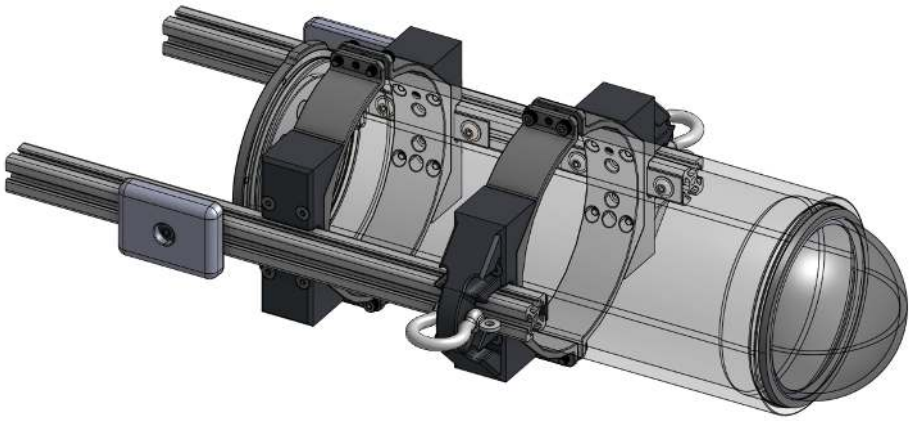


Figure 4.4: CAD model of Buoy

4.2 Subsystem One - Communications

This is the subsystem that will enable the buoy to transmit data, collected by the REEF system, back to shore. The communication method is critical for IRIS' core functionality. A reliable communication system is necessary for scientists and marine biologists to receive timely data from the REEF system and ensure confidence in successful and accurate data transfer. Furthermore, the communication method drives the structural design of the buoy.

4.2.1 Design Drivers

- **Transmission Frequency:** Must be able to transmit weekly
- **Volume:** Infrastructure able to fit within the surface facing part of the system
- **Schedule:** Should be self-reliant on a 6-month service rotation
- **Environmental durability:** Durable in protecting against water ingress, UV-stable materials.
- **Sea-keeping ability:** Transmit to surface conditions of the Beaufort sea state of 4.
- **Must not require infrastructure outside of the Palos Verdes Reef:** Cannot infringe on land not owned by Vantuna Research Group
- **Communication Range:** Must have a minimum communication range of 1km.

4.2.2 Proposed Solutions

- **Cellular (Typical)**

  A communication method using mobile networks such as LTE or 5G on land (horizontally from the buoy) to transmit and receive data when the buoy's antenna is above the water and within range of a cell tower. Figure 4.5 below shows Boron 404x currently being used in the IRIS development.

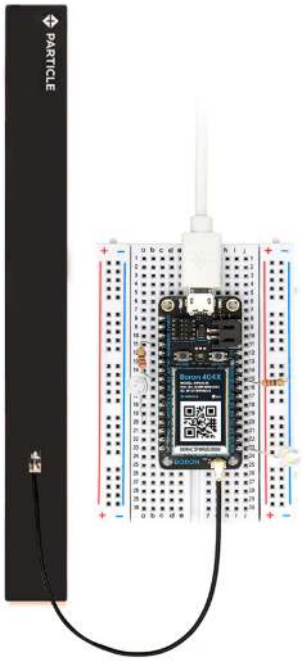


Figure 4.5: Particle Boron 404x Kit [14]

- **Satellite**

  A system that uses satellite networks directly above the buoy, such as Iridium or Inmarsat, to send and receive data between the buoy and a ground station, enabling communication over large distances and with less wave interference. Potential options include the Iridium Certus 9704 shown below in Figure 4.6. Other satellite considerations include satellite Wi-Fi options like Starlink.



Figure 4.6: Iridium Certus 9704 [15]

- **Acoustic**

  An underwater communication system that transmits data using sound waves, one of the few communication mediums with the capability of transmitting through water. This requires a boat to be deployed within range of the system. An illustration demonstrating the operating principle of acoustic communication is shown below in Figure 4.7.

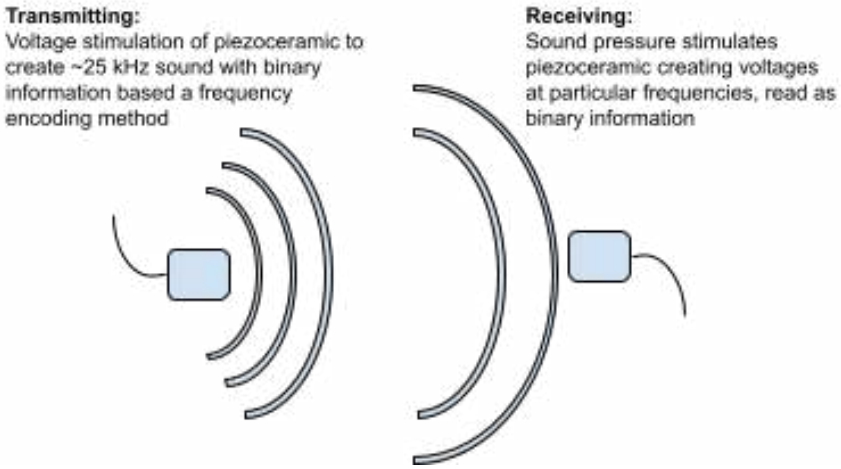


Figure 4.7: Operating principle of acoustic communication

- **Free Space Optical**

  A short-range communication method using visible or infrared light, typically from LEDs or lasers, to transmit data through light beams. Requires a clear line of sight from the buoy. Optical communication utilizes a transmitter and a receiving sensor, encoding information in light frequency, amplitude, or simple On/Off.

- **LoRa**

A low power, long range radio communication protocol operating in low frequencies. It transmits small amounts of data over large distances but requires an additional unit acting as a receiver or gateway within range.

4.2.3 Communication Trade-Off Drivers

A. **Ease of Data Retrieval (25%)**

This driver considers how complex it is for stakeholders to receive the data from the system. A score of 0 indicates that retrieval difficulty actively prevents the action of retrieving data, whereas a score of 3 represents semi-hands-off data access. A score of 5 reflects simple integration of fully automated data download. With a weight of 25%, this factor is heavily emphasized because ensuring accessible, straightforward data delivery is essential for effective system use.

B. **Communication Speed (10%)**

This category assesses the rate of data transfer. A score of 0 reflects communication occurring only at a bits-per-second scale, while a score of 3 corresponds to kilobits-per-second speeds. A score of 5 denotes megabits-per-second capability. Weighted at 10%, this driver captures the importance of timely transfer without being the dominant factor in system performance.

C. **Surface Condition Effects (25%)**

This driver evaluates how susceptible the communication protocol is to waves particularly with protocols limited to horizontal communication constraints. A score of 0 means waves consistently obstruct communication or require an unfeasible buoy structure. A score of 3 indicates the system may still be affected by waves but under reasonable circumstances it transmits successfully. Finally, a score of 5 reflects consistent and accurate transmission despite low elevation out of the water. With a 25% weight, this factor is critical due to the system's need to operate reliably in dynamic surface conditions.

D. **Range (10%)**

Range refers to the distance over which a signal or message can be effectively transmitted and received. A score of 0 represents patchy or nonexistent performance at 1 km, a score of 3 indicates stable connection beyond 1 km, and a score of 5 corresponds to global reach. Weighted at 10%, this driver acknowledges transmission distance as important but not overwhelmingly dominant in the decision framework.

E. **Upfront Cost (10%)**

This driver captures the direct costs associated with purchasing the communication related components required to transmit in the particular method. A score of 0 reflects costs greater than $3000, a score of 3 represents an approximate range of $500–$1000, and a score of 5 indicates costs below $100. With a weight of 10%, upfront cost is considered in balancing performance with financial feasibility.

F. **Ongoing Cost (20%)**

This category focuses on the recurring cost of data transmission services or supporting infrastructure. A score of 0 indicates costs exceeding $10/kB for weekly transmissions, a score of 3 corresponds to $1–$5/kB, and a score of 5 reflects a cost of $0/kB. Weighted at 20%, ongoing cost plays a significant role due to the multi-month operational nature of the system and the need to keep recurring expenses manageable.

4.2.4 Trade-Off Analysis

Table 4.2: Communication Trade-Off

Driver Code	A	B	C	D	E	F	Total
Cellular	5	5	2	4	4	4	3.30
Satellite	5	3	4	5	3	3	3.39
Acoustic	0	2	2	1	2	1	1.03
Optical	0	3	1	3	0	5	1.59
LoRa	3	3	2	4	4	5	2.87

4.2.5 Trade-Off Conclusion - Buoy to Shore Communication

For communication between the buoy and shore, the acoustic and optical methods can be removed from consideration, having scored 1.03 and 1.59, respectively. Their performance in the key metrics of data retrieval and surface conditions are extremely lower than the alternatives, which alone removes them from consideration.

The cellular and satellite options emerged as the top candidates, scoring 3.30 and 3.39, respectively. The LoRa option scored moderately high as well at 2.87, but despite this, it is not an ideal solution due to its practical limitations in underwater communication and dependency on proximity to shore-based receivers. Furthermore, it is only slightly cheaper than cellular while being slower and less easy to retrieve the data.

Between the two leading options, the choice largely depends on the tradeoff between performance stability and cost. Cellular communication faces challenges when waves obstruct the antenna, potentially requiring structural adjustments to the buoy to maintain a consistent signal. Satellite communication potentially avoids these issues because it communicates with satellites overhead, rather than horizontally toward cell towers. As such, it provides more reliable connectivity in open-water conditions, but it comes with higher hardware and operating costs, as well as lower data rates. This means that satellite is the better option if cost were not a factor, but since cellular may be suitable and

cheaper, it must also be considered. Additionally, for the same reasons, LoRa can again be eliminated from consideration since cellular is effectively the "cheaper" option that we have to satellite, and LoRa only provides the benefit of being cheaper than cellular and satellite. Additionally LoRa requires a gateway on land, which may not be feasible based on the ownership at the coastline according to Professor Williams.

While both satellite and cellular communication were initially considered as equal front runners, our analysis indicated that cellular is the preferred approach for this project based on cost, availability, and ease of integration. Since we already have access to cellular hardware, it will be practical to conduct small-scale tests to evaluate the effects of wave interference on signal performance and determine realistic operational limits.

Satellite communication, while technically viable, introduces substantially higher equipment and subscription costs and would require specialized antennas unsuited to our current constraints. Direct testing of satellite hardware is also cost-prohibitive, so any further investigation into satellite performance will rely on literature review or consultation with domain experts rather than empirical testing.

Overall, cellular should be treated as the baseline communication method moving forward, with remaining work focused on validating its performance in dynamic sea-state conditions and finalizing antenna placement and housing design. Satellite will remain a secondary contingency option but is no longer considered the primary communication path.

4.2.6 Trade-Off Conclusion - Subsea Electronics to Buoy

As discussed in sections 3.6–3.7, it was determined that spooling and managing two cables simultaneously was suboptimal. Despite early prototypes and concepts to support this management, including a separate tensioning buoy for the data cable, with advice from Bill Kirkwood, and a desire to streamline the cable management, the decision was made to move away from a separate data cable.

Though numerous load capable data tethers exist, such as the Blue Robotics Fathom ROV tether seen below in Figure 4.8, they all face notable concerns when it comes to winding on a traditional winch. The stationary end of the cable, as it exits the winch will continue to twist as the winch winds the cable in.



Figure 4.8: Blue Robotics Fathom ROV Tether [16]

The traditional solution for a spinning cable assembly is to install a slip ring. Slip rings are able to maintain data communication without the twisting force being applied to the cable. The diagram from MOFLON in Figure 4.9 demonstrates the working principle of a slip ring, in which brushes have constant contact with a rotating body, resulting in electrical continuity, without the wires on either end being rigidly connected.

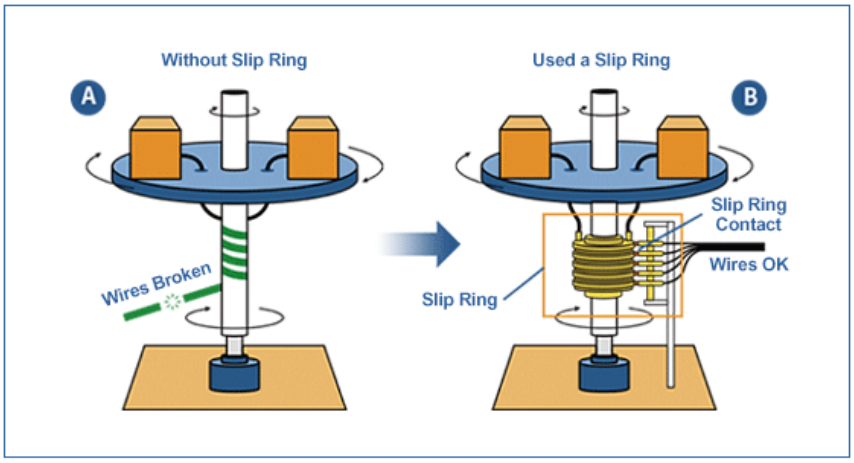


Figure 4.9: MOFLON Slip Ring Working Principle [17]

While waterproof slip rings do exist, and would be a feasible architecture when developing on the concept of a traditional winch, they are very expensive and specialized. Under the intention to build a low cost system, this idea was abandoned in favor of using a wireless method to communicate the data.

Though acoustic was vetoed for the buoy to shore communication method due to its slow speed, it had a distinctly strong fit to solve this particular issue of getting data into the buoy. Since the buoy spends the majority of its time

submerged, there is no restricting time limit on the acoustic modems to transfer all of the collection period's sensor data to the buoy prior to cellular communication. The Robotic Systems Laboratory had purchased extremely low cost acoustic modems in the expectation of enabling an intermediate solution to the problem that IRIS intends to solve, by allowing a boat to transmit and receive data from the REEF system, without sending a diver down. IRIS instead utilizes these acoustic modems, purchased from a company called Succorfish (Figure 4.10), to communicate the data from the subsea electronics attached to the REEF system wirelessly to the buoy. The protocol and functionality of this communication will be further explained in the following section.



Figure 4.10: Succorfish Delphis Acoustic Modem [18]

4.2.7 End to End Electrical

The entire IRIS system utilizes multiple communication protocols that will be explained in this section. The beginning of the communication is understood to be the sensor nodes of REEF collecting data and transmitting. The termination of the communication is understood to be the publishing of the data to a website online.

Upon receiving signal from the master node, the REEF sensor nodes record data and transmit it over an RS485 network. RS485 is a particularly robust communication protocol that uses a differential voltage to transmit serial data over long distances. By connecting to the A/B differential data lines that connect the REEF network, IRIS can access the data that the REEF sensors record without needing to interrupt the REEF microcontroller functionality or edit the firmware.

As the REEF system records data, the IRIS system is able to listen and store all data matching the desired string format to an SD card using an SPI data protocol. SPI is a particularly useful high speed, short distance communication protocol that allows a primary device to communicate with peripherals. The IRIS subsea microcontroller, at the desired schedule, then communicates with the acoustic modems through a MAX 3232 UART to RS232 converter.

RS232 is another serial data communication protocol, similar to UART or RS485, however uses higher voltages to represent "1" bits (up to +15V) in order to protect against noise. Using particular commands transmitted through serial, the IRIS subsea microcontroller transmits the sensor data packages through the acoustic modem, which is received at

the buoy.

The buoy's reception of the data happens as a mirror of the process on the seafloor. The acoustic modem of the buoy receives the transmission, and then forwards it through RS232 converted to UART to the microcontroller in the buoy. The buoy microcontroller stores the data via SPI into an SD card.

Once the desired measurement duration has been achieved, IRIS will begin its transmission state, transmitting all of the data recorded from the REEF sensors to the buoy. The buoy will then be released to the surface, and the data stored on the buoy SD card from the previous data transfer will be transmitted over cellular, and then published to a website online for access.

4.2.8 Final Software Flow

The final software flow consists of two programs, which are run from two microcontrollers: The Feather Adalogger M0 in the subsea electronics enclosure, and the Particle Boron 404x on board the buoy. Combining the two, the overall software flow is as follows.

The Boron keeps an internal clock with the deployment interval, and is counting down until it reaches that time. While this is happening, the Adalogger is intermittently receiving data from REEF and storing it on its SD card in the proper transmission format. When the deployment interval is reached, the Boron sends a deploy command via acoustic to the Adalogger. When the Adalogger receives this command, it begins the deployment cycle by sending all of its currently stored data to the Boron via acoustic where it's stored on the Boron's SD card. Once it's finished sending, the Adalogger opens the brake servo, and then drives the winch servo to deploy the buoy. Upon full deployment of the tether, the Boron connects to cellular and transmits the data to the cloud. From there it's grabbed by a Particle webhook and loaded into a google sheet, which is embedded within our website. Once data has been successfully sent over cellular, the Boron gives the retract command via acoustic and upon receiving this the Adalogger retracts the tether back to it's resting positions. From here both microcontroller reenter their idle/listening states and the Boron starts counting for another deployment interval. The source code for these software flows can be found in Appendix B.

4.3 Subsystem Two - Winch

This is the subsystem designed to address the fundamental motion tradeoff done in section 4.1. This is primarily concerned with the configuration of the motor such that it can transmit power to the spool for retracting the buoy without being exposed to the marine environment. Successful isolation includes a dry side electronics and motor region which is critical for the long term use of the motor, resisting corrosion, and preventing electronic shorts, while

maintaining the torque required for retraction.

4.3.1 Design Drivers

- **Waterproofing:** The drive shall be isolated from the marine environment at a depth of 30m for a 6 month deployment period.
- **Power:** The interface shall be able to handle the torque required to spool the buoy cable.
- **Corrosion Resistance:** The functionality of the components in the drive isolation system shall remain functional after 6 months of marine exposure.
- **Cost:** The system must fit within the budget.

4.3.2 Proposed Designs

- **Commercial Subsea Winch (Typical)**

  A system built and waterproofed by an external company that specializes in the field such as that seen in Figure 4.11 by Okeanus.



Figure 4.11: Okeanus Subsea Winch [19]

- **Submerged Airpocket**

  Winch within a submerged pocket of air, enclosure on sides and above keeps the majority of air in. Water pressurizes the air, reducing volume, but not reaching the winch seen in Figure 4.12.

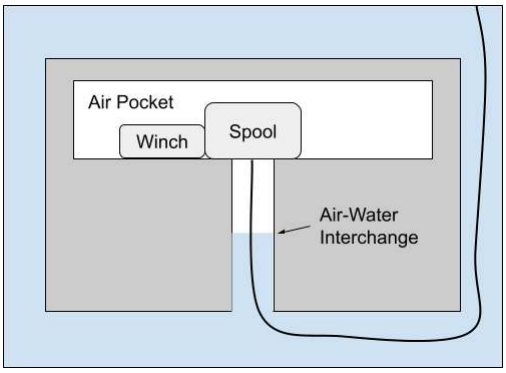


Figure 4.12: Theoretical submerged air pocket with dry winch

- **Shaft Seal (Secondary typical)**

  Rotary seal on shaft to prevent water ingress, potentially also requiring oil pressurized vessel for motor to sit in.

- **Magnetic Torque Coupling**

  Torque is transferred across an air gap and through an enclosure wall from the motor shaft to the spool via magnets arranged on either side as seen in Figure 4.13.

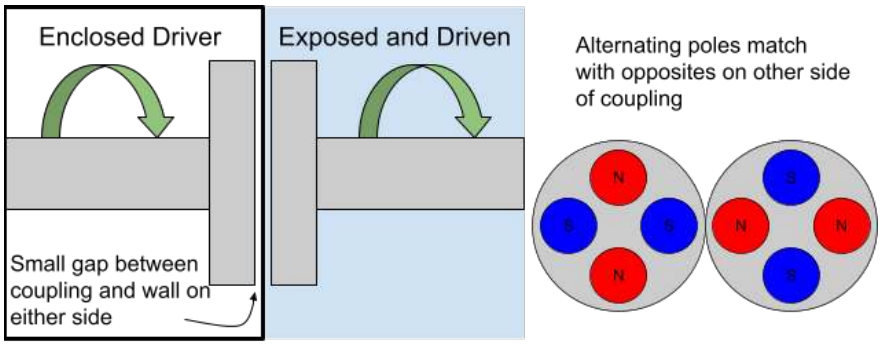


Figure 4.13: Magnetic coupling diagram

- **Submersible Motor**

  An existing brushless DC motor rated for submersion beyond required depth, such as that seen in Figure 4.14



Figure 4.14: Blue Robotics M200 submersible brushless DC motor [20]

4.3.3 Trade-Off Analysis

Though a formal full-team tabular rating was not completed for this subsystem, multiple solution architectures were analyzed for this subsystem.

The commercially available subsea winches that exist are well beyond budget, having gotten quoted orders of magnitude beyond what money is available for this subsystem. While the solution would be more "plug and play" it is not an option for this particular design.

The submerged air pocket mechanical configuration was deemed to be too complex as it requires maintaining an upright configuration throughout deploying the system and through the full six month schedule. Additionally, air-water interface introduced the concern of mixing and potentially losing air volume over time.

Shaft seals, while a notably straightforward approach, require extremely precise machining and potentially fixturing design and engineering work beyond the time scale of this project to be confident that there would not be failure.

The submersible motor market is not fully mature, though some frontrunners from reliable companies include the M200 submersible motor from Blue Robotics [20]. A motor like this could be a potential option, though torque becomes a significant concern.

Magnetic couplings provided an initially strong opportunity, as it fully isolates the driven part of the system from the driver. While many off the shelf options exist, few of them are made in a material suitable for continuous subsea applications (notably 6061 Al, 316L SS, or Titanium). Additionally, while less of a problem for axial couplings, the traditional magnetic face coupling requires a significant amount of mechanical configuration to have a precise air gap between the boundary and the coupling face. This still proved to be a strong frontrunner, and a significant amount of

development time was spent developing a solution.

4.3.4 Trade-Off Conclusion

Though magnetic couplings were the strongest initial option, facing the mechanical configuration and cost of a custom material coupling outweighed the benefits. The strongest contender became the submersible motor architecture, as it innately allowed full isolation, but required less unique configuration and customization to make it functional. The initial submersible motor was a gift from Ulyssess, though had too high of a power draw and no positional feedback, as well as no specification sheet to verify its ratings and requirements. A more appropriate solution was found and will be explained in the following section.

4.3.5 Submersible Servo-Motor

A convincing submersible motor option was discovered in the Blue Trail Engineering servo lineup. Specifically, the SER-2130, pictured below in Figure 4.15.



Figure 4.15: Blue Trail SER-2130 Waterproof Servo [21]

This servo utilizes an input voltage that is naturally within the range that the REEF system already uses (25.6V Nominal LiFePO4 Battery, and as high as 32V servo input). Traditional servos utilize PWM command to control the position of the motor to a precise angle, using a potentiometer to track the position. The SER-2130 communicates via Controller Area Network (CAN) to control the motor, and a magnetic encoder to track position. These features enable the servo to rotate ±32760 turns with programming. More specifically, the magnetic positional encoder can be utilized to program an arbitrary number of turns, as well as request the current position back to confirm. Since the servo uses a CAN bus interface, it is programmable and has numerous beneficial commands in its firmware to allow precise use.

The servo has a maximum torque value of 5.9 N*m which, based on the dimensions of the winch and size of the buoy, exceeds the value required. The calculations for this required torque are explained and shown below.

$$T = F_b + F_d - W \tag{4.1}$$

$$F_d = \frac{1}{2} C_d \rho v^2 A \tag{4.2}$$

$$F_b = \rho V g \tag{4.3}$$

$$W = mg \tag{4.4}$$

$$\tau_{drum} = T \cdot r_{drum} \tag{4.5}$$

$$\tau_{drum} = 2.85 \text{ Nm} = 25.18 \text{ in·lb} \tag{4.6}$$

The above calculations function to estimate the required torque by first calculating the required rope tension to lower the buoy (T). This tension balances the upwards buoyancy force (F_b) and the drag force (F_d) against the downward weight (W). The drag force depends on the water density (ρ), the projected cross-sectional area of the buoy (A), the relative water velocity (v), and the drag coefficient (C_d). Buoyancy is simply the displaced water weight, governed by the buoy's volume (V), while its dry weight is determined by its mass (m) and gravity (g). Then multiplying this tension by the maximum effective radius of the drum (r_{drum}) we get the final torque required (τ_{drum}). These calculations make some assumptions about the system and environment conditions that could affect the actual torque requirements. For example, it was assumed that the maximum effective radius of the drum and tether would be 2.75 inches, which would mean that the tether had wrapped around itself three times in the same location. Furthermore, the buoy-tether system was assumed to have a drag coefficient of 1.2 based on an academic reference of drag coefficients for the 2-D projection of a cylinder [22]. However, this value could change based on design if we determine that drag reduction is necessary through additions such as feathering of the tether. Additionally, drag force was calculated assuming that a current was acting directly against the motion of the buoy during retraction with a water speed of 1 m/s, which was determined to be the fastest current that might be experienced in this location based on the experience of Prof. Williams. This current was assumed to be acting perpendicular to the buoy cylinder as to calculate for the worst case scenario for our drag force. All of these assumptions and calculations lead to a factor of safety of around 2 which gives confidence that our system will have enough torque for most applications.

Despite exceeding the torque required for most applications, it is important to have a large factor of safety to account for the outlier events, including biofouling on the rope, wave and current motion, and static friction. Additionally, the tether drag was ignored in the calculations above, but later validated in section 4.3.9. For that reason, IRIS

utilizes a 2:1 gearbox manufactured from marine safe plastic, to further increase the maximum torque output.

4.3.6 Acetal Gearbox Speed Reduction

The gearbox functions to reduce the speed and increase the speed of the drive servo with a gear ratio of 2:1. This ratio provides ample torque output, doubling our torque safety factor to 4. The box is not fully watertight but it is fully enclosed to protect the spur gears and shafts from debris and from as much biomatter as possible. The enclosure is manufactured out of acetal (polyoxymethylene) for its excellent marine application, low cost, and high machinability. Acetal has excellent corrosion resistance, high strength, low moisture absorption, and low friction in salt water environments, making it the ideal material for this subsea gearbox. Both the enclosure and the gears are made from acetal. The enclosure was chosen to be made from acetal for its low cost, high machinability, and corrosion resistance, while the low friction played a higher role in the decision to choose acetal spur gears. The shaft bushings for the gearbox were chosen to be made from a Moisture-Resistant Ultra-Low-Friction Plastic Blend to provide low friction for the rotating shaft while being protected against swelling or warping from the salt water environment. Finally, the gearbox was designed so that the it is separate from the rest of the drive system. This was done so that future applications of IRIS with higher torque requirements can use a higher gear ratio without redesigning the whole system. The final gearbox design can be seen below in Figure 4.16.

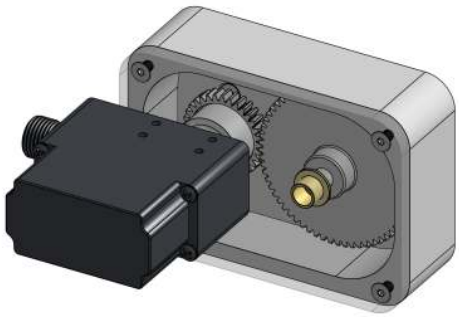


Figure 4.16: Gearbox Design

4.3.7 Ratchet & Pawl Braking

Since our buoy is positively buoyant, it must be held in place when it is in its fully retracted state. One option that is considered is to use a worm gear with a high gear ratio which is self-locking when the motor is not being actuated. This option is not viable for material and design considerations. First, there are not cheap, easily obtainable worm wheels and gears that are manufactured with marine grade materials. Second, the worm gear is self-locking due to high friction between the gears at a high gear ratio, however, since our servo motor provides sufficient torque without

significant gear reduction, an extremely high torque worm gear box is unnecessary for our application. Instead, we utilize a ratchet and pawl system to lock the drum in place. The pawl rests within the notches of the ratchet, being pulled into place by a spring. The ratchet gear is fixed to the drum and as the buoy is retracted, the pawl slides freely over each tooth. Once the buoy is fully retracted, the spring holds the pawl in place on the ratchet and prevents rotation in the drum in the ascension direction. When it is time for the buoy to ascend, the second servo disengages the pawl from the ratchet and allows the drive servo to raise the buoy. This system is beneficial since it has limited power draw due to the pawl servo only being actuated during the ascension process, then while the buoy is retracted the servo can be turned off to reduce power consumption. The ratchet and pawl were chosen to be cut out of titanium. We choose this over other marine enduring materials, such as acetal, because it is significantly stronger. This means it is less susceptible to chipping and we are able to make a thinner gear which is easier to interface with our design. The primary drawback of using titanium is that it is difficult to machine. Another disadvantage is that titanium is expensive. While it is within our budget, but we also considered cheaper options such as 316 stainless steel. The drawback with stainless steel is that our servo shaft is made from titanium, so there would be galvanic corrosion between the stainless steel pawl and the titanium shaft. We opted to go with the more expensive titanium pawl option for its integration simplicity since we do not have to design a method to conductively insulate a stainless steel pawl from a titanium shaft.

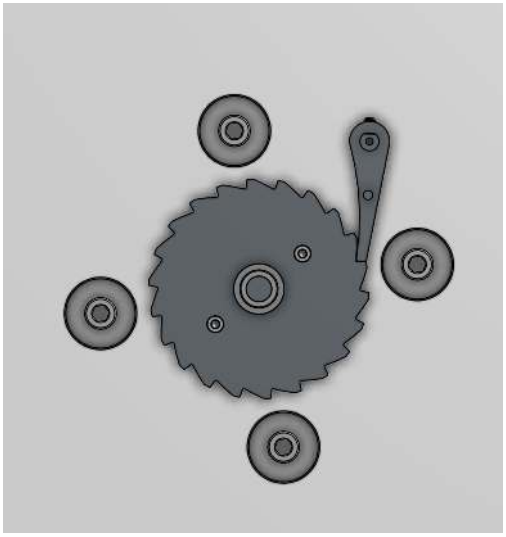


Figure 4.17: Ratchet & Pawl integration

4.3.8 Winch Design

The winch is the mechanism responsible for drawing the buoy underwater to protect it from surface hazards such as vessel strikes and wave loading, and it represents one of the central technical challenges of the IRIS system. Designing

a winch capable of surviving a 6-month deployment in harsh marine conditions presents a significant engineering challenge, as the system must maintain reliable operation despite continuous exposure to saltwater, biofouling, and cyclic mechanical loading, any of which can degrade moving parts or seize an improperly specified mechanism. Because the winch must actuate repeatedly throughout the deployment, every component was evaluated not only for its initial performance but for its ability to retain that performance after months of submersion. Figure 4.18 provides an overview of the design, which consists of a flanged drum supported by structural columns with a rope guide mounted on top to ensure consistent line layering during spooling. Torque is transmitted from the gearbox (Figure 4.16) to the drum, which wraps a strand of polypropylene rope to pull the buoy downward toward the ocean floor. Polypropylene was selected for its neutral buoyancy, chemical resistance, and low water absorption properties that make it well suited for long-term submerged use, ensuring the rope itself does not become a source of added load or degradation over time. To ensure that the winch survives the full 6-month deployment, all structural components were manufactured from acetal (Delrin), while all hardware including fasteners, shafts, and fittings was specified in 316 stainless steel. Acetal was selected for its excellent resistance to moisture absorption, dimensional stability in wet environments, and ease of machining, making it well suited for structural components that must maintain tight tolerances during extended submersion, as even small amounts of swelling could cause binding between rotating parts. 316 stainless steel was chosen for its superior corrosion resistance in saltwater compared to more common grades, due to its molybdenum content which provides enhanced protection against chloride-induced pitting, the primary mechanism by which marine hardware fails over long deployments. Together, these two materials form a complementary pairing that has precedent in marine robotics applications. Blue Robotics, a widely used platform in the underwater vehicle industry, employs the same acetal and 316 stainless combination throughout their thruster and enclosure hardware, lending confidence that this pairing will perform reliably in the IRIS application. The rope, as discussed above, is polypropylene, selected for its neutral buoyancy and low water absorption. This deliberate material selection strategy ensures that no component in the winch assembly represents a corrosion weak point, reducing the risk of mechanical failure over the deployment period and removing the need for any sacrificial or replaceable components.

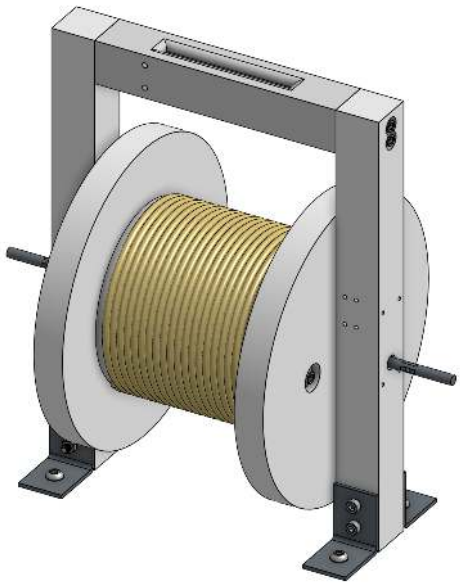


Figure 4.18: Winch Overview

The supporting structure, shown in Figure 4.19, consists of two vertical acetal columns joined by a top crossbar and anchored to the system frame by bolted foot brackets at their base. This frame establishes the rigid reference between the two drum shafts, holding them in coaxial alignment so that the drum rotates freely without binding even as the tether load shifts across the spool during retraction. Maintaining this alignment is critical, since any misalignment between the shafts would introduce side loading that increases friction and accelerates wear over the deployment period. The columns are spaced to fully capture the flanged drum between them, and the foot brackets distribute the reaction loads from the buoyant tether into the mounting structure rather than into the columns alone, keeping bending stresses in the acetal well below levels that would risk deformation. Mounting the entire winch as a self-contained frame also simplifies integration and servicing, as the assembly can be removed and reinstalled as a single unit without disturbing the surrounding subsea structure, which is valuable both during testing and during any future refurbishment between deployments.



Figure 4.19: Winch Structure

Torque transfer from the drive train to the drum is illustrated in the exploded view of Figure 4.20. As described in Section 4.3.8, the constraints of the machine shop did not allow for bolting through the acetal drum, so the rotational motion of the shaft and drum is instead mated together with a set of machine bolts whose hexagonal heads lock into the drum flanges and lock its rotation to the shaft. This approach was adopted as a practical workaround that achieves a positive mechanical connection without requiring the through-holes that the available tooling could not reliably produce in the acetal. These flanges are then connected to the drum via 4 bolts as seen in Figure 4.20, ensuring that the full torque path from shaft to flange to drum is carried entirely through mechanically fastened joints rather than friction alone. The shafts are supported by each column, with the gearbox output coupling to one end to deliver the 2:1 reduced torque developed in Section 4.3.6. This arrangement keeps the load path short and fully within the corrosion-resistant material set, transmitting the servo torque into the spool while allowing the drum and shaft to rotate as a single rigid body, which provides consistent and predictable behavior across every actuation cycle.

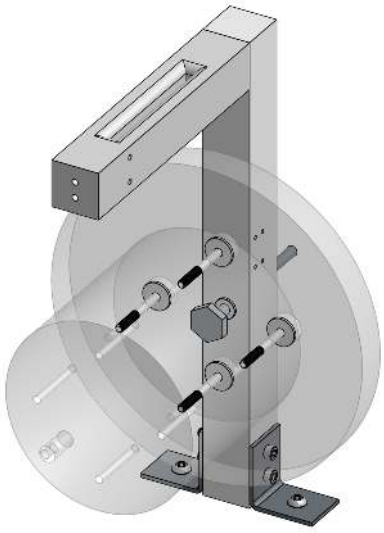


Figure 4.20: Torque Transfer through the Winch

The rope guide, shown in Figure 4.21, is a slotted block mounted across the top crossbar of the structure directly above the drum. As the spool rotates, the rope is fed through the narrow slot, which constrains the line's lateral position and lays it onto the drum in consistent, evenly distributed wraps rather than allowing it to pile unevenly toward one flange. This prevents the tether from bird-nesting or jumping the flanges during deployment and retrieval, a failure mode that is particularly important to eliminate over the thousands of actuation cycles expected across the deployment period, since a fouled or jumped line could jam the winch and disable the entire system. By maintaining an orderly wrap pattern, the rope guide also keeps the effective drum radius predictable, which directly supports the torque margin assumptions made in the sizing calculations of Section 4.3.5. In this way the rope guide contributes not only to the mechanical reliability of the winch but also to the validity of the analysis used to specify the drive system.



Figure 4.21: Winch Rope Guide

4.3.9 Drag Force Validation

The calculations done in section 4.3.6 make multiple assumptions surrounding the drag force component. Specifically, the water speed was assumed to be 1 m/s and the tether drag was ignored. While those assumptions were deemed appropriate for validating the purchase of the Blue Trail SER-2130 Waterproof Servo [21], it is necessary to perform

a more in-depth drag calculation to verify our survival requirement of wave heights up to 1.5 m and wind speeds up to 15 m/s without mechanical failure. In order to do this, calculations were performed to obtain more accurate water speed via Linear Wave Theory. From this water speed, a more accurate drag calculation could be done on the buoy and the tether to confirm whether or not our system can output the required torque.

Linear Wave Theory

Linear Wave Theory, also known as Airy Wave Theory, is a fluid mechanics model used to describe the kinematics and dynamics of surface waves. It provides the necessary mathematical framework to determine the exact water particle velocities acting on the buoy and its tether at various depths.

The theory relies on a few core assumptions to simplify complex fluid behavior. It assumes that the wave amplitude is extremely small compared to both the wavelength and the water depth. Furthermore, the water is treated as an ideal fluid, meaning it is inviscid, incompressible, and irrotational [23]. While real-world ocean waves can exhibit nonlinear behavior, especially as they approach the 1.5 m survival height requirement, Linear Wave Theory provides a sufficiently accurate baseline for engineering worst-case drag calculations without requiring expensive computational fluid dynamics simulations.

Applying this theory to the system requires determining the wavelength and the corresponding horizontal water velocities. The wavelength (L) is calculated using the dispersion relation, which depends on the water depth (d), the gravitational acceleration (g), and the wave period (T) [23]. Because the wavelength appears on both sides of the relation, it must be solved iteratively.

$$L = \frac{gT^2}{2\pi} \tanh\left(\frac{2\pi d}{L}\right) \tag{4.7}$$

Once the wavelength is established, the horizontal velocity u of the water particles can be calculated at any depth y and time t, where H is the wave height. This horizontal velocity profile governs the dynamic drag applied to the tether and the buoy [23].

$$u(x, y, t) = \frac{\pi H}{T} \frac{\cosh\left(\frac{2\pi(y+d)}{L}\right)}{\sinh\left(\frac{2\pi d}{L}\right)} \cos\left(\frac{2\pi x}{L} - \frac{2\pi t}{T}\right) \tag{4.8}$$

Drag Calculations

With the accurate water speed profile established, the standard drag equation (4.2) can be applied to both the spherical buoy profile and the cylindrical tether cross-section to determine the force required to submerge the buoy. From these

calculations, it was found that the water speed created by 1.5m high waves and a 15 m/s wind speed was 1.3 m/s in the horizontal direction and 0.23 m/s in the vertical direction. These speeds were then used to find a total drag on the buoy and tether, which was around 70 N. Since our system was able to provide 135 N of force to the tether with our drum output torque, we validated that our system is able to survive our requirement without mechanical failure.

A complete breakdown of these specific variables, the iterative wavelength calculations, and the resulting force profiles is detailed in the MATLAB script and extended calculation section in Appendix D.

4.4 Subsystem Three - Buoy

In order to develop a subsea electromechanical system, it is critical that the electronics stay isolated from the marine environment and maintain their ability to perform as expected throughout the duration of the deployment. This subsystem outlines the structural components of the IRIS system to ensure the continued functionality of all the components.

4.4.1 Design Drivers

- **Waterproofing:** Enclosures shall be rated for submersion greater than 30 meters.
- **Longevity:** Structural materials shall not corrode or lose functionality due to marine exposure over the deployment period.
- **Environmental Impact:** Structural materials shall not leach into the environment over time, negatively impacting marine life.
- **Serviceability:** Components within watertight enclosures shall be easily accessible and the enclosures themselves shall be retrievable.
- **Cost:** The cost of the structural components shall not exceed the budget.

4.4.2 Buoy Exploration

The buoy is one of the most critical components of the IRIS system, allowing for cellular communication to permeate out through the shell, while housing the electrical components to facilitate that process without water permeability. Through discussion with Kortney Opshaug of Blue Ocean Gear, it was determined that polycarbonate was one of their tested best materials to allow cellular communication permeability and was easily accessible. This is backed up by a study by Curbell Plastics [24], that found polycarbonate to be in the desirable range for plastics utilized in protective structures for cell tower antennas.

The first buoy structure intended to be used was a donation of a Blue Ocean Gear buoy shell in support of the development of IRIS. This buoy, pictured in Figure 4.22, was desirable because it had been tested to function in a similar capability. A dissimilarity, however, was that this buoy was not intended to be anchored from the bottom, nor was there an expectation to have data penetrating through. The buoy's intended use was to rise to the surface and float behind a larger buoy, communicating positional information via satellite and radio [25]. Though the communication verification was appreciated, the concern with boring a hole to bring data into the buoy while ensuring the interchange was watertight was significant. Additionally, due to the small size of the buoy, it was not guaranteed that it would be able to withstand surface conditions when directly anchored.



Figure 4.22: Blue Ocean Gear Buoy [25]

A Blue Robotics Watertight Enclosure was chosen as the alternative to the Blue Ocean Gear Buoy. Along with the benefit of experience with the modularity of the system, there was also an option to have a polycarbonate dome as one side of the enclosure. By adhering the antenna within this dome, the cellular transmission will be directed through a material that has been tested and verified in existing systems and in literature. A Blue Robotics Enclosure with a polycarbonate dome is pictured below in Figure 4.23.



Figure 4.23: Blue Robotics Watertight Enclosure [26]

4.4.3 Final Buoy Design

The final buoy integrates the Blue Robotics watertight enclosure into a modular structural assembly designed to keep the housing upright, corrosion-resistant, and serviceable throughout the deployment period. The completed buoy is shown in Figure 4.24, and an exploded view identifying its constituent components is provided in Figure 4.25.

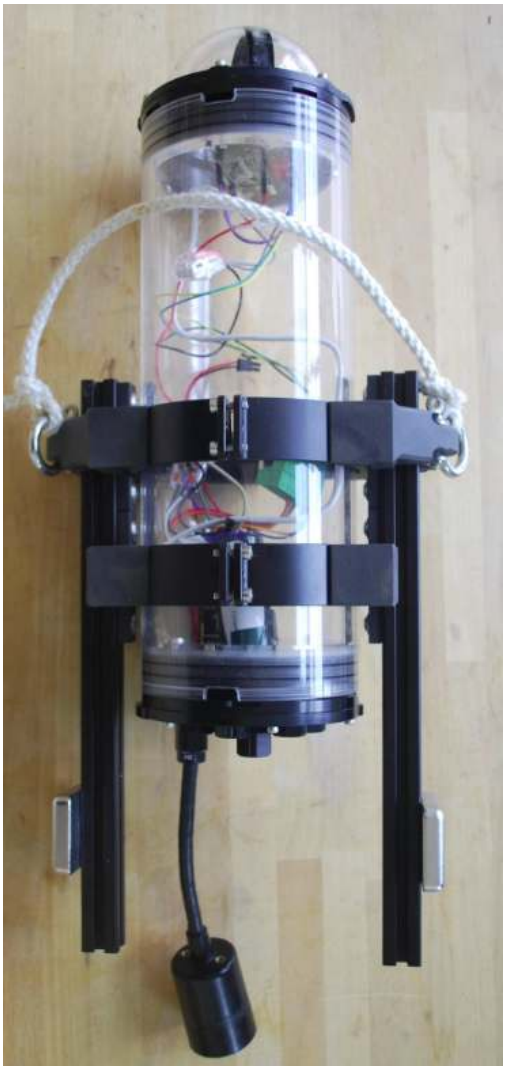


Figure 4.24: Final IRIS buoy assembly

At the core of the assembly is the Blue Robotics pressure vessel: a 300 mm cast acrylic tube sealed at each end by an end cap. The polycarbonate dome cap occupies the upper end, elevating the cellular antenna above the waterline once the buoy is upright, while an aluminum flat cap occupies the lower end and accommodates the Wet-Link penetrators through ten M10 ports. Each cap seals against the tube with dual radial O-rings retained by locking cords.

The cast acrylic tube is the depth-limiting component of the assembly, rated to 100 m, whereas the polycarbonate dome and aluminum cap are rated to 500 m and 1000 m respectively. Measured against the 30 m deployment requirement, the acrylic tube establishes a factor of safety of 3.33 on depth. This margin comfortably satisfies the waterproofing requirement and provides headroom for the gradual creep of acrylic under the sustained external pressure experienced across the six-month maintenance interval.

The pressure vessel is secured within a frame of anodized 6560 aluminum T-slot extrusion that serves as the structural spine of the assembly. Two Blue Robotics enclosure clamps wrap the acrylic tube and fasten it to the frame. Each clamp is machined from 6061-T6 anodized aluminum, lined with Buna-N rubber, and assembled with 316 stainless steel hardware. The clamps connect to the frame via two selective laser sintering (SLS) mounts. Both mounts secure the clamps to the T-slot rails, but one also carries the tether attachment hardware. SLS nylon was selected for these brackets because their geometry is not easily machined and the sintered material does not require additional treatment to withstand marine exposure. The tether interfaces with the assembly through two 316 stainless steel screw-pin shackles rated to 700 lb. The consistent use of anodized aluminum and stainless steel throughout the structure satisfies the longevity requirement, as both resist the corrosion and galvanic attack characteristic of prolonged immersion in seawater.

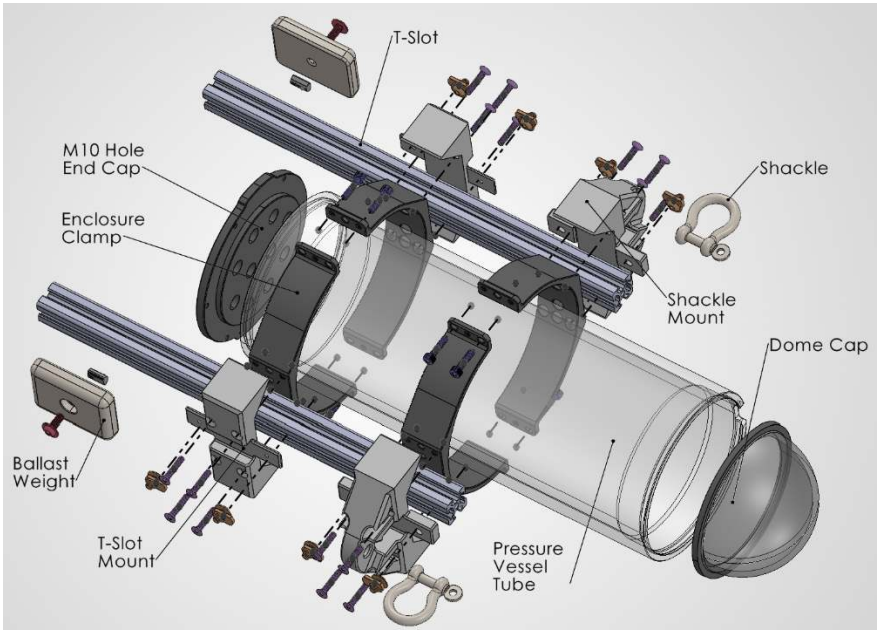


Figure 4.25: Exploded view of the IRIS buoy assembly

To ensure the buoy returns to vertical after being displaced by surface conditions, two 316 stainless steel ballast weights are mounted at the base of the T-slot frame. The buoy's self-righting behavior is governed by the relative

positions of its center of gravity and center of buoyancy, as illustrated in Figure 4.26. When the buoy tilts, the submerged geometry changes and the center of buoyancy shifts laterally. The point at which the resulting line of buoyant force intersects the buoy's central axis is the metacenter. So long as the metacenter remains above the center of gravity, the offset between the weight and buoyancy forces produces a restoring moment that rights the buoy. The vertical distance between the center of gravity and the metacenter is the metacentric height, and its positive value is what guarantees stability. For a slender vertical body such as this enclosure, the metacentric radius contributed by the waterplane geometry is small, so the dominant source of stability is the separation between the center of gravity and the center of buoyancy. Mounting the ballast as low as possible on the frame draws the center of gravity well below the center of buoyancy, maximizing the metacentric height and the resulting restoring moment. Because the ballast fastens to the modular T-slot nuts, its height remains adjustable, allowing the stability margin to be tuned during integration.

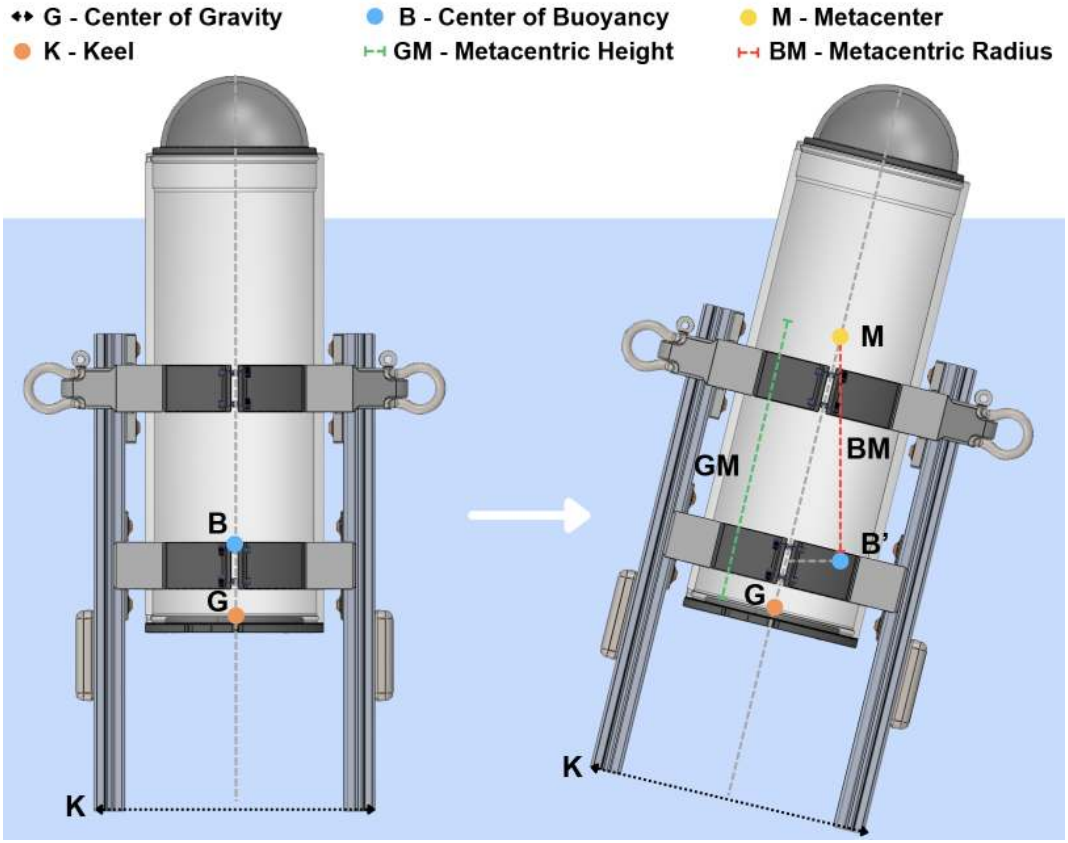


Figure 4.26: Metacentric stability of the IRIS buoy. In the upright orientation (left), the center of gravity (G) and center of buoyancy (B) lie on the central axis. When tilted (right), the submerged geometry shifts the center of buoyancy laterally to B′, and the line of buoyant force intersects the body axis at the metacenter (M). Because M sits above G, the offset between the weight and buoyancy forces generates a restoring moment that returns the buoy to vertical. The vertical distance from G to M is the metacentric height (GM). Not to scale; lateral shift of B exaggerated for clarity.

Buoy exposure was verified both experimentally and through the hydrostatic stability analysis documented in Appendix E. This analysis is most beneficial for calm ocean conditions where the tether is slack, as increased lateral drag from ocean currents will introduce a vertical component pulling the buoy down based on the tether geometry. We were advised that net ocean currents are typically under 0.5 m/s, and that instantaneous wave speed is less important due to its cyclical nature. Linear wave theory for additional particle motion of the waves was also completed, but due to the cyclical nature, it was deemed more relevant to tether forces than buoy motion. A very early feasibility study was completed in MATLAB including tilt of the buoy and submersion due to drag forces. This study is summarized in Appendix C and was primarily done as a proof of concept due to the quantity of assumptions made. The modular nature of the buoy allows for buoyancy and stability to be adjusted based on experimental results from observations in real ocean conditions.

4.4.4 Subsea Electronics Enclosure

Initially, it was considered that this project would be able to use the master REEF node to house the electronics needed. In order to contain IRIS to a separate but adjacent addition, this plan was discouraged. Instead, IRIS utilizes an additional Blue Robotics Watertight enclosure in order to house the electronics that are proprietary to the system. This enclosure is similar to that used for the buoy, except it will not necessitate the use of a polycarbonate dome to support transmission. Since this product is already well established and rated beyond our depth requirement, it is not a concern of whether it will function.

The electronics that will be housed in this enclosure can be seen in the component block diagram, including but not limited to an M0 Adalogger microcontroller, CAN Bus, RS232 and RS485 converters, as well as a buck converter. The modular nature of the Blue Robotics enclosures allows IRIS to use the Wet-Link Compression Gland Penetrators to pass through cables of various sizes. A mechanical diagram of these penetrators is shown below in Figure 4.27, where by tightening the nut on the wet side of the assembly, the gland is compressed around the cable creating a watertight seal rated for upwards of 1000 meters.

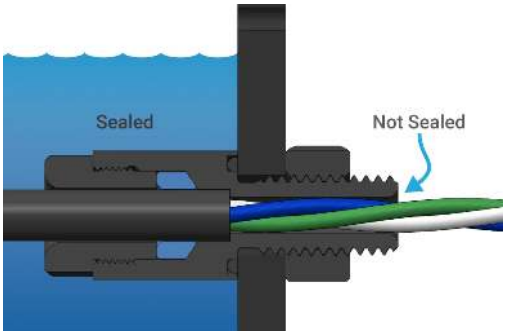


Figure 4.27: Blue Robotics Compression Gland Penetrator [27]

The servo connections utilize the proprietary Cobalt Connector by Blue Trail Engineering. This connector is not wet-mateable; however, it can be connected above ground and is watertight when submerged after connection.

The power and data cabling from REEF utilizes Sub-Conn micro-circular style connectors that are industry standard and can be plugged and unplugged while underwater. This allows battery replacement without needing to retrieve the whole system if maintenance is not required. Note that while Sub-Conn connectors allow for submerged mating, great care must be taken to ensure that the battery power line is broken prior to disconnecting the cable, which is a feature to be implemented in the power system on REEF.

4.4.5 Subsea Platform and Anchoring

An important component of the deployment process of IRIS is anchoring the structure to the ocean floor. Due to project scope and timeline constraints, a finalized anchoring subsystem was not fully developed during this phase of the project. However, the planned approach involves the use of a custom concrete anchor incorporating integrated mounting brackets designed specifically to interface with the IRIS winch structure. This concept would provide both the necessary ballast and a stable, semi-permanent mechanical attachment point for long-term deployment.

Chapter 5

System Evaluation and Timeline

5.1 Timeline

The following Figure 5.1 shows the timeline of the development of IRIS from the initial project decision in June of 2025 to final documentation and wrapping up in June of 2026. This year-long project had numerous facets and stages as shown in the figure, each of which were a unified or parallel effort across the team.



Figure 5.1: IRIS Project Timeline (June 2025 – June 2026)

5.2 Overview

To evaluate the successful completion of the requirements outlined in Table 3.1, testing was completed in three separate stages. The first stage was laboratory bench-top testing, where the primary intention was to validate electrical and software connections, as well as ensuring essential hardware actuation worked as expected. The second stage of testing utilized test tanks and pools, in which various aspects of the system were validated with the primary focus on baseline functionality. The third stage, which happened concurrently with the previous stage was ocean deployment, where additional deployment level functionality was tested. These stages, along with the learnings and adjustments from each will be outlined in the following sections.

5.3 Bench-top (Unit) Testing

As mentioned in the system overview, the primary goal of bench-top testing was to validate baseline elements of the system throughout software, electrical, and mechanical aspects. The majority of this testing involved direct connection to the computer for monitoring MCU activity or simplified code to specifically test single functions. With direct monitoring access, it was trivial to perform root cause analysis and ensure robust functionality before moving to a less controlled environment.

5.3.1 Software

To validate software on the bench, the primary goal was to integrate the electromechanical actuation while ensuring proper throughput of data from the REEF sensor network to the cloud database. In order to complete this validation, numerous collection events were scheduled on the sensor network at various timestamps. With IRIS attached and scheduled to deploy, these data packets were successfully recorded followed by transmission to the buoy and finally published to the cloud. During this transmission, actuation of the winch mechanism was also validated. Over the course of multiple rounds of data events from the sensor network, and multiple full cycles of IRIS including complete data transmission and actuation, the software was validated in a bench-top environment.

5.3.2 Hardware

Hardware validation was a multi-stage process, from single component to full integration and satisfying requirements. Starting at the component level, the first aspect was verifying wiring and single component functionality for every electronic part of the system. Primary electrical component testing included acoustic and cellular connectivity through the system, as well as RS-485 connectivity with REEF. Additional testing was completed independent of the entire

system to ensure robust actuation and control of the drive and brake servos with their respective protocols (CAN and PWM), and programming to set limits and desirable responses.

As mentioned in the software testing, end to end data flow alongside appropriate winch actuation was completed to verify the system's functionality. While the actuators themselves were robust and consistent, an issue that arose during this testing was the gearbox, specifically the set screws in the acetal gears stripping. This resulted in low torque output and eventually free-spinning of the gears. Initial attempts to solve this problem were sufficient to enable in-water testing, however the persistence of the problem led to a re-design of the gearbox such that set screws were not relied on for torque transmission. Additionally, it was discovered that an issue that had led to the browning out of the sub-sea electronics stemmed from the brake servo getting jammed on the side of the spool, drawing an uncharacteristically high current. This was remedied by reducing material that protruded out from the horn of the servo.

Prior to going into the water, the pressure vessels were all tested to ensure that they were watertight. This was done using a hand pump, pulling a vacuum in the enclosures. The tests were completed at a vacuum of 10-15 inHg and the enclosures were considered to be sufficiently watertight if they dropped less than 0.5 inHg over the course of 10 minutes. The buoy was successfully watertight as assembled, but additional verification of O-Ring and compression gland seals was needed for the sub-sea electronics to be able to pass this test.

The final stage of dry testing was verifying the repeatability of the tether spooling. This was done by hanging a weight off of a balcony, and having the winch raise and lower it a repeated number of cycles. While a complete deployment, at 4 actuations per month and a 6 month duration would only require 24 cycles, a full 100 cycles was finished. This built confidence in the systems ability to repeatedly spool without the tether becoming unkempt. The winch following its 100th retraction can be seen in Figure 5.2.



Figure 5.2: Repeatability Test - 100th Repetition

5.4 Tank and Pool Testing

The intentionality of testing in the tank and pool was to verify deployment level functionality in a heavily controlled environment. Final deployment verification was intended to be done at an additional pool location, the Monterey Bay Aquarium Research Institute (MBARI) test tank due to its depth and closely monitored parameters, though due to scheduling constraints this was not possible. Pool and test tank verification therefore sought to validate individual subsystems initially as well as a final end-to-end test. Initial tests in the smaller test tank within the Robotic Systems Laboratory demonstrated some initial capabilities, but due to the size constraints made it very difficult for the acoustic communication to function. Therefore, we also pursued the use of the Santa Clara University Malley Center pool, where we deployed and tested the entire system.

Two key tests were performed to validate requirements, including the transmission of a large quantity of data over cellular while the buoy was in the water, as well as an end to end test in the water, where both data flow and actuation were tested. Additional testing included confirming the buoy's flotation characteristics, including it's tendency to right itself, and its emersion above the water, tuned by adjusting the ballast weights on the buoy rails.

5.4.1 Cellular Payload

In order to test the functionality of the buoy's cellular transmission when in the water, a test of the buoy transmitting a significant number of sample sensor data packets was completed while the buoy was floating in the test tank. The buoy was set to transmit upward of 200 lines of sensor data, and did so having only lost one packet in transmission. Additional code safeguards such as including numbered lines of data in the transmission may be useful for ensuring retrieval of lost packets, however for initial payload testing, this result was promising given the lean implementation. Figure 5.3 shows the buoy in the test tank and the cellular data appearing in the Google Sheets file that the data gets forwarded to during this test.



Figure 5.3: Buoy transmitting cellular payload in tank

5.4.2 End to End IRIS Testing

The culminating test of IRIS was completed in the Malley Pool. We tested the functionality of IRIS system to be able to operate on its independent schedule to deploy the buoy, transmit a payload of data once surfaced, and retract. This test had decent success, and proved the ability for IRIS to complete all necessary stages of its operation without intervention. The buoy autonomously gave the command to the subsea electronics, recieved a data payload, was released to the surface, transmitted the data, and was retracted. The one caveat is that this test did not entirely validate the REEF-Cloud pipeline, as the payload was stored on the onboard SD card of the subsea electronics. Due to technical wiring difficulties with the REEF sensor network itself, independent of our development of IRIS, the onboard payload was deemed the best option for verification of IRIS given the time slot available. Regardless, of not being able to test the REEF integration itself, the nature of the subsea electronics acting as a state machine awaiting input from the buoy makes it fair to assume data points collected by REEF over a few collection cycles would have resulted in similar success. Certain insights were gained for future work regarding forcing the connection/disconnection from cellular at the certain deployment stages, to make the cellular connect and publish once surfaced more seamless.

5.5 Bay Deployment

This deployment was done while waiting to get access to the Malley Center pool, but provided crucial verification for both generic deployment level considerations as well as informing late stage design considerations and actuation functionality. The location chosen for this deployment was Foster City, California, where the walkway along the water provided easy access to shallow water for preliminary submerged actuation verification. IRIS can be seen in Figure 5.4 about to be deployed off the side of the walkway. From these tests, we were able to adapt and prepare for the final

end-to-end test completed in the pool.

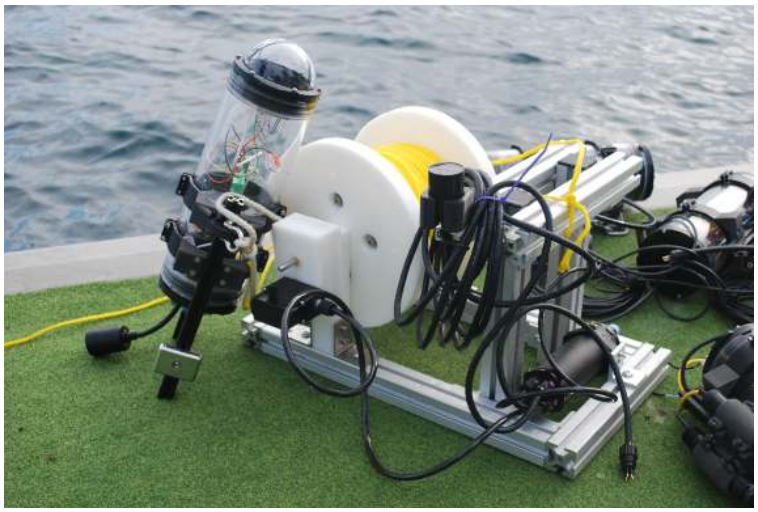


Figure 5.4: IRIS ready for deployment in Foster City

In this test, 5 complete cycles of actuation (without data throughput) were completed successfully proving the ability of the winch to retract over multiple cycles. The only reason for the end of the test was the modem getting unseated in the sub-sea system and no longer facing upward to the buoy, which was fixed with a rigid attachment system.

Additional useful verification was the self-righting ability of the buoy and its natural tendency to ebb and flow with the waves. While the Foster City deployment spot didn't necessarily have a complete open ocean swell profile, there was enough small period wind-swell waves and reflection off the side of the walkway that the buoy was challenged with staying righted through continuous water motion and stayed well emerged for the entire duration of being on the surface.

A final stage of verification that came from Foster City deployments was the deployment method. Though the winch and sub-sea electronics aren't in their final form factor, and don't include a final anchor, it was relatively easy to deploy with a single person, just by attaching a rope and lowering the system off the side. This method made deployment rather trivial, especially if keeping the REEF sensor network dry. In true deployment scenarios, REEF would get submerged alongside, but since the two systems interface with a wet-mate connector, they can be deployed separately.

5.6 Evaluation Results and Next Steps

Throughout the testing we were able to validate numerous requirements, while some others will require future testing or work to ensure robust success. A brief overview of our key requirements and our validations for each will be outlined in the following sections.

5.6.1 Validated Requirements

1. The housing shall be waterproof and withstand submersion at 30 m depth without leakage, with a 6 month maintenance cycle.

   This requirement was satisfied by design, utilizing enclosures rated for the depths expected. Prior to deployment and after each opening of the enclosures, vacuum tests are to be performed to continually ensure that the requirement remains validated. Testing structures such as the T-Slot that the winch was bolted to are not considered to be a final deployment-level design and therefore not counted against this requirement.

2. The system shall resist corrosion and bio-fouling in seawater for a minimum deployment period of 6 months without loss of function.

   This requirement was satisfied by design, utilizing marine enduring materials including acetal, acrylic, 316 Stainless Steel, 6061 Aluminum, or Titanium. Contact areas between dissimilar metals are designed to be minimal.

3. The tether shall retract repeatedly to the correct depth without tangling for the number of cycles in a deployment.

   This requirement was verified by a repeatability test outlined in section 5.2, in which the tether unwound and rewound the deployment distance successfully 100 times.

4. The system shall be constructed using materials rated for UV exposure $\geq$ 1 year in a marine environment.

   The primary concern for this requirement is the degradation of polymers used. Acetal, polycarbonate, and acrylic are the polymers used in IRIS, all of which have reasonable UV stability, particularly acetal and acrylic. The less stable polycarbonate is still expected to easily last the duration given it's nature of only being surfaced for a short period of time in the system. It will be noted for future work that the polycarbonate dome is to be assessed for replacement on the maintenance cycle.

5. The power system shall operate safely between 12°C - 22°C while including charge and short-circuit protection.

   Both the onboard communications buoy battery and the REEF battery that the sub-sea electronics utilize have onboard battery management systems (BMS) that limit charge and discharge rates and keep individual cells

protected. The LiFePO4 chemistries used are also relatively more stable than the traditional rechargeable LiPo chemistry. Charging is designed to be done in a controlled manner, either on shore or on a boat, plugged into 120V AC with voltage regulated DC adapters using direction specific Molex connectors.

6. The system shall transmit sensor data at least once a week.

   This requirement is satisfied by design for battery usage by the communication buoy along with repeatability testing to confirm the winch can maintain functionality over the necessary cycles.

7. All sensor readings shall be time stamped with ±1 second accuracy.

   This requirement is satisfied by design of the transmission payload to include the RTC timestamp as collected by the REEF sensor network. The payload of data recorded is preceded by time, then sensor node address, and then the data itself. Additionally, the time of cellular transmission is also recorded as it arrives to shore.

8. There shall be less than 1% data loss as it transmits through the system.

   This requirement was tested in the bulk data payload test, where 200 lines of sensor payloads were transmitted, losing only 1 packet. This is a 0.5% loss of data, and can be addressed easily by implementing requests for missing data.

9. There shall be an accessible interface by the end user for scientific purposes.

   This requirement is satisfied by the Google Site endpoint where the data is forwarded to. The data from here can easily be accessed including viewing previous instances of transmitted data.

10. The system shall be capable of 32 kb/week of data delivery.

    This requirement is satisfied dually by design and the bulk payload test. The buoy was capable of transmitting upward of 15 kb of data in a few minutes in this test and the system is designed to remain on the surface until all of the data has been transmitted.

11. The total system weight should not exceed 45 kg, allowing two-person deployment and recovery.

    This requirement is satisfied by the buoy weighing roughly 5 lbs and the winch and sub-sea electronics weighing around 35 lbs (including the test T-Slot apparatus). Together, this provides large margins for the weight requirement. It is assumed that a permanent anchor would be deployed separately and is inherently heavy, though likely wouldn't need to be greater than 45 kg in itself.

12. The system shall use non-toxic, reef-safe coatings and materials for any part in water contact.

This requirement is satisfied because no coatings were used on materials, and all of the surfaces are expected to be minimally bio-reactive for their own longevity. No lubrication was used in the drivetrain that could leak into the environment, relying instead on the self-lubricating properties of the plastic gears.

13. The system shall be able to operate continuously for ≥180 days on DC battery.

    This requirement was designed for when sizing the battery. A cohesive power study, including BMS parasitic discharge, was completed for the buoy battery, leaving a large factor of safety. The power for the winch drive comes from the modular REEF sensor network power system and can be increased in capacity arbitrarily.

14. The system shall cost less to use per data collection event than the existing solution.

    Each deployment and data collection from the existing REEF sensor network would require a crewed boat for at least a day of work, including two divers and two people to remain on the boat. With the labor cost of environmental technicians set at $24/hr [28], with two divers, and two boat crew for an 8 hour day, rough cost per data retrieval without IRIS could be found as follows. $\$24/hr * 8\ hr * 4 = \$800$ per deployment. With the price of gas for a multiple hour boat trip, the price per deployment well exceeds $1000.

    This requirement is satisfied because with only three data transmissions not requiring onsite divers, IRIS has paid off the cost off all materials as assembled in the BOM, and would continue to provide data without need for additional diver intervention for at least 20 additional cycles through 6 months at no additional cost.

5.6.2 Untested Requirements

1. The buoy shall survive wave heights up to 1.5 m and wind speeds up to 15 m/s without mechanical failure.

   While this requirement was designed against for drag and torque requirements, the externalities introduced in true oceanic conditions would suggest that an onsite test would provide the strongest evidence of survival.

   This requirement is inconclusive because it may well be satisfactory in its current state, but our scope did not allow for testing against it.

2. The system shall transmit under normal oceanic conditions, and shall repeat the transmission until successful, if not received.

   Transmission successfully was tested in controlled water as well as proven cellular connectivity in somewhat choppy water in the bay, but full ocean swell connectivity was not tested.

   For a similar reason as the previous requirement, this remains inconclusive, though the system may well be capable.

5.6.3 Incomplete Requirements

1. Buoy shall have high visibility coloring, as well as RSL identifying markings

   High visibility coloring was not used in the system, nor are RSL markings currently attached. This will be an important next step prior to a full deployment.

2. The system may be capable of wireless firmware updates over the cellular link.

   This was not designed for though may be possible with additional software implementation. The cellular microcontroller is capable of wireless firmware updates, and since the scheduling is done on the buoy, this could include updating buoy schedule for optimal transmission times.

5.7 Project Risks

The following Table 5.1 outlines primary risk mechanisms in a deployed state of our system. These are categorized by probability, severity, and impact. The most significant possibility is that cables are unplugged underwater leading to complete failure of the system and potential damage to components. With the mitigation methods of strain relief on critical cables, the chance of this occurring is very low, especially with careful deployment. Other notable risk areas include biofouling or marine life entanglement, which both are risks that come with the location. These problems are mitigated by reducing exposure, both from sunlight at the surface causing organism growth and by keeping the tether length short the majority of the time to reduce chances of entanglement with marine life. Data transmission failures are a high likelihood occurence, but due to the ability to implement failsafes and re-transmission requests, it is not catastrophic.

Table 5.1: Risk Assessment Table

Risk	**Probability**	**Severity**	**Impact**	**Consequences**	**Mitigation**
Biofouling on tether and winch	0.3	3	7	The winch could stop properly working, making the buoy not able to rise to the surface to transmit.	We are at a depth where this shouldn't be an issue, but we are factoring this risk into our design to accommodate it should it happen. Consistent motion will help prevent buildup.
Data isn't transferred properly	0.5	3	2	We lose one cycle of data, and have a gap in our data.	If we are doing consistent transmission, missing one cycle shouldn't be very impactful, but we will implement checks to make sure data is being transmitted properly and can try multiple times if not.
An animal or other sea debris gets caught in tether	0.3	5	8	The tether gets caught up and is unable to be retracted properly.	To mitigate this we plan on having the tether in its unretracted state for as little time as possible.
Physical data cables get caught and unplugged	0.05	10	10	This would cause our system to completely fail, and require a diver to retrieve the network in order to repair it.	We will arrange the cables in a way that when pulled, they won't put tension on the actual connections and won't create risk of being unplugged.

Chapter 6

Constraints and Standards

6.1 Constraints

There are areas of our project that are affected by different factors and cause setbacks or design changes in some ways. We have done our best to address these constraints so that we can deliver the absolute best version of our system that we are able to.

- **Waterproofing**: Since all electrical parts will be underwater, everything needs to be completely waterproof. This caused us to choose certain types of connectors, which added extra complexity to the wiring. It also caused us to implement certain protocols like acoustic communication, to bridge the underwater gap between parts of our system.
- **Power**: Being an autonomous system kept underwater, the size and capacity of our battery determines how much power we are able to consume. This sets a limit on how often and how much data we can transmit. To resolve this, we came up with ways for components to draw less power when able, and to efficiently budget the power between all components while still giving enough room for error.
- **Cellular coverage**: For our system to work, it needs to be in a location where it can establish a strong cellular connection. If it were to be deployed in an area that doesn't have good cell coverage, it would not work. We have tested the coverage in our area of deployment and it is strong enough, but this is a notable constraint for further iterations of this project in different locations.
- **Communication bandwith**: Our data transmission is constrained by the bandwith of the components in our system. For example, the acoustic modems can only transmit 64 Bytes per packet [18], so we have to split up our data into smaller chunks, sending them one at a time. This makes a cycle of data transmission take longer, and uses more power, which has an effect on the other components needed.

6.2 Standards

Our system uses many software and hardware protocols that are considered standards. In order for IRIS to operate as efficiently and robustly as possible, we have adhered to these standards and implemented them in the necessary areas. We have ensured that our system properly utilizes existing protocols that best suit our goals and vision for this project.

- **TIA-485**: The TIA-485 standard [29] covers RS-485 serial communication, which we use to transfer data from the Feather Adalogger M0 in REEF's master node to the Feather Adalogger M0 in our subsea electronics enclosure. This allows us to efficiently move data through a waterproof cable between REEF and the subsea electronics. This method of data transfer also serves as a power-efficient way of having our system be "listening" constantly through serial input. Another benefit of serial communication is that it integrates easily with the Feather Adalogger M0 by using its TX and RX serial pins.

- **TIA-232**: The TIA-232 standard [30] outlines the protocol for RS-232 serial communication. This is what we use in our subsea electronics enclosure to convert serial output from the Feather Adalogger M0 to become a voltage signal that our Succorfish Delphis acoustic modem can read, and visa versa for getting output from the acoustic modem into the serial of the Adalogger. It's also used in our buoy to accomplish a similar task, but in this case the communication is between our Boron 404x and the Succorfish Delphis acoustic modem.

- **ISO 11898**: The ISO 11898 standard [31] covers the protocols for dealing with a Controller Area Network (CAN) Bus. This is what we use to communicate between our Feather Adalogger M0 and the Blue Ocean Trail SER-2130 servo motor that spins our winch. Using this standard gives us the ability to efficiently control exactly when, and how much we want the servo to move. It also provides us with integration with our Feather Adalogger M0 so that we can wire them directly. The servo-controlled winch is one of the most important parts when it comes to demonstrating the ability and performance of our project. Because of this, we made sure to implement a robust and reliable protocol for dealing with the movement.

- **SPI**: The Serial Peripheral Interface (SPI) standard [32] covers the protocol for interacting with the microSD cards on board our Feather Adalogger M0 and our Boron404x. SPI involves driving clock and chip select signals to read and write data to and from the microSD card. On the Adalogger, this lets us gather data from REEF over time, and then read it when it's ready to be transmitted via the acoustic modem. For the Boron, we use SPI to temporarily write incoming data to the microSD card so that it can pull from this cache when publishing to the cloud. Another benefit of having the ability to read and write to an SD card is that we are able to save the current state of our system, so in the event of a power malfunction it knows what state to return to to finish its cycle.

Chapter 7

Professional Considerations

7.1 Ethics

Our project follows the engineering code of ethics and puts the welfare of the world above all else. Our driving factor is to increase researchers' ability and efficiency in monitoring reefs to help with restoration and preservation efforts. This helps the researchers do their job, helps the reef thrive and be taken care of, and provides the rest of humanity with a natural world that is doing well around them.

Ethical Justification of IRIS

The development of IRIS is ethically justified by its direct contribution to environmental stewardship and the public good. The Palos Verdes Reef is a shared natural resource whose restoration depends on continuous, reliable ecological data, yet the current retrieval model requires diver intervention roughly once per year, creating dangerous working conditions and significant gaps in monitoring continuity. IRIS addresses this by enabling autonomous, scheduled data transmission, improving both researcher safety and the quality of information available to NOAA and the Vantuna Research Group for evaluating restoration progress. The ASME Code of Ethics is directly applicable here. The first canon calls on engineers to hold paramount the safety, health, and welfare of the public, while the eighth canon specifically addresses the engineer's responsibility to consider environmental impact [33], both of which IRIS directly serves. Beyond individual safety, IRIS is committed to environmental justice, working to address degraded reef ecosystems that disproportionately harm coastal communities, fisheries, and marine biodiversity that cannot advocate for themselves. By making long-term ecological monitoring safer and more practical, IRIS helps ensure that decisions for these entities are grounded in real data rather than infrequent snapshots.

Ethical Issues with IRIS

The primary ethical concern for IRIS is the entanglement risk posed by the tether mechanism. Sea lions, fish, and invertebrates native to the Palos Verdes Reef could become trapped during buoy actuation cycles or around the deployed tether line. This risk is partially mitigated by the retractable design, the tether is stowed in a minimal configuration subsurface during the majority of operation, but the risks cannot be fully eliminated. Secondary risks include potential for enclosure failure, introducing electronic waste to the marine environment. Another secondary ethical consideration is the use of lithium based battery compositions, which have had notable ethical concerns in the past. Use of Lithium Iron Phosphate in the IRIS system as opposed to Li-Po notably avoids critical materials such as cobalt which have been linked to high environmental and labor ethic concerns. Because IRIS is designed for autonomous 180-day deployments with no opportunity for real-time human intervention, pre-deployment risk disclosure to all stakeholders the Vantuna Research Group, NOAA, and relevant California coastal regulatory bodies is especially critical. Any residual entanglement or material hazard risks should be fully communicated and appropriate permits secured before unmonitored deployments occur. The reef ecosystem itself, along with the researchers and divers who periodically work the site, constitute the population most directly exposed to these risks and should be the primary focus of safety considerations.

Ethics in Engineering

Working on IRIS reinforced that good engineering requires more than technical skill. It also requires good judgment, communication, and a shared sense of purpose across an interdisciplinary team. Because the system sits at the intersection of mechanical, electrical, and software domains, no single team member could own the full design. Progress depended on communication beyond the technical scope of each team member and mutual trust. Design decisions around reef-safe materials, tether geometry, and autonomous operation were not made in isolation, requiring the full team to align on decisions and awareness of the impact that each individual decision would make. That awareness, sustained through honest team discussion rather than individual initiative, is what shaped IRIS into a system built with genuine responsibility as a key driver.

7.2 Science Technology and Society

IRIS is mainly designed for marine researchers and environmental agencies that need continuous, real time data from the seafloor without the massive costs of research vessels or diver deployments. By automating the telemetry link to the surface, the system makes deep sea monitoring much more accessible for smaller organizations and schools. However, if these buoys become the industry standard, jobs focused on manual sensor retrieval will probably decline.

Another potential issue to consider is the environmental impact of 'sensor density.' If we end up deploying thousands of autonomous winches, the cumulative effects on local marine life and the risk of cable entanglements definitely need to be managed. Ultimately, IRIS shifts ocean data collection from labor intensive 'snapshot' missions to a continuous, remote presence.

7.3 Civic Engagement

To avoid being a navigation hazard, IRIS has to comply with COLREGs (International Regulations for Preventing Collisions at Sea) for marking and lighting the surface buoy. Also, deploying the system in US coastal waters means it falls under Coast Guard and NOAA jurisdiction. For NOAA specifically, this means we have to show that our winch operations won't disrupt any protected marine sanctuaries. On the design side, we are sticking to ASME standards for the winch mechanics and IEEE standards for the subsea electronics and telemetry. The current prototype hits these safety requirements using a mechanical braking system and a pressure tested enclosure built to handle the specific hydrostatic loads at our target depth.

7.4 Economic

We have kept costs in mind while designing our project, but have made sure to make the necessary financial decisions to ensure that our project will be successful. Where we could safely cut costs without sacrificing the potential of our project we made those decisions, as well as using donations or reusing materials where it made sense. For the benefit that this project provides, we feel that the cost makes sense and outweighs the cost that would accumulate from diver retrieval of data from the REEF sensor network without our system. A complete list of costs absorbed to develop IRIS can be found in the BOM in Appendix A.

7.5 Manufacturability

Our project has been designed to use off the shelf components or due consideration has been put into the manufacturability of custom components. Apart from the titanium ratchet and pawl, which were custom ordered, Santa Clara University School of Engineering had sufficient resources to manufacture and build every other custom component. Our product would also be able to be manufactured at scale, since it is fairly modular and would be able to be built in parts and then assembled. The software also allows for changes to be made easily if expanding the project, such as differences in ocean depth, transmission frequency, or type of data being collected.

7.6 Environmental Impact

We designed IRIS to have a long functional lifespan, using corrosion-resistant materials like 316 stainless steel and marine grade polymers to handle the harsh saltwater environment. The system produces zero operational emissions since it runs entirely on a rechargeable battery bank. For the system's end-of-life, we made sure that most of the winch structure is recyclable and that the electronics are easy to pull out for proper e-waste disposal. As a note, acetal is technically recyclable but in practice it is very uncommon. Environmental impact was also a huge factor in our power source selection; by optimizing the winch duty cycle, we cut down on the total battery capacity needed, which helps lower the footprint that comes with lithium-ion battery production.

7.7 Usability

Our product is built to be very user friendly and easily used by the people it is providing. While it is deployed, the system does not require intervention and will run autonomously for the six month deployment period. On the receiving end, a researcher or scientist can visit the IRIS website anytime and instantly receive the most recent data in a user-friendly format. The people that our project is directly benefiting won't have to do anything to actually run the system, they can just access it when they choose and get all of the information they need on their own.

7.8 Lifelong learning

This project has given us the experience of creating a product from start to finish that will be used to make the world a better place. It has opened our eyes to how important it is that the work we do has a positive impact and is something that we find interesting and important. It is a strong motivator that the things we build can and will be beneficial to society. Working on this project has also opened us to the world of hardware, software, and mechanical integration to make complete systems. It has an engaging challenge to integrate electrical and mechanical knowledge with programming to create a cohesive system. By working with an interdisciplinary team, we have gained the skills to collaborate and communicate beyond our technical specialization.

Chapter 8

Conclusion

8.1 Summary and Discussion

Over the course of the 2025–2026 school year, our team designed and built the system we set out to create. Many aspects of the final product differ from our original plans, but the system as a whole achieves our desired outcome. Reviewing our needs synthesis across the three categories of survivability, data delivery, and efficacy, we have satisfied nearly all of our criteria. On the survivability side, testing and design confirmed that IRIS is waterproof, corrosion resistant, and UV resistant, and that its tether can be repeatedly deployed without issue. For data delivery, we demonstrated that the system meets the desired transmission frequency, can transmit in normal sea conditions, receives timestamped data, and allows easy data access. On efficacy, the system came in within our weight limit, is non-toxic, runs on DC battery power, and is cost beneficial overall. We also met the majority of our key requirements, including depth rating, battery capacity, repeatable deployment to the correct depth without tangling, data transmission, and system weight. Some requirements could not be fully verified, such as open-water seaworthiness, which would require a proper offshore deployment. These remain in progress but do not diminish the success the product has demonstrated. Other requirements, such as high-visibility coloring and OTA firmware updates, were deprioritized as out of scope for this phase, and are left for future teams preparing the system for final deployment.

8.2 Future Work

8.2.1 Near Term (1-3 Years)

The most immediate priority for teams in the near future is hardening IRIS for real open-water deployment. This includes validating seaworthiness in offshore conditions beyond protected nearshore environments, and refining electrical connections and software safeguards to further minimize the effect that externalities of a true deployment could have. Power management is another near-term focus for true longevity, implementing relays to the drive servos for

true power-off behavior as opposed to idle modes on the servos. Prior to deployment, another final stage of work will be preparing deployment level anchor design, including the brackets needed to secure the existing spool.

8.2.2 Long Term (5-10 Years)

Over a more generous time period, the vision for IRIS shifts from a single deployable unit at Palos Verdes Reef to a more readily integratable system to be used in multiple monitoring systems. Adding capabilities for larger sensor networks to use a single buoy, but multiple acoustic data outputs is a recognizable possibility. Further increasing the buoy's capability by introducing onboard sensing to attain water-profiling ability is another possibility. The long term goal of IRIS is to enable the flourishing of marine monitoring systems by being able to be seamlessly integrate and provide low-footprint connectivity with critical data for scientists.

8.3 Reflection

We learned important lessons while working on this project that we will each carry with us into our future careers. On the hardware engineering side, we gained experience in designing for manufacturing, designing for assembly, material specific sourcing, and a wide variety of mechatronic skills in conjunction with different forms of data transmission. On the software engineering side, coordinating scheduling, implementing state behavior, and handling data protocols and payloads. Overall, we learned lessons about avoiding analysis paralysis when designing, how to efficiently communicate with an interdisciplinary team, and how to balance our workload as full-time students.

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Appendix A

Expected Budget and Bill of Materials

The IRIS project has successfully secured funding through the Undergraduate Programs Senior Design Grants. The team has been awarded a total of $2453 for the 2025-2026 academic year. Furthermore we will receive more funding from the Robotic Systems Laboratory (RSL) of Santa Clara University, this funding will be of or around $2500. This funding will be allocated primarily toward marine-grade materials, waterproofing hardware, and the cellular communications hardware required for the final deliverable.

Here is a breakdown of all of the materials we have already purchased. We believe that we have the vast majority of the materials that we need to build this project. The future materials that we will need to purchase will be a mix of brackets and fasters to hold everything together, or they will be purchased to replace things we've already bought if the systems aren't working correctly.

Table A.1: Bill of Materials

Item	Vendor	Part #	Item Cost	Notes
M3 20mm SHCS - Low profile	McMaster	90666A108	$7.98	
Black Polypropylene Rope	Uline	S-12863BL	$29.00	
316 Stainless Steel Rod	McMaster	9298K14	$63.62	
McMaster Item	McMaster	35355T14	$11.16	
Watertight Enclosure Cylindrical Locking Series	Blue Robotics	WTE-VP	$127.00	TUBE: 2 in diameter 3.9 inch length acrylic, ENDCAP 1: Flange Cap Aluminum - Blank, ENDCAP 2: Flange Cap Aluminum - 2xM10, VENT: PRV
Epoxy Structural Adhesive	McMaster	7513A1	$89.09	Same as Suburban Marine Epoxy, but Suburban Marine is out of stock
Insulated Male Female Bullet Quick Splice Wire Terminals (Yellow, 12-10 Gauge)	Amazon	B01D9DHQLA	$7.69	
Blue Robotics Watertight Enclosure	Blue Robotics	WTE-VP	$499.52	4 in diameter tube + corresponding clamps
Blue Robotics Enclosure Clamps (4in)	Blue Robotics	BR-102710-004	$61.60	
Adalogger FeatherWing - RTC + SD + CR1220	Adafruit	2922	$19.80	RTC battery option please (95 cent addon below add to cart)!
Underwater Servo SER-2130 CAN, Brushless, High Torque	Blue Trail Engineering	SER-2130	$525.00	
DPC-20 Servo Programmer	Hi-Tec	44428	$26.99	
Oversized 316 Washers (pk 10)	McMaster	95211A164	$5.11	
316 Stainless SHCS (pk 10)	McMaster	92290A340	$16.47	
Plastic Bushing	McMaster	2705T135	$15.84	
Acetal Disk	McMaster	8582K72	$299.06	
Acetal Drum	McMaster	8582K48	$165.86	
Acetal Bar (2 foot)	McMaster	8739K93	$49.10	
316 Brackets	McMaster	2313N13	$33.44	

Bill of Materials (continued)

Item	Vendor	Part #	Total Cost	Notes
30 toothed Acetal Gear	McMaster	2662N41	$7.57	
60 toothed Acetal Gear	McMaster	2662N43	$14.09	
#3 316 threading screw (pk 25)	McMaster	99461A315	$15.87	
Shaft bushing for 10mm ID	McMaster	2705T131	$2.23	
Shaft bushing for 12mm ID	McMaster	2705T133	$4.07	
M3 316 SHCS (pk 50)	McMaster	92290A122	$17.00	
Acetal Bar	McMaster	8739K67	$61.70	
M3 316 Set Screws (pk 50)	McMaster	92029A101	$11.08	
SER 2010 Blue trail servo	BlueTrail	SER-2010	$375.00	
Adafruit CAN Bus Featherwing	Adafruit	5709	$12.50	
Cobalt 4 Pin Bulkhead	BlueTrail	COB-1140-SS	$55.00	
Cobalt 4 Pin Cable Double Ended (1 Meter)	BlueTrail	COB-1341-SS	$115.00	
Cobalt 3 Pin Bulkhead	BlueTrail	COB-1130-SS	$42.00	
Cobalt 3 Pin Cable Double Ended (1 Meter)	BlueTrail	COB-1331-SS	$83.00	
Sparkfun Transceiver Breakout - MAX 3232	Sparkfun	BOB-11189	$14.52	
Custom LiFePO4 26650 Battery: 9.6V 3300 mAh with PCB	BatterySpace	6007 CU-J433	$47.09	
316 SS M4 Screws	McMaster	93395A263	$13.36	
FLIPSKY 75100 Pro V2.0 75V 100A ESC	Amazon	B0CNLJ6ML3	$119.99	
10 Pairs Amass MR60 Connector Female Male 3.5mm	Amazon	B07X223LK3	$14.99	
		Total Cost	**$ 2874.12**	

Appendix B

Source Code

The complete repository of source code, including final implementation and test code, can be found at:

https://github.com/colinfriedel/IRIS-Senior-Design

Below are the loop() functions for both microcontrollers.

B.1 Buoy Loop Function

```
void loop() {
    // ---- FAILSAFE: retry pending RETRACT from previous cycle ----
    // If a retract was never confirmed last cycle, retry it before deploying again
    if (pendingRetract && millis() - lastDeploy >= DEPLOY_INTERVAL && retractConfirmed
         == false) {
        Serial.println("Failsafe: retrying pending RETRACT from previous cycle...");
        Serial1.print("$B07RETRACT");
        retractRetryCount = 1;
        waitingForRetracted = true;
        retractedSentAt = millis();
        pendingRetract = false;
        return;
    }

    // Auto-deploy on schedule     unchanged
    if (millis() - lastDeploy >= DEPLOY_INTERVAL && !cycleInProgress &&
        retractConfirmed) {
        triggerDeploy("");
    }

    static unsigned long lastHeartbeat = 0;
    if (millis() - lastHeartbeat > 10000 && !cycleInProgress && retractConfirmed) {
        lastHeartbeat = millis();
        if (millis() - lastDeploy >= DEPLOY_INTERVAL) {
            Serial.println("Status: Deploy due!");
        } else {
            Serial.print("Status: Waiting... Time until next deploy: ");
            Serial.print((DEPLOY_INTERVAL - (millis() - lastDeploy)) / 1000);
```

```
            Serial.println("s");
        }
    }

    // Read incoming acoustic modem data     unchanged
    while (Serial1.available()) {
        char c = Serial1.read();

        if (c == '\n' || c == '\r') {
            serialBuffer.trim();
            if (serialBuffer.length() > 0) {
                Serial.println("Raw from modem: " + serialBuffer);

                if (serialBuffer.startsWith("#B") && serialBuffer.length() > 7) {
                    String data = serialBuffer.substring(7);

                    if (data == "DONE") {
                        Serial.println("All data received. Waiting for DEPLOYED...");
                        allDataReceived = true;
                    } else if (data == "DEPLOYED") {
                        Serial.println("Deployment confirmed! Starting publish...");
                        deployedConfirmed = true;
                        waitingForDeployed = false; // Failsafe: confirmed, stop
                            watching
                    } else if (data == "RETRACTED") {
                        Serial.println("Retract confirmed.");
                        retractConfirmed = true;
                        waitingForRetracted = false; // Failsafe: confirmed, stop
                            watching
                        retractRetryCount = 0;
                        pendingRetract = false;
                        lastDeploy = millis();
                    } else if (data.length() > 0) {
                        Serial.println("Saving to SD: " + data);
                        appendToSD(data);
                    }
                } else {
                    Serial.println("Ignored: " + serialBuffer);
                }
                serialBuffer = "";
            }
        } else {
            serialBuffer += c;
        }
    }

    // ---- FAILSAFE: DEPLOY timeout and retry ----
    // Only triggers if DEPLOYED was never received     happy path sets
        waitingForDeployed=false
    if (waitingForDeployed && millis() - deployedSentAt >= DEPLOYED_TIMEOUT) {
        if (deployRetryCount < MAX_RETRIES) {
            deployRetryCount++;
            Serial.print("Failsafe: No DEPLOYED received, retry ");
            Serial.print(deployRetryCount);
            Serial.print("/");
            Serial.println(MAX_RETRIES);
```

```
            Serial1.print("$B06DEPLOY");
            deployedSentAt = millis();
        } else {
            Serial.println("Failsafe: Max DEPLOY retries reached    sending RETRACT and going idle.");
            Serial1.print("$B07RETRACT");
            waitingForDeployed = false;
            retractRetryCount  = 1;
            waitingForRetracted= true;
            retractedSentAt    = millis();
            retractSent        = true;
            retractConfirmed   = false;
            resetCycleFlags();
            retractSent        = true;        // keep retractSent so idle waits for confirmation
            waitingForRetracted= true;
            retractedSentAt    = millis();
        }
    }

    // ---- FAILSAFE: RETRACT timeout and retry ----
    // Only triggers if RETRACTED was never received     happy path sets waitingForRetracted=false
    if (waitingForRetracted && millis() - retractedSentAt >= RETRACTED_TIMEOUT) {
        if (retractRetryCount < MAX_RETRIES) {
            retractRetryCount++;
            Serial.print("Failsafe: No RETRACTED received, retry ");
            Serial.print(retractRetryCount);
            Serial.print("/");
            Serial.println(MAX_RETRIES);
            Serial1.print("$B07RETRACT");
            retractedSentAt = millis();
        } else {
            Serial.println("Failsafe: Max RETRACT retries reached    going idle, will retry next cycle.");
            waitingForRetracted = false;
            retractRetryCount   = 0;
            pendingRetract      = true;  // flag to retry retract before next deploy
            retractConfirmed    = false; // keep false so auto-deploy won't fire
            lastDeploy          = millis();
        }
    }

    // Only publish once BOTH data received AND deployment confirmed     unchanged
    // Gap is 2000ms     safely within Particle's 1 publish/sec rate limit
    if ((allDataReceived || cycleStartPublished) && deployedConfirmed && millis() - lastPublish >= 2000) {

        if (!Particle.connected()) {
            Serial.println("Waiting for cell connection...");
            Cellular.on();
            if (!waitFor(Particle.connected, 600000)) {
                // Timed out     give up and retract
                Serial.println("Cell timeout    sending RETRACT and resetting.");
                Serial1.print("$B07RETRACT");
                retractSent         = true;
```

```
                retractRetryCount   = 1;
                waitingForRetracted = true;
                retractedSentAt     = millis();
                allDataReceived     = false;
                deployedConfirmed   = false;
                publishingComplete  = false;
                cycleStartPublished = false;
                cycleEndPublished   = false;
                retractConfirmed    = false;
                return;
            }
        }

        // Publish cycle_start FIRST before any data rows
        if (!cycleStartPublished) {
            if (Particle.publish("cycle_start", "", PRIVATE)) {
                Serial.println("cycle_start published.");
                cycleStartPublished = true;
            } else {
                Serial.println("cycle_start failed, retrying next tick.");
            }
            lastPublish = millis();
            return;
        }

        // Publish one data row per tick     gap enforced by millis() check above
        if (!publishingComplete) {
            publishNextFromSD();
            lastPublish = millis(); // ensure gap resets after every row
            return;
        }

        // Publish cycle_end after all rows     waits one full gap after last row
        if (!cycleEndPublished) {
            if (Particle.publish("cycle_end", "", PRIVATE)) {
                Serial.println("cycle_end published.");
                cycleEndPublished = true;
                lastPublish = millis();
            } else {
                Serial.println("cycle_end failed, retrying next tick.");
                lastPublish = millis(); // back off before retry
            }
            return;
        }
    }

    // Send RETRACT once cycle_end confirmed sent     unchanged trigger condition
    if (publishingComplete && cycleEndPublished && !retractSent) {
        Serial.println("Sending RETRACT...");
        Serial1.print("$B07RETRACT");
        retractSent         = true;
        cycleInProgress     = false;
        allDataReceived     = false;
        deployedConfirmed   = false;
        publishingComplete  = false;
        cycleStartPublished = false;
```

```
187         cycleEndPublished   = false;
188
189         // Failsafe: start watching for RETRACTED
190         retractRetryCount   = 1;
191         waitingForRetracted = true;
192         retractedSentAt     = millis();
193         retractConfirmed    = false;
194     }
195 }
```

B.2 Subsea Electronics Loop Function

```
void loop() {
    switch (currentState) {

        case STATE_LISTEN:
            Brake.detach();
            handleRS485Logging();

            if (Serial1.available()) {
                String cmd = Serial1.readStringUntil('\n');
                Serial.println("Received: " + cmd);
                // Strip modem wrapper if present
                if (cmd.startsWith("$B") && cmd.length() > 7) {
                    cmd = cmd.substring(7);
                }
                if (cmd.indexOf("DEPLOY") >= 0) {
                    Serial.println("DEPLOY received, moving to DUMP state.");
                    currentState = STATE_DUMP;
                }
                if (cmd.indexOf("RETRACT") >= 0) {
                    Serial.println("Power cycled, retracting");
                    currentState = STATE_RETRACT;
                }
            }
            break;


        case STATE_DUMP:
            dumpDataToAcoustic();
            currentState = STATE_DEPLOY;
            break;


        case STATE_DEPLOY:
            Serial.println("Opening brake...");
            Brake.attach(9);
            Brake.writeMicroseconds(OpenMS);
            smartWait(5000);


            Serial.println("Deploying...");
            Turn(DeployedTurns);
            smartWait(DeployedTurns * turnDelay);


            readReg(0x18);
            freespin();
            Serial.println("Closing Break...");
            //Brake.writeMicroseconds(CloseMS);
            //smartWait(5000);
            Brake.detach();


            // Send DEPLOYED confirmation acoustically
            Serial.println("Sending DEPLOYED confirmation...");
```

```
            sendAcoustic("DEPLOYED");


            currentState = STATE_WAIT_RETRACT;
            break;


        case STATE_WAIT_RETRACT:
            if (Serial1.available()) {
                String cmd = Serial1.readStringUntil('\n');
                Serial.println("Read: " + cmd);
                if (cmd.indexOf("RETRACT") >= 0) {
                    currentState = STATE_RETRACT;
                }
            }
            break;


        case STATE_RETRACT:
            Serial.println("Retracting...");
            Turn(RetractedTurns);
            smartWait(DeployedTurns * turnDelay);
            Serial.println("Sending RETRACTED confirmation...");
            sendAcoustic("RETRACTED");

            //Serial.println("Closing Break...");
            //Brake.detach();

            readReg(0x18);
            freespin();

            updateFileName();
            Serial.println("Done. Returning to Listen State.");
            currentState = STATE_LISTEN;
            break;
    }
}
```

Appendix C

Buoy Tilt Equations

Note that these calculations were completed for very early demonstration of principle of buoy motion, as these equations don't fully represent our buoy's final geometry. Final design level calculations were also completed for drag force (see Appendix D) and Buoyancy (See Appendix E) that apply some similar concepts.

C.1 Force Balance

C.1.1 Net Buoyancy Force (Skyward - Z axis)

The net buoyancy force is a simple subtraction of weight from the net buoyancy force.

$$F_{B,Net} = F_{B,total} - Weight = V_{Sub}\rho_{Ocean}g - mg \tag{C.1}$$

V_{Sub} is the submerged volume, defined by the draft, T (distance from surface to lowest submerged point) and a cylindrical volume.

C.1.2 Drag Force (Horizontally - X axis)

Drag force is a function of the projected area, which for a cylindrical object is a square.

$$F_D = \frac{1}{2}\rho_{Ocean}C_D U^2 A_{Proj} \tag{C.2}$$

A high, conservative estimate for drag coefficient was used at 1.2, however with further literature review can be lowered if margins are slim. The variable U represents the current speed in the water, and primarily has been based on a 0.5–1 m/s commonplace surface current reported by numerous weather agencies.

C.1.3 Tether Force (X and Z components)

The tether force is a multicomponent force that is based on an attachment point at a point along the cylindrical buoy's length, reflecting the design choice to make a movable attachment. The direction of the tether force is enforced rigidly based on an assumption that the tether will be pulled in some direction from the current. The direction can then be assumed based on the depth of the water and length of the tether with right angle triangle geometry.

The following force balances were created for the X and Z directions.

$$0 = F_D - F_{Tether} \cdot T_x \tag{C.3}$$

$$0 = F_{B,Net} - F_{Tether} \cdot T_z \tag{C.4}$$

C.1.4 Force Balance Equation

The above equations can be solved for the overall tether force as well as the draft, T.

C.2 Moment Balance

Following the solution of the force balance equation, a moment balance can be created. This moment will be solved using a metacentric buoyancy analysis, which is commonplace for ships and other semi-submerged bodies.

Assumptions: A metacentric analysis in this form assumes that the buoy doesn't experience excessive tilt angles, and that when it rotates, it does so freely like a ship, not constrained by the tether attachment. With the appropriate pivot point near the water surface, this effect is insignificant, however with a pivot far from the surface, the effective lever of the tether is more significant.

C.2.1 Metacentric Righting Moment Analysis

This form of analysis uses the combination of buoyancy force as well as weight distribution to determine the righting moment based on the displaced volume of water.

For small angles, the center of buoyancy height, KB can be said to be at $\frac{1}{2}$ the draft distance. The metacentric radius, BM, is defined as the ratio of the second moment of area of the waterplane that the cylinder cuts through and the volume that is displaced.

$$BM = \frac{I_{WP}}{V_{Sub}} = \frac{R^2}{4T} \tag{C.5}$$

The metacentric height, GM, is a combination of KB, BM, and the center of gravity height from the bottom.

$$GM = KB + BM - KG \tag{C.6}$$

The lever arm for the restoring moment then becomes $GM \sin(\theta)$ at small angles, for a force dictated by the displaced volume of water.

C.2.2 Drag Moment Analysis

This is the force acting against the righting moment, which is based on the drag force and the submerged area. The lever arm can be found as follows:

$$L_D = (z_{drag} - z_{pivot}) \cos(\theta) \tag{C.7}$$

C.2.3 Moment Balance Equation

The moment balance then becomes:

$$0 = \underbrace{(-\rho g V \cdot GM \cdot \sin(\theta))}_{\text{Righting Moment}} + (F_D L_D) \tag{C.8}$$

C.3 Methodology

This analytical analysis was done using a MATLAB code with various input parameters, outputting forces, tilt angle, and submersion draft. A custom code was chosen as a strong method for creating a rough analysis of the buoy's dynamics to assess critical design choices and limitations that could then be refined with physical tests.

One key difficulty is the commitment to a certain buoy geometry, as changing this geometry can result in large changes to the required calculations.

C.4 Parametric Study

Table C.1: Current Speed Sweep

Current Speed (m/s)	Tilt Angle (deg)	Draft (m)	Tether Magnitude (N)
0.1	-1.08	0.1537	0.12
0.3	-8.08	0.1603	1.13
0.5	-15.41	0.1752	3.43
0.7	-16.42	0.2034	7.79
0.9	-7.56	0.2584	16.33

A current speed greater than ~1.0 m/s results in the buoy submerging, breaking the model. This set of outputs came from holding all inputs the same apart from current speed. These constant inputs can be altered based on design requirements.

Appendix D

Drag Calculations

D.1 Force Balance

The goal of these calculations is to determine the tension required to retract the buoy beneath the surface of the water.

$$T = (F_b - W) + F_d \tag{D.1}$$

Tension T is a combination of drag force F_d and net buoyancy $(F_b - W)$ which is the buoyancy force minus weight, which was determined to be 4.9 N. For these calculations, it was assumed that the drag force was acting opposite of tension, which yields the highest required tension.

D.1.1 Drag Force

$$F_d = \frac{1}{2} C_d \rho v^2 A \tag{D.2}$$

Drag force is a product of the the drag coefficient C_d, the area A, and the fluid velocity v. The drag coefficient was assumed to be 1.2 for both the tether and the buoy, as both can be approximated as cylinders. It also was assumed that the fluid flow was always acting perpendicular to both objects to simplify calculations. In reality, the fluid flow will often act at an angle, decreasing the overall drag force.

D.2 Linear Wave Theory

Since the drag force is proportional to the velocity squared, changes in velocity drastically affect overall drag, so it is important to accurately model the water speed, which can be done with linear wave theory calculations.

D.2.1 MATLAB Code

```
clear; clc;

%inputs
d = 20;% water depth m
T = 5.16;% wave period s
g = 9.81;% gravity m/s^2
H = 1.5; %wave height m

% deep-water wavelength
L0 = g*T^2/(2*pi);

% fixed-point iteration on X = d/L
X0 = d/L0;      % known
X  = X0;        % initial guess (equiv. L = L0)

tol = 1e-10;
maxIter = 100;

for n = 1:maxIter
    Xnew = X0 / tanh(2*pi*X);    % X_{n+1} = X0 / tanh(2*pi*X_n)
    if abs(Xnew - X) < tol*max(1,abs(X))
        X = Xnew;
        break
    end
    X = Xnew;
end

L = d / X;% wavelength m
k = 2*pi / L;% wavenumber rad/m
kd = k*d;
```

```
L
k
kd

u = @(x,y,t) (pi*H/T)*(cosh(2*pi*(y+d)/L))/(sinh(2*pi*d/L))*cos(2*pi*(x/L-t/T));
v = @(x,y,t) (pi*H/T)*(sinh(2*pi*(y+d)/L))/(sinh(2*pi*d/L))*sin(2*pi*(x/L-t/T));

t = linspace(0, 2*T, 1000);
t = transpose(t);
us = u(0,0,t);
vs = v(0,0,t);

wavedir = 0; % deg CCW from +East
uWave = us * [cosd(wavedir) sind(wavedir)]; % Nx2 (E,N)

% Wind-Driven Current Calculations
wind_ms = 15; % m/s
wind_dir = 0; % Wind blowing East (matching wave direction)

% Calculate Current Speed (3% of wind speed)
Ucur = 0.03 * wind_ms;

% Calculate Current Direction (Assuming Northern Hemisphere, 30 deg right of wind)
% Note: negative degrees is clockwise.
curdir = wind_dir - 30;

% Calculate the current velocity vector
uCur = Ucur * [cosd(curdir) sind(curdir)];

uNet = uWave + uCur;
```

```
plot(t,uNet(:,1),t,uNet(:,2),t,sqrt(sum(uNet.^2,2)));
legend('u_E','u_N','|u_{net}|'); grid on; xlabel('t (s)'); ylabel('m/s');
s = "Swell = "+H+" m East, "+"Period = "+T+" s --- " + "Mean Surface Current = "+ Ucur +" m
title(s)
```

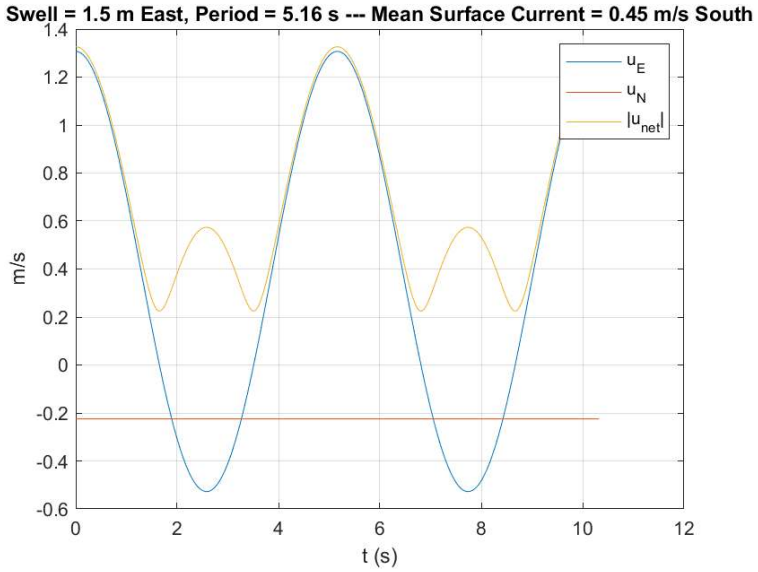


Figure D.1: Water Speed for 1.5 m Wave Height and 15 m/s Wind Speed

D.3 Drag Values

The linear wave theory calculations yielded a horizontal surface water speed of 1.3 m/s and vertical speed of 0.23 m/s. When calculating the drag force, the worst case scenario was calculated, where the water is its maximum speed. Because ocean waves are cyclical, the water particle velocities oscillate continuously over the wave period, reaching their maximum speeds only at the moment a wave crest or trough occurs. For the vast majority of the wave cycle, the instantaneous water velocity is significantly lower than these calculated peaks. By evaluating the system at these maximums, we ensure that the winch motor and tether can safely supply the necessary torque to withstand the harshest instantaneous wave loads without stalling or experiencing mechanical failure.

The drag force values were calculated as follows:

$$F_{Dx,\text{tether}} = \frac{1}{2}(1025)(1.3)^2(1.2)(18.8 \times 0.006)\frac{1}{3} = 39.1 \text{ N}$$

$$F_{Dx,\text{buoy}} = \frac{1}{2}(1025)(1.3)^2(1.2)(0.3 \times 0.1) = 31.2 \text{ N}$$

$$F_{Dy,\text{tether}} = \frac{1}{2}(1025)(0.23)^2(1.2)(18.8 \times 0.006)\frac{1}{3} = 1.2 \text{ N}$$

$$F_{Dy,\text{buoy}} = \frac{1}{2}(1025)(0.53)^2(1.2)(0.3 \times 0.1) = 5.2 \text{ N}$$

$$F_{Dx,\text{total}} = 70.3 \qquad F_{Dy,\text{total}} = 6.4 \text{ N}$$

$$\text{Total Tension} = \sqrt{(4.9 + 6.4)^2 + (70.3)^2} = \boxed{71.2 \text{ N}}$$

These equations come from the drag force equation D.2, but it is important to note the following: $F_{Dy,\text{buoy}}$ has a velocity of 0.53 m/s instead of 0.23 m/s. This is because we are retracting the buoy at a speed of 0.3 m/s, increasing the relative velocity of the buoy. Additionally, the $1/3$ scalar comes from the velocity profile throughout the water column. This affects the tether but not the buoy because the tether spans the entire water column while the buoy resides at the surface during the period of maximum velocity. It was assumed that the water's surface has the speeds of 1.3 m/s and 0.23 m/s but then decreases linearly to 0 m/s at the ocean floor. The velocity at any height y (measuring from the floor up to the surface) is:

$$v(y) = v_{max}\frac{y}{L}$$

To find the total drag, you have to integrate the drag force along the entire length of the tether from 0 to L:

$$F_{D_{total}} = \int_0^L \frac{1}{2}\rho C_D D\left(v_{max}\frac{y}{L}\right)^2 dy$$

Pulling the constants out of the integral:

$$F_{D_{total}} = \frac{1}{2}\rho C_D D\frac{v_{max}^2}{L^2}\int_0^L y^2 dy$$

The integral of y^2 is $\frac{y^3}{3}$. Evaluating this from 0 to L gives $\frac{L^3}{3}$:

$$F_{D_{total}} = \frac{1}{2}\rho C_D D\frac{v_{max}^2}{L^2}\left(\frac{L^3}{3}\right)$$

Simplifying this leaves the $1/3$ scalar:

$$F_{D_{total}} = \frac{1}{2}\rho C_D (D \cdot L) v_{max}^2\left(\frac{1}{3}\right)$$

Appendix E

Weight – Buoyancy Balance

This appendix documents the hydrostatic stability analysis summarized in the Final Buoy Design subsection of Section 4.4. The calculation verifies that the IRIS buoy possesses a positive metacentric height in its operating configuration, demonstrating that it will self-right after being displaced from vertical.

E.1 Method

The analysis follows the standard metacentric approach for a small floating body. The reference frame is fixed at the keel (K), defined as the bottom of the T-slot frame, with the vertical coordinate z measured upward in millimeters. Each component is characterized by its mass, displaced volume, and centroidal height above the keel.

The waterline is located by enforcing Archimedes' principle: the submerged volume of the assembly must equal the total mass divided by the seawater density, $V_{\text{sub}} = m_{\text{total}}/\rho_{\text{seawater}}$, with $\rho_{\text{seawater}} = 1025$ kg/m^3. The four governing quantities are then computed as follows:

$$\overline{KG} = \frac{\sum_i m_i z_i}{\sum_i m_i} \tag{E.1}$$

the mass-weighted average height of all components, representing the center of gravity above the keel.

$$\overline{KB} = \frac{\sum_i V_{\text{sub},i}\, z_{B,i}}{\sum_i V_{\text{sub},i}} \tag{E.2}$$

the volume-weighted average centroid of the submerged portion of each component, representing the center of buoyancy above the keel.

$$\overline{BM} = \frac{I_{wp}}{V_{\text{sub}}} = \frac{\pi R^4/4}{V_{\text{sub}}} \tag{E.3}$$

the metacentric radius, where I_{wp} is the second moment of area of the circular waterplane about the tilt axis and R

is the tube outer radius.

$$\overline{GM} = \overline{KB} + \overline{BM} - \overline{KG} \tag{E.4}$$

the metacentric height. The buoy is stable when $\overline{GM} > 0$.

E.2 Inputs

The geometric constants and seawater properties used in the analysis are given in Table E.1. The component inventory, including mass, displaced volume, and centroidal height above the keel, is documented in Table E.2. Vertical positions are taken from CAD measurement of the final assembly. The battery and microcontroller positions are engineering assumptions consistent with internal mounting practice, and are noted in the inventory.

Table E.1: Constants and geometry used in the stability analysis.

Quantity	**Value**	**Units**
Gravitational acceleration, g	9.81	m/s^2
Seawater density, ρ	1025	kg/m^3
Tube outer diameter, D	114.3	mm
Tube length	300	mm
Tube bottom above keel, $z_{\text{tube,bot}}$	113.8	mm
Waterplane radius, $R = D/2$	57.15	mm
Waterplane second moment, $I_{wp} = \pi R^4/4$	8.38×10^6	mm^4

Table E.2: Component inventory used in the stability analysis. Heights z are measured from the keel (bottom of the T-slot frame) to each component's centroid. Asterisks denote assumed positions.

Component	**Qty**	**Mass each (g)**	**Disp. vol. each (mL)**	z **(mm)**
Acrylic tube (cast)	1	746.00	—	263.8
Aluminum flat cap	1	212.00	—	110.0
Polycarbonate dome cap	1	81.00	—	435.0
Battery (inside PV)	1	275.00	—	150.0*
Microcontroller (inside PV)	1	30.00	—	250.0*
T-slot extrusions	2	149.46	55.36	152.4
316 SS ballast weights	2	191.00	25.00	50.0*
Acoustic modem (external)	1	200.00	87.96	60.0
Lower enclosure clamp pair	1	192.00	61.49	156.8
Lower T-slot mounts	2	50.00	45.82	156.8
Upper enclosure clamp pair	1	192.00	61.49	269.8
Stainless screw-pin shackles	2	32.58	4.07	269.8
Rope T-slot mounts	2	70.00	70.47	269.8
Pressure vessel cylinder envelope	1	—	3078.2	submerged portion
Pressure vessel dome envelope	1	—	390.9	435.2 (above water)

E.3 Results

The hydrostatic equilibrium and metacentric calculation results are summarized in Table E.3. The metacentric height $\overline{GM} = 31.5$ mm is positive, confirming static stability. Decomposing the result reveals that the pendulum contribution $\overline{KB} - \overline{KG} \approx 28.6$ mm dominates over the waterplane contribution $\overline{BM} \approx 2.95$ mm, accounting for roughly 91% of the total metacentric height. This is consistent with the expected behavior of a slender vertical body, in which stability is established primarily by ballast placement rather than waterplane geometry.

Table E.3: Hydrostatic equilibrium and metacentric results.

Quantity	**Value**	**Units**
Total mass, m_{total}	2914.1	g
Total weight, $W = mg$	28.59	N
Required submerged volume, V_{sub}	2843.0	mL
Maximum available displacement	4081	mL
Reserve buoyancy	30	%
Waterline above keel	331.2	mm
Center of gravity, $\overline{KG}$	183.1	mm
Center of buoyancy, $\overline{KB}$	211.7	mm
Metacentric radius, $\overline{BM}$	2.95	mm
Metacentric distance, $\overline{KM}$	214.6	mm
Metacentric height, $\overline{GM}$	**31.5**	**mm**
Stability status	$\overline{GM} > 0$: stable (self-rights)	

The buoy displaces approximately 2.84 L of seawater at equilibrium against a maximum displacement capacity of 4.08 L, corresponding to a reserve buoyancy of roughly 30%. The resulting waterline leaves the polycarbonate dome and approximately 83 mm of the acrylic tube above the surface, providing adequate clearance for the cellular antenna housed within the dome.

E.4 Ballast Position Sensitivity

The 316 stainless steel ballast weights mount on the modular T-slot rails, so their centroidal height is adjustable during integration. A sweep of ballast position from the base of the frame upward is presented in Table E.4. The buoy remains stable for any ballast centroid below approximately 327 mm above the keel, beyond which the metacentric height crosses zero and the configuration becomes unstable.

Table E.4: Ballast position sensitivity. The design-intent position (50 mm) is shown in bold.

z_{ballast} (mm)	$\overline{KG}$ (mm)	$\overline{KB}$ (mm)	$\overline{GM}$ (mm)	Status
0	176.6	211.0	37.4	stable
25	179.9	211.4	34.4	stable
50	183.1	211.7	31.5	stable
75	186.4	212.1	28.6	stable
100	189.6	212.4	25.7	stable
150	196.2	213.2	19.9	stable
200	202.7	213.9	14.2	stable
250	209.3	214.6	8.3	marginal
300	215.9	215.4	2.5	marginal
327	219.4	215.8	0.0	threshold
350	222.4	216.1	−2.9	unstable

Appendix F

Safety Form

Student Project Hazard Assessment

A Hazard Assessment is designed to help students and project advisors recognize hazards associated with student projects at the early planning stages to find ways to minimize the chance of injury, loss, or harm while students are working on the project. A detailed assessment will also prevent the potential for project interruption due to unanticipated safety issues. This form is intended for projects where the primary hazards are associated with engineering work (physical, mechanical, electrical, etc.). Chemical and biological focused projects require a separate project assessment form.

Each student project must complete a Hazard Assessment form, even if the project is not expected to have any hazards. All required approvals must be obtained before proceeding with the project. It is important that all team members participate in the process, with close supervision of your advisor. To help ensure that hazards and risks associated with your project are not overlooked or underestimated, you are encouraged to contact additional university staff (lab directors/managers, EHS, etc.) with relevant knowledge or experience for guidance.

The Hazard Assessment process usually involves these five steps below, with an example:

Step:	Example:
1. Identify the specific tasks that must be completed to reach your project goals	One of your project tasks involves testing a live electrical circuit
2. Determine if there are hazards associated with completing the tasks	On the form, you select the "Electrical parts and assemblies > 50V or high current", under the Hazardous Conditions/Processes/Activities, Electrical Hazard section
3. If hazards exist, identify the risks connected with the hazards of each task. Ask yourself, what could go wrong? If you are not already familiar with the risks, do a quick internet search	After some research, you learn that there is potential of electrical shock from accidental contact with exposed live components
4. Develop a list of controls (things you can do) to eliminate the hazard or reduce the risks. Refer to Hierarchy of Controls on the next page	To minimize the risk identified above, you could: ○ De-energize and isolate the system or ○ Guard live components to prevent accidental contact

5. Create a safe working procedure. Describe how you will safely complete each task	You write a detailed procedure for testing a live electrical wire, that includes all the information from your hazard/ risk assessment and which controls you will use to reduce the risks
6. Submit your hazard assessment form for approval (see section below for details)	

Definitions

A hazard is something that has the potential to cause harm (injuries, accidents, or other undesirable effects). Hazards can be eliminated but not reduced. A hazard can be in the form of an agent, condition, process or activity.

Risk is the likelihood or probability of a hazard causing harm to people, property, or the environment. Risks associated with a hazard can be reduced. Put another way,

Risk = Hazard x Exposure

A hazard assessment is the process of identifying anything that can cause harm (hazardous agents, conditions, processes or activities). A risk assessment is the process of determining how great the chance is of harm occurring from a given hazard.

Hierarchy of Controls

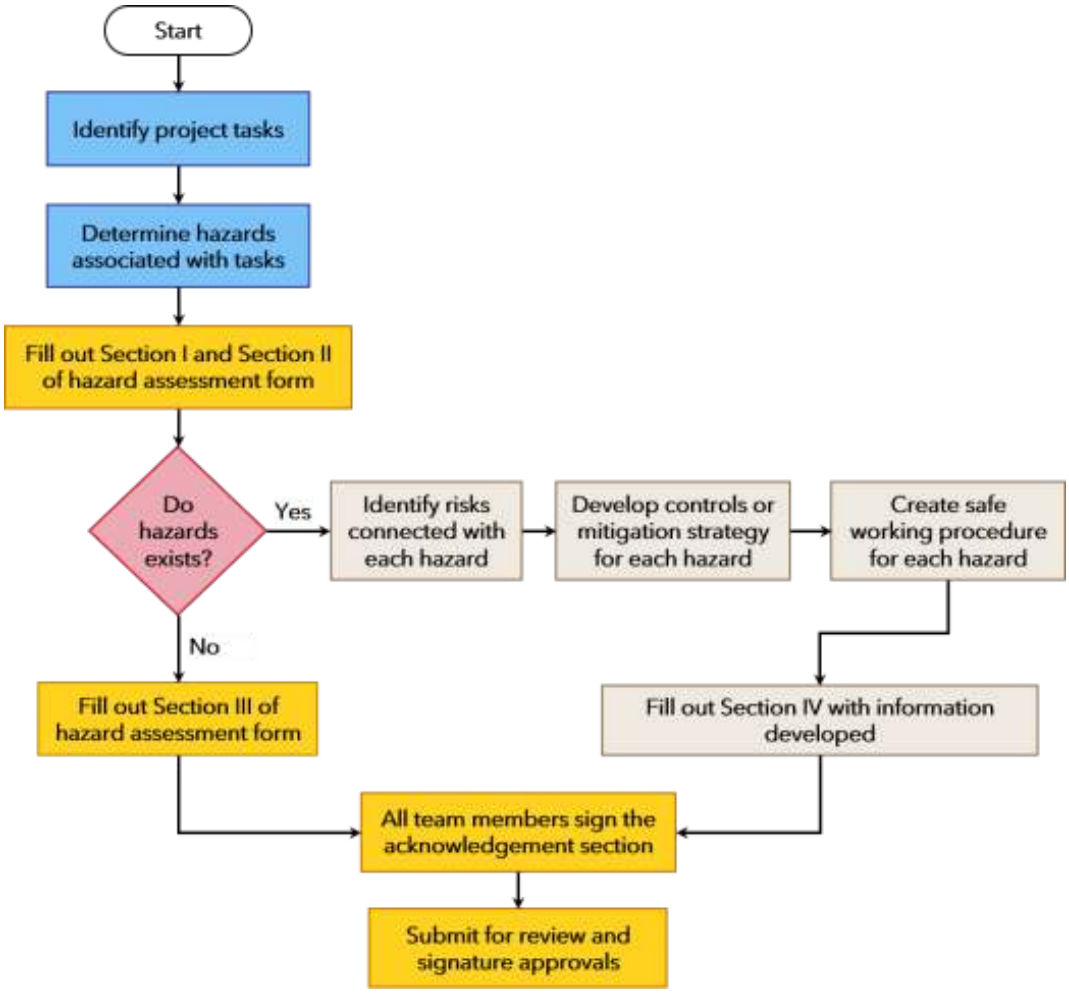
Unless the hazardous agent, condition, or activity is removed, hazards cannot be eliminated. However, risks from the hazard can be minimized by employing the proper control measures and safe work practices that will have been identified from completing a hazard assessment.

Some controls are more effective than others at eliminating hazards or reducing risk. Use the hierarchy of controls chart below to evaluate the controls measures you plan to use. Priority should be given to the most effective controls at the top of the hierarchy (elimination and substitution) and moving down, rather than start with the easiest one. While personal protective equipment (PPE) should always be used, it should be considered the last line of defense from potential hazards.



Revised Sept 2025 scu.edu/ehs 2

Hazard Assessment Form Process



1. After completing the form, all team members must sign the acknowledgement section.
2. Your advisor and the department chair must approve this form prior to obtaining formal approvals from other university staff.
3. If your project involves the use of lab or shop space or equipment, the form must be routed to the appropriate lab/shop manager(s) for review and approval.
4. Your faculty advisor or department chair will make the determination whether EHS approval is additionally required.

Depending on the scope of your project, the faculty advisor, department chair, lab/shop manager(s), or EHS may request further approval from other relevant faculty, staff, or department.

Student Project Hazard Assessment Form

This form is to be used for student projects where primary hazards are associated with engineering

work (physical, mechanical, electrical, etc.). Chemical and biological focused projects require a separate form.

Complete this form and obtain all the required approvals (Faculty Advisor, Department Chair, Laboratory Manager, EHS, etc.) before proceeding with the project. Please refer to the hazard assessment guide for assistance in filling this form.

SECTION I: Project Description

Project Title: IRIS - Integrated Reef Information System

Project Team Members:

Colin Friedel [CSEN]
Cooper McCarthy [MECH]
Zach Henderson [MECH]
Thatcher Steele-Maley [MECH]
Elias Pedroza [MECH]

Project Faculty Advisor

Name: Dr. Christopher Kitts

Department: Mechanical Engineering

Phone:

Email: ckitts@scu.edu

Name: Dr. Michael Neumann

Department: Mechanical Engineering

Phone:

Email: maneumann@scu.edu

Proposed Project Location(s) (Department, building, room#): Mech. Dep. SCDI, 1205, the RSL

Anticipated Dates of Project Duration: 9/22/25 - 5/18/2026

Summary of Project Objectives:

Our project is to create a marine data-transmission system for the IRIS buoy capable of higher-frequency, low-power data delivery from a benthic monitoring platform. The system must reliably transmit sensor packets through varying water conditions, maintain stable performance over multi-month deployments, and satisfy environmental monitoring requirements for the Palos Verdes Reef project. We will evaluate the system through bench tests, range and latency measurements, and staged tank trials that replicate real deployment conditions.

SECTION II: Hazard Checklist

Identify all the tasks that must be completed for your project. Carefully evaluate each task to determine if there are any associated hazards. After identifying the hazards of your project, you will be asked to assess the risk connected to each hazard and to identify control measures that will either eliminate the hazard or reduce the risk to an acceptable level. Safe work procedures for each step involving a known hazard will need to be developed.

HAZARDOUS CONDITIONS / PROCESSES / ACTIVITIES

Electrical Hazards:
- ☐ Electrical parts and assemblies > 50V or high current
- ☑ Batteries
- ☐ Control Panels

Mechanical Hazards:
- ☐ Power tools and equipment
- ☑ Machine guarding/power transmission: gears, rotors, wheels, shafts, belt/chain drives, rotating parts, pinch points
- ☐ Robotics
- ☐ Sharp Objects
- ☐ Stored Energy (springs, gravity, pneumatic, hydraulic, pressure)

Physical Hazards:
- ☐ Extreme temperature (high temp fluids: water > 160 °F, steam, hot surfaces > 140 °F, cryogenic fluids
- ☐ Material handling of heavy objects
- ☐ Elevated heights (scaffolding, ladders, roofs, lifts, etc.)
- ☐ Overhead falling objects (cranes, hoists, drones, projectiles, etc.)
- ☐ Confined Spaces
- ☐ Airborne Dusts
- ☑ Bonding / Grounding
- ☑ Electrostatic Discharge

Reaction Hazards:
- ☐ Explosive
- ☑ Exothermic, with potential for fire, excessive heat, or runaway reaction
- ☐ Endothermic, with potential for freezing solvents, decreased solubility or heterogeneous mixtures
- ☐ Gases produced
- ☐ Hazardous reaction intermediates/ products
- ☐ Hazardous side reactions

Hazardous Processes:
- ☐ Generation of air contaminants (gases, aerosols, or particulates)
- ☐ Heating chemicals
- ☐ Large mass or volume
- ☐ Pressure > Atmospheric
- ☐ Pressure < Atmospheric
- ☐ Scale-up of reaction
- ☑ Metal fabrication (soldering, welding, cutting, drilling, etc.)
- ☐ Construction/Assembly, etc.

Other Hazards:
- ☐ Noise > 80 dBA
- ☑ Vehicle traffic
- ☐ Hazardous waste generation
- ☐ Other (list):

☐ Check here if project will not involve any hazardous conditions, processes, or activities

Hazard Checklist (continued)

HAZARDOUS AGENTS

Physical Hazards of Chemicals:	Health Hazards of Chemicals:	Non-Ionizing Radiation:	Biohazards:
☐ Compressed Gases ☐ Cryogens ☐ Explosives ☐ Flammables ☐ Oxidizers ☐ Peroxides or Peroxides Formers ☐ Pyrophorics ☐ Water Reactives	☐ Acute Toxicity ☐ Carcinogens ☐ Nanomaterials ☐ Reproductive Toxins ☐ Respiratory or Skin Sensitization ☐ Simple Asphyxiant ☐ Skin Corrosion/ Irritation ☐ Hazards Not Otherwise Classified	☐ Lasers ☐ Magnetic Fields (e.g. NMR) ☑ RF/Microwaves ☐ UV Lamps	☐ BSL-2 Biological Agents ☐ rDNA ☐ Human Cells, Blood, BBP ☐ Animal Work ☐ Other (List):

☐ Check here if project will not involve any hazardous conditions, processes, or activities

SECTION III: Projects with No Hazards

IF your project meets all these conditions:

□ No hazardous conditions, processes, or activities
□ No hazardous agents
□ Will not use any laboratory or shop space

You can stop at this section. Acknowledge below and obtain the approval from your faculty advisor.

If your project involves hazardous processes, activities, conditions or agents are, please continue to SECTION IV to complete the rest of the form.

ACKNOWLEDGEMENT (for projects with no hazards)

By signing, I verify that the project will not involve any hazards listed in Section

II Name of Project Team Member Signature Date

APPROVALS (for projects with no hazards)

This document must be reviewed and approved by the people below before any project work can begin. A copy of the approved document must be kept with the project team

Faculty Advisor (required)

Name:

Department: Signature Date

SECTION IV: Hazard Assessment and Control

Description of Potential Hazards

Provide a summary of the procedure and describe the risks associated with each hazard that you have identified above or on the previous page. Use one box below per hazard. You may add additional pages if needed. Define the hazard control measures that will be employed to minimize the risks based on the hierarchy of controls (elimination, substitution, engineering controls, administrative controls, PPE), and then describe specific control measures you will use (e.g. work on system de-energized, receive hazard specific training, shield hot surfaces, guard pinch points, relieve stored energy, wear protective equipment, use less hazardous chemical, etc.). Refer to "Hierarchy of Controls" in the instructions sheet for more information to decide which hazard controls measures are most appropriate.

Hazardous Activity, Process, Condition, or Agent (identified from previous page): Batteries, Exothermic
Summary of Procedure or Tasks: We will use lithium-polymer (LiPo) batteries to power the REEF sensor network, underwater winch mechanism, and communication system during testing and operation.
Describe Hazards (why is the procedure hazardous or what can go wrong – what is the risk): LiPo batteries are sensitive to water and moisture; exposure can cause internal short circuits, fires, or electrical failure. High-pressure conditions may deform the battery casing and cause leakage. Improper charging or discharging can also lead to overheating or thermal runaway, posing fire and equipment-damage risks.
Hazard Control Measures (what you will do to eliminate the hazard or minimize risks): All batteries will be enclosed in sealed, waterproof housings to prevent water contact and inspected before use for pressure and integrity compliance. Charging and discharging will be done with approved LiPo chargers equipped with over-voltage and temperature protection. Damaged or swollen batteries will be removed from use, and all handling will follow manufacturer and institutional LiPo safety protocols to minimize the risk of short circuits or combustion. Furthermore, each member of the team will go through LiPo battery training through the SCU Maker Lab in order to fully understand safe handling practices, and minimize risks.

Hazardous Activity, Process, Condition, or Agent (identified from previous page): Bonding/Grounding
Summary of Procedure or Tasks: We will use bonding and grounding practices to maintain equal electrical potential among metal components in our system. Because the IRIS buoy does not have access to an earth grounding point, all components will share a common reference using the negative battery terminal as the grounding node.
Describe Hazards (why is the procedure hazardous or what can go wrong – what is the risk): Improper bonding can create voltage differences between exposed metal parts, leading to electrical shock or unintentional current flow through sensitive equipment. Unequal potentials can also promote galvanic corrosion, especially in a saltwater environment where conductive pathways accelerate metal degradation. Inadequate grounding increases the risk of fault currents, which may cause equipment failure or unpredictable system behavior. These issues can compromise system reliability, reduce component lifespan, or create unsafe electrical conditions for operators during integration and testing.
Hazard Control Measures (what you will do to eliminate the hazard or minimize risks): All conductive components will be bonded to a common electrical reference to eliminate voltage differences and prevent galvanic current pathways. The system will use the negative battery terminal as the dedicated grounding point to ensure stable electrical potential throughout the device. Materials will be selected with corrosion resistance in mind, and all bonding connections will be regularly inspected for integrity. Proper grounding and bonding practices will be followed during assembly and operation to minimize the risk of electrical shock, equipment damage, and corrosion-related failures.

Hazardous Activity, Process, Condition, or Agent (identified from previous page): Machine guarding/power transmission: gears, rotors, shafts, pinch points on the underwater winch system.
Summary of Procedure or Tasks: We will integrate a commercially sourced marine-grade submersible winch into the IRIS system and operate it during testing, deployment simulations, and routine maintenance procedures. These tasks will involve interacting with the winch's mechanical components, including the motor, cable spool, and transmission system.
Describe Hazards (why is the procedure hazardous or what can go wrong – what is the risk): The winch contains rotating components such as gears, shafts, and the cable spool, all of which present significant pinch-point and entanglement hazards. Accidental contact with these moving parts can lead to cuts, crushed fingers, entanglement, or more severe injuries such as lacerations or amputations. Loose clothing, tools, or hair may also become caught in the mechanism, especially during active spooling or maintenance. Operating the winch in wet or confined environments increases the difficulty of maintaining safe clearance from moving components, elevating the overall risk of personal injury.
Hazard Control Measures (what you will do to eliminate the hazard or minimize risks): All rotating and moving winch components will be fully enclosed using the manufacturer-supplied guard or an equivalent protective housing integrated during assembly. Clear markings and visual indicators will identify hazardous pinch-point areas, and an exclusion zone will be established around the winch during all operational and testing activities. Only trained team members will operate or service the winch, and they will maintain secure clothing and tool management practices to eliminate entanglement risks. The winch will remain powered off and locked out during inspection or maintenance to prevent accidental activation.

Hazardous Activity, Process, Condition, or Agent (identified from previous page): Electrostatic Discharge
Summary of Procedure or Tasks: We will handle, integrate, wire, and test sensitive electronic components—including control boards, sensors, and power-supply modules—during the assembly of the buoy and the winch's control enclosure. These tasks require direct contact with electronics that can be damaged by electrostatic discharge.
Describe Hazards (why is the procedure hazardous or what can go wrong – what is the risk): Electrostatic discharge (ESD) can occur when static electricity accumulates on a person or object through common actions such as walking, handling plastic materials, or working in low-humidity environments. When a charged individual touches a sensitive electronic component, the sudden transfer of charge can permanently damage internal circuitry. This damage may be immediate and catastrophic, rendering the component nonfunctional, or it may weaken internal structures in ways that cause premature failure during later deployment. Because the IRIS system relies on low-power electronics for critical sensing and communication tasks, ESD-related failures pose a significant reliability and operational risk.
Hazard Control Measures (what you will do to eliminate the hazard or minimize risks): Team members will use ESD wrist straps connected to a common grounding point, such as a grounded mat or bonded metal chassis, whenever handling or installing circuit boards. Sensitive components will remain in their original anti-static bags or containers until the moment of integration to prevent unnecessary exposure. Work surfaces will be kept clean and free of materials that generate static, and components will be handled by their edges to minimize contact with circuitry. Together, these practices will reduce the likelihood of ESD events and protect the integrity of the system's electronics.

Hazardous Activity, Process, Condition, or Agent (identified from previous page): Metal Fabrication
Summary of Procedure or Tasks: Our construction process will involve the use of various metal fabrication tools, including drilling, soldering, and potentially welding, to assemble structural components for the IRIS system. These tasks require direct handling of powered equipment and heated materials.
Describe Hazards (why is the procedure hazardous or what can go wrong – what is the risk): Metal fabrication involves tools and processes that can cause injury if used improperly or without the correct protective measures. Drilling and cutting operations can generate sharp metal shavings, rotating pinch points, and kickback forces that pose risks of cuts, punctures, or hand injuries. Soldering and welding introduce high temperatures and molten material that can cause burns or ignite nearby objects, while also producing fumes that may be harmful if inhaled without proper ventilation. Inadequate clothing or missing personal protective equipment increases susceptibility to burns, eye injuries, and respiratory irritation. These hazards can result in significant bodily harm or equipment damage if not properly controlled.
Hazard Control Measures (what you will do to eliminate the hazard or minimize risks): All team members will follow established shop safety procedures and manufacturer guidelines for every tool used during fabrication. Appropriate personal protective equipment—such as safety glasses, gloves, closed-toe clothing, and respiratory protection when needed—will be worn at all times. Hot-work activities such as soldering or welding will be conducted only in designated, well-ventilated areas to minimize fume exposure and fire risk. Tools will be inspected prior to use to ensure they are in proper working condition, and team members will receive supervision or training as required for each fabrication process. When soldering we will only use lead free solder, all team members have been trained to solder through the maker lab. Furthermore team members tasked with fabrication have taken MECH 101L - Machine Shop and have been trained on the machines that will be used. These tools include but aren't limited to bandsaw, mills, and lathes. If welding is needed we will work with Rod the on campus machinist and other experienced parties for training.

Hazardous Activity, Process, Condition, or Agent (identified from previous page): Vehicle traffic
Summary of Procedure or Tasks: We will transport project equipment and team members by vehicle to multiple testing sites and deployment locations throughout the development of the IRIS system. These tasks require driving with specialized equipment and loading or unloading materials near active traffic areas.
Describe Hazards (why is the procedure hazardous or what can go wrong – what is the risk): Traveling to test locations introduces the risk of vehicle accidents caused by distraction, fatigue, adverse weather, or heavy traffic conditions. A collision can result in injuries to team members as well as damage to sensitive project equipment being transported. Loading and unloading items near traffic lanes also increases the risk of being struck by passing vehicles, especially in areas with limited visibility. Transporting heavy or awkward equipment can further cause injuries if objects shift unexpectedly during vehicle movement or handling.
Hazard Control Measures (what you will do to eliminate the hazard or minimize risks): Only rested, sober, and authorized drivers will operate vehicles carrying project materials. All equipment will be securely stored and tied down to prevent shifting or damage during transport. When loading or unloading, team members will establish a safe work zone away from active traffic and use designated parking or drop-off areas whenever possible. Weather and road conditions will be evaluated before each trip to ensure that travel is conducted safely.

Hazardous Activity, Process, Condition, or Agent (identified from previous page): RF/Microwaves
Summary of Procedure or Tasks: Our buoy will transmit data using an RF communication system that operates via satellite or cellular networks, and we will test the antenna while it is actively transmitting during bench and field evaluations. These tasks require working in close proximity to an energized RF antenna.
Describe Hazards (why is the procedure hazardous or what can go wrong – what is the risk): Active RF antennas emit non-ionizing radiation that can cause localized heating of skin or tissue if personnel stand too close during high-power transmission. Prolonged or repeated exposure at close range may increase the risk of overheating, especially around the face or hands. High-power RF sources can also induce heating in nearby electronic components, including LiPo batteries, if they are placed too close to the antenna's radiating elements. Improper handling, accidental direct exposure, or failure to maintain safe distances can lead to equipment damage or thermal effects on personnel.
Hazard Control Measures (what you will do to eliminate the hazard or minimize risks): Transmit power levels will be kept as low as possible during bench testing or when personnel need to work near the antenna. For full-power tests, all team members will maintain the minimum safe separation distance specified in the antenna's manufacturer documentation. Personnel will avoid standing directly in front of or looking into the antenna's radiating aperture at any time. RF transmitters and LiPo batteries will be physically separated to prevent unintended heating or interference. These precautions will reduce the risk of thermal exposure and protect both personnel and equipment.

Hazardous Activity, Process, Condition, or Agent (identified from previous page): Marine Robot Deployment & Water-Adjacent Testing
Summary of Procedure or Tasks: We will conduct system tests and deployment simulations near bodies of water, including shorelines, docks, and watercraft, to evaluate buoy performance, winch operation, and communication range under realistic environmental conditions. These activities require handling equipment in wet environments and working in close proximity to open water.
Describe Hazards (why is the procedure hazardous or what can go wrong – what is the risk): Working near or above water introduces the risk of accidental slips, falls, or immersion, especially when handling heavy or awkward equipment on wet or unstable surfaces. Electronic components and battery modules may be damaged if dropped into the water, posing electrical hazards or environmental contamination risks. The presence of waves, tides, or sudden vessel movement can cause unexpected shifts in balance or equipment position, increasing the likelihood of injury. Cold water temperatures or strong currents can also create dangerous conditions if a team member enters the water unintentionally, possibly leading to drowning. Additionally, deployment lines, cables, or winch mechanisms can become entangled during lowering or retrieval operations, potentially leading to pinch injuries or uncontrolled loads.
Hazard Control Measures (what you will do to eliminate the hazard or minimize risks): All team members working near water will wear appropriate personal protective equipment, including non-slip footwear and, when required, personal flotation devices. Equipment will be transported and handled using stable platforms, and tasks such as lowering or retrieving hardware will be assigned to trained individuals using controlled, deliberate motions. Electronics and batteries will remain secured in waterproof housings and tethered during deployment to prevent loss or contamination. Clear verbal communication and designated roles will be maintained throughout field operations to minimize confusion or unsafe movement around the water. Environmental conditions—such as wave height, weather, and tide cycles—will be reviewed prior to testing, and operations will be halted if conditions become unsafe. All of these control measures have been taught to our team members through ENGR 180 - Marine Operations. Relating to boating here these are what we will do to minimize risk: drivers will have California Boating License (or equivalent from their home state), no bow riding, ensuring properly fitting life preservers are available for everyone on board, throwable flotation device on board and available, flares or signals to draw attention, and lastly a working radio to communicate with

t

SAFETY EQUIPMENT and PPE

Select the appropriate PPE and safety supplies you will need for the project (Check all that apply)

- ☑ Appropriate street clothing (long pants, closed-toed shoes)
- ☐ Gloves; indicate type: _____
- ☑ Safety glasses/ goggles
- ☑ Face shield and goggles
- ☐ Lab coat
- ☑ Hearing protection
- ☑ Fire extinguisher
- ☐ Eyewash/safety shower
- ☐ Spill kit
- ☐ Other (list):

TRAINING REQUIREMENTS

Identify the appropriate training (check all that apply)

- ☐ Biology & Bioengineering Lab Safety Camino Course – contact Lab Manager or EHS to enroll
- ☐ Chemistry & Biochemistry Lab Safety Camino Course – contact Lab Manager or EHS to enroll
- ☐ Dry Lab Safety Camino Course – contact Lab Manager or EHS to enroll
- ☐ Electrical Safety for Engineering Camino Course – contact EHS to enroll
- ☑ LiPo Battery Safety Training – contact MAKER Lab to enroll
- ☐ Review of SDS for chemicals involved in project – access SDS library at: chimeracloud.org/sds/
- ☐ Laboratory Specific Training – contact Lab or Shop Manager/Director
- ☑ Project Specific Training – contact Faculty Advisor
- ☑ Other (describe below):
- ☑ Marine Operations (ENGR 180)

EMERGENCY PROCEDURES

Life-threatening emergencies (police, fire, explosion, hazardous material spill/release, gas leak, etc.)

1. Call or text 911, then call SCU Campus Safety at 408-554-4444
2. Alert people in the vicinity and if necessary, activate the local alarm systems
3. Evacuate to your emergency assembly point (EAP). Fill in your EAP here: Fountain on Sobrato Mall (Northwest corner of SCDI building)
4. Remain nearby to advise emergency responders

For medical emergencies:

1. Stay with the individual if it is safe to do so. Do not move the individual unless absolutely necessary 2. Call SCU Campus Safety or 911
3. Administer first aid if you are trained to do so

If person received an electrical shock:

1. Do not touch the person
2. Shut off power and/or rescue the person with non-conductive object IF it is safe to approach
3. Call SCU Campus Safety or 911
4. Administer CPR or AED if trained to do so

If person exposed to hazardous materials:

1. Flush contamination from eyes/skin using the nearest emergency eyewash/shower for a least 15 minutes. Remove any contaminated clothing
2. Call SCU Campus Safety or 911

For building maintenance emergencies (power outages, plumbing leaks etc.): call Facilities 408-554-4742

REPORT all injuries and incidents (including near misses) to faculty advisor and EHS

Project work outside of SCU campus

Generally, all project work must be performed at SCU campus. However, there may be few cases where work cannot be completed on campus (e.g. competition venue outside of SCU, no appropriate campus location to field-test). These exceptions must be described in the hazard assessment section above. Specific location(s) and their scope of work must have approval from the faculty advisor. Additional approvals may be required.

Check the appropriate project location scope:

All project work will be performed on SCU campus

X All project work cannot be completed on SCU campus. Provide more details below: List offsite location(s) and provide description:

Alviso Boat Ramp - boat ramp into the Guadalupe River / Alviso Slough 10 minutes away from campus.

MBARI Test Tank - 10 m salt water test tank at the Monterey Bay Aquarium Research Institute.

Bethany Reservoir - reservoir ~1 hour away from campus. Tests may be done off of the dock, or using one of the RSLs boats.

Monterey Bay - testing may be performed off of a dive boat during the Marine Operations course deployment(s) in spring.

Describe supervision arrangements and emergency procedures for each offsite location:

Testing at MBARI or using boats (Bethany Reservoir, Monterey Bay, and Alviso Boat Ramp) will have an RSL staff or faculty member present to supervise. At MBARI, an MBARI employee will also host the visit to ensure that their safety protocols are followed.

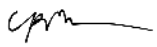
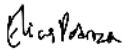
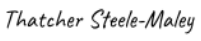
All off-site testing and field deployments will be discussed in advance with the faculty advisor to confirm safety plans, environmental conditions, and team readiness. After each deployment or retrieval, the team will check in with the advisor to confirm safe return and report any unexpected issues. Before heading out, the team will review site-specific risks—such as weather, tides, and equipment handling—and ensure conditions meet the project's go/no-go criteria. For any marine or waterfront work, PFDs will be worn and all electrical components will be verified for waterproofing and isolation.

ACKNOWLEDGEMENT

By signing, I verify that:

1) I am aware of the hazards and risks of all the tasks associated with the project 2) I have received, or will receive all the necessary safety training, have read the safety manual and/or safety data sheets (SDSs) relevant to the project before performing any hazardous tasks.

3) I will follow all required safety precautions while working on this project, including but not limited to use of engineering controls, following safe work practices, and wearing appropriate personal protective equipment, as well as all emergency procedures

Name of Project Team Member	Signature	Date
Cooper McCarthy		11/17/2025
Elias Pedroza		11/17/2025
Thatcher Steele-Maley		11/17/2025
Zach Henderson		11/17/2025
Colin Friedel		11/17/2025

APPROVALS
This document must be reviewed and approved, in the order listed below, before any project work can begin. A copy of the approved document must be kept with the project team.
1. Faculty Advisor (required)
Name Signature Date Micahel Neumann Michael Neumann 2025-11-17
2. Department Chair (required)
Name Signature Date
3. Laboratory Director/Manager (required if project involves use of lab or shop space)
Name Signature Date
4. EHS (required as determined by faculty advisor, department chair, or lab director/manager)
Name Signature Date
5. Other (if requested by faculty advisor, department chair, lab director/manager, or EHS)
Name Signature Date

Appendix G

Presentation Slides

SANTA CLARA UNIVERSITY
SCHOOL OF ENGINEERING

I.R.I.S.

Integrated Reef Information System

Cooper McCarthy (MECH), Zach Henderson (MECH), Elias Pedroza (MECH),
Thatcher Steele-Maley (MECH), & Colin Friedel (CSEN)

Faculty Advisors: Dr. Michael Neumann & Dr. Christopher Kitts

www.scu.edu/engineering

Agenda

- Motivation
- System Overview
- Technical Subsystem Analyses
  - Buoy
  - Drive Train
  - Software
- Testing
- Conclusions

Motivation - Importance of Reefs

- Biodiversity
- Medical Research
- Economic Incentive


Spanish Shawl Nudibranch


Tectitethya Crypta - Source of Anticancer Drug


Wave Energy Dissipating Over a Coral Reef

Motivation - Current Issues

- Inaccessible Data
- Expensive Divers
- Infrequent Updates
- Disturbances
- Destroyed Buoys


Ecological Monitoring Diver

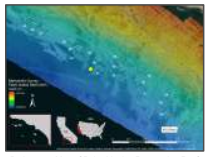

Smashed Buoy


Palos Verdes Kelp Forest

Motivation - Context


Restoration Reef Barges


REEF - Past RSL project


Bathymetry Map of Palos Verdes Restoration Reef

Motivation - Objective Statement

To design, build, and test a retractable surface communication buoy that integrates with an existing benthic reef monitoring system to improve the frequency and ease of data delivery for researchers and environmentalists.

Customer & Stakeholder Needs

- Survivability
- Data Transmission
- Efficacy

7

Needs Synthesis

Survivability	Data Delivery	Efficacy
Depth Rating	Frequency	System Weight
Corrosion Resistant	Transmit in Normal Conditions	Non-Toxic
Tether Repeatability	Timestamped Data	DC Battery Powered
Seaworthiness	Data Robustness	Cost Benefit
UV Resistance	Data Accessibility	High Visibility Coloring
Power Protection	Data Quantity	Wireless Firmware Updates

8

Key Requirements

The housing shall be **depth rated to 30 m** without leakage, with a 6 month maintenance cycle
The system shall be able to operate continuously for **≥180 days** on DC battery.
The system shall **maintain function** through **corrosion & biofouling** in seawater for 6 months.
The tether shall retract repeatedly to the **correct depth without tangling**.
The system shall transmit sensor data at least **once a week.**
Data shall be **easily accessible** by the end user for scientific purposes.
The total system weight should **not exceed 45 kg**, allowing two-person deployment & recovery.

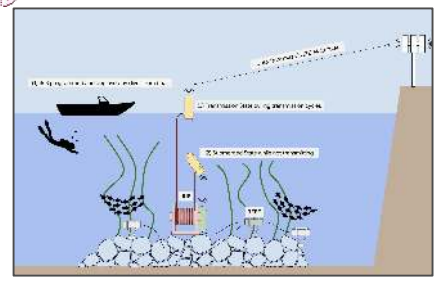
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Agenda

- Motivation
- **System Overview**
- Technical Subsystem Analyses
  - Buoy
  - Drive Train
  - Software
- Testing
- Conclusions

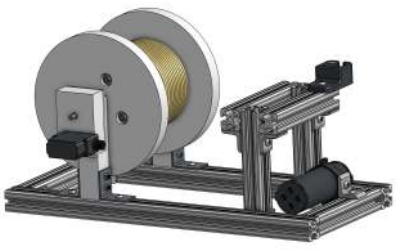
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Overview - Concept of Operations



11

Overview - Proposed Solution



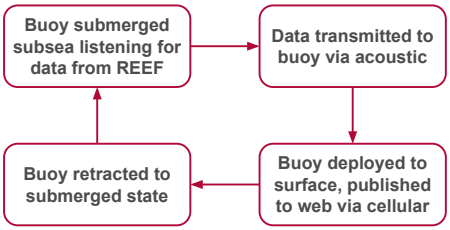


12

Overview - Component Block Diagram

13

Overview - High Level Software Flow



14

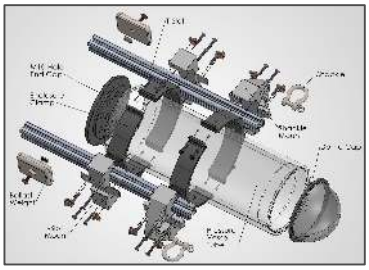
Agenda

- Motivation
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  - Buoy
  - Drive Train
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- Conclusions

15

Buoy - Overview





16

Buoy - Buoyancy Analysis

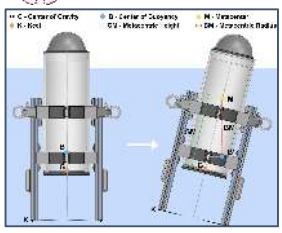


Diagram Not to Scale



Metacentric Stability Condition, GM > 0

GM = KB + BM - KG = 31.5 mm

17

Buoy - Electronics & Power Study

Power Draw	Acoustic Modem	2.52 Wh
	Particle Boron 404x	3.90 Wh
	BMS Parasitic Draw	6.22 Wh
	Net	**12.64 Wh**
Charge Capacity	Required Charge	1.32 Ah
	Battery Capacity	3.3 Ah
	Factor of Safety	**2.5**

Buoy dome with antenna



Particle Boron 404x Cell MCU





Batteryspace 9.6V LiFePO4



Succorfish Delphis Acoustic Nano Modem

18

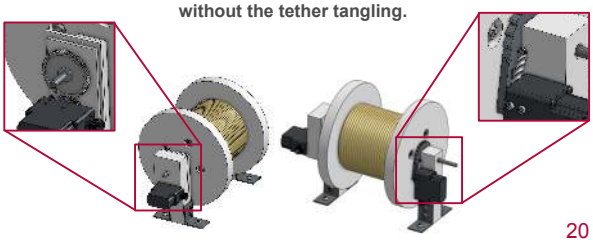
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  - Drive Train
  - Software
- Testing
- Conclusions

19

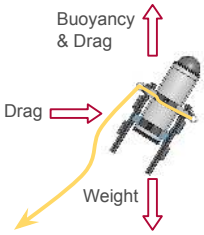
Drivetrain - Motivation & Overview

The buoy needs to retract to the required depth repeatedly without the tether tangling.



20

Drivetrain - Force Analysis



Buoyancy: 25.9 N

Weight: 21 N

Drag:

Horizontal: 1.23N

0.2 m/s surface currents

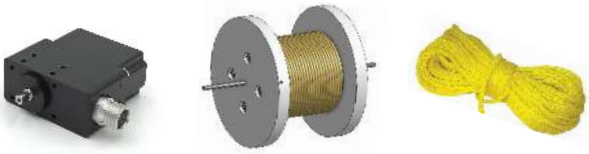
Vertical: 1.66 N

Descent speed of 0.3 m/s

Tension Required: 6.7 N

21

Drivetrain - Capability



5.9 Nm → **x2** → **11.8 Nm** → **÷ 0.09 m** → **135 N**

22

Drivetrain - Drag Analysis

Sea State	Description	Wind Speeds	Wave Height	Tether Drag	Buoy Drag	**Tension Req.**
4	Moderate Breeze	11-16 kts	3 ft (max 5)	23 N	7 N	**31 N**
7	Near Gale	28 - 33 kts	13 ft (max 19)	174N	44N	**218 N**

23

Drivetrain - Brake

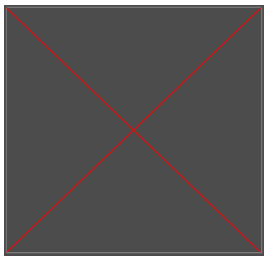


Submerging:
Servo off → Brake Locking

Surfacing:
Servo on → Brake Open

24

Drivetrain - Torque Transfer











25

Drivetrain - Environment Constraints

- Material Corrosion
- Galvanic Corrosion
- Biofouling



Rust - Material Corrosion



Biofouling on Outboard Propeller



Galvanic Corrosion Between Bolts & Structure

26

Drivetrain - Material Selection

		Strengths	Weaknesses	Manufacturability
	Stainless Steel - 316	Good Corrosion Resistance High Strength	Crevice Corrosion Expensive	Difficult Work Hardens
	Delrin® Acetal	Low Moisture Absorption High Fatigue Resistance	Sensitive to pH Vulnerable to UV	Excellent
	Grade 5 Titanium	Great Corrosion Resistance	Expensive	Challenging Water Jet

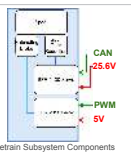
27

Drivetrain - Actuation and Control

Drive: SER 2130 CAN Servo

DC 25.6V, 3A Max (10mA idle), Multi-Turn

Protocol: Controller Area Network (CAN) 2.0, Read/Write



Drivetrain Subsystem Components

High Level Control:

- Deploy and retract the buoy a set repeatable number of turns
- Neutral (idle) when not deploying or retracting
- Live power and position readings



Blue Trail Engineering Waterproof Servo

Brake: SER 2010 Servo

DC 5V, 2A Limit (30mA idle), +/- 90°

Protocol: Standard servo PWM

High Level Control:

- Active (open) only during buoy deployment
- Servo detached (idle) during retraction and hold state

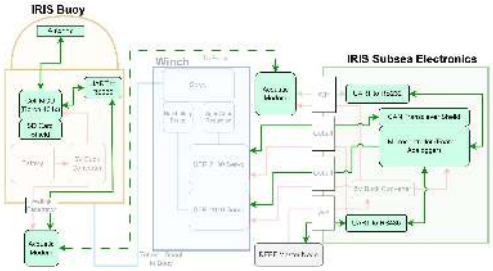
28

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29

Software - System Level Overview

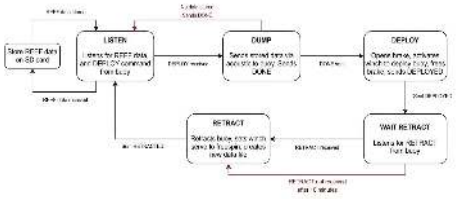


30

Software - Subsea MCU

Feather Adalogger M0 - Reef Data and Winch Actuation

- State loop behavior
- C++ via Arduino IDE

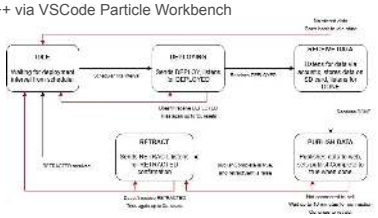


31

Software - Buoy MCU

Particle Boron 404x - System Control and Cellular

- Continuous Loop
- Onboard Schedule via RTC
- C++ via VSCode Particle Workbench



32

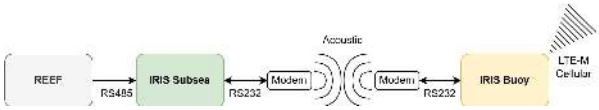
Software - Web Endpoint



33

Software - Data Protocol

RS485 Serial	RS232 Serial	Acoustic	LTE-M Cellular
REEF to IRIS	MCU to Modem	Subsea to Buoy	Buoy to Cloud
Long Range	Short Range	1km range	> 10km range
Noise Robust	High Voltage	Slow (600 bps)	Low Power



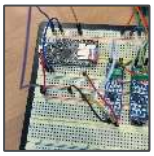
34

Agenda

- Motivation
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  - Software
- **Testing**
- Conclusions

35

Testing - Benchtop



Validations & Verifications

- REEF Data Integration
- Initial Acoustic Connectivity
- Initial Cellular Connectivity

Issues & Adjustments



RS485 throughput	Component testing, validation, and replacement.
Vacuum Leak	O-Ring inspection, re-greasing, and retightening of penetrators.

36

Testing - Repeated Deployment





Validations & Verifications

- Tethering Reliability
- Drivetrain Reliability
- Torque Output

Issues & Adjustments

Gear Slippage	Stripped set-screw threads, cut threads with set screw rather than tapping

37

Testing - Ocean/Reservoir





Validations & Verifications

- Submerged Repeatability
- Deployment
- Seaworthiness

Issues & Adjustments

Deployment Method	Lower system via rope
System Repeatability	Acoustic handshakes if not heard
Modem stability	Threaded mount for robust positioning

38

Testing - Test Tank/Pool





Validations & Verifications

- Visual of Underwater Movement
- End to End Data Transfer
- Deployment with REEF & IRIS together

Issues & Adjustments

Acoustic throughput	Excess noise/signal absorption in small tank → move to larger area (bigger pool/ocean)
Buoy Attachment	Bridle style harness of rope
Balance and emersion	Moving external ballast weights on buoy to tune

39

Agenda

- Motivation
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40

Conclusions - Needs Synthesis

Survivability	Data Delivery	Efficacy
Waterproof	Frequency	System weight
Corrosion Resistant	Transmit in Normal Conditions	Non-toxic
Tether Repeatability	Timestamped Data	DC Battery Powered
Seaworthiness	Data Robustness	Cost Benefit
UV Resistance	Data Accessibility	High Visibility Coloring
Power Protection	Data Quantity	Wireless Firmware Updates

41

Conclusions - Key Requirements

The housing shall be **depth rated to 30 m** without leakage, with a 6 month maintenance cycle
The system shall be able to operate continuously for **≥180 days** on DC battery.
The system shall **maintain function** through **corrosion & biofouling** in seawater for 6 months.
The tether shall retract repeatedly to the **correct depth without tangling**.
The system shall transmit sensor data at least **once a week.**
Data shall be **easily accessible** by the end user for scientific purposes.
The total system weight should **not exceed 45 kg**, allowing two-person deployment & recovery.

42

Conclusions - Next Steps (May 2026)

Deployment

- Full end to end data transfer in water
- Code failsafes

System

- Power Transmission
  - Potential gearbox redesign
- Acetal spool guide

43

Conclusions - Future Work

Deployment

- Protocol
- Site study
- Anchor design
- Full scale ocean conditions testing

System

- Drag reduction
- Power saving for servos
- Reduced cost per unit
- Plug and play modularity

44

Conclusions - Ethical Considerations

Strengths

- Documentation
- Repairability
- Fail safes

Concerns

- Lithium based batteries
- Sacrificial components
- Potential for remote failure

45

Conclusions - Impact of IRIS

Enables

- Continuous Monitoring
  - Accessible Data
  - Informed decisions

Avoids

- Polluting the Environment
  - Intensive Maintenance
  - Agitating Marine Life



Palos Verdes Kelp Forest

46

Conclusions - Lessons Learned

Engineering

- Design for Manufacturing
- Design for Assembly
- Material Specific Sourcing

Team & Project

- Avoiding Analysis Paralysis
- Interdisciplinary Communication
- Balancing Workload



47

Acknowledgements

Dr. Christopher Kitts
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Dr. Hannah Yang
Dr. Michael Abbott
All the SoE Staff

MBARI
Dr. Geoff Wheat
Bill Kirkwood

Blue Ocean Gear
Dr. Kortney Opshaug

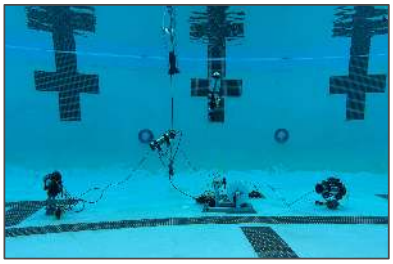
LIQUID ROBOTICS
Will Anderson

Jonathan Williams

ULYSSES
Jamie Wedderburn

48

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Eng. Requirements - Survivability

** Primary Need
* Secondary Need

**	The housing shall be **waterproof** and withstand submersion at **30 m depth** without leakage, with a 6 month maintenance cycle
**	The system shall **resist corrosion and biofouling** in seawater for a minimum deployment period of **6 months** without loss of function.
**	The tether should **retract repeatedly** to the correct depth without tangling.
**	The buoy should survive **wave heights up to 1.5 m** and **wind speeds up to 15 m/s** without mechanical failure.
*	The system shall be constructed using materials rated for **UV exposure ≥ 1 year** in a marine environment.
*	The power system shall operate safely between 12°C - 22°C while including **charge and short-circuit protection**.

50

Eng. Requirements - Data Delivery

** Primary Need
* Secondary Need

**	The system shall transmit sensor data at least **once a week.**
**	The system should transmit under **normal oceanic conditions**, and shall repeat the transmission until successful, if not received.
**	All sensor readings shall be time stamped with **±1 second accuracy**.
**	There should be **less than 1% data loss** as it transmits through the system.
**	There shall be an **accessible interface by the end** user for scientific purposes.
*	The system should be capable of **32 kb/week** of data delivery

51

Eng. Requirements - Efficacy

** Primary Need
* Secondary Need

**	The total system weight should not exceed **45 kg**, allowing two-person deployment and recovery.
**	The system shall use **non-toxic**, reef-safe coatings and materials for any part in water contact.
**	The system shall be able to operate continuously for **≥180 days** on DC battery.
**	The system should **cost less** to use per data collection event than the existing solution.
*	The buoy shall have **high visibility** coloring, as well as RSL identifying markings
*	The system may be capable of **wireless firmware updates** over the cellular link.

52

Defense of Acoustic Modem

a) One buoy, multiple networks



b) Subsea slip ring ($2.5k+ and lead times)



Novasub underwater slip ring

c) Torque rating not limited to structural data tethers

d) Long submerged time allows plenty of time to transmit data, can be called to deploy from a boat

53

Buoy Weight/Balance



54

Data Payload

Average Data per Measurement (1 Nodes, RTD, DO, Ph, Conductivity)	0.0635479798	kb
Deployment Length	180	days
# of Nodes	3	Ct.

Msmt Interval	**Daily Data (kB)**	**Weekly Data (kB)**	**Deployment Data (MB)**	**Measurements/Deployment (Ct.)**
Daily	0.191	1.335	0.034	540
Hourly	4.575	32.028	0.824	12960
/15 Min	18.302	128.113	3.294	51840
1/4 sec bursts (1minute/day, 15 minutes intervals otherwise)	64.056	448.395	11.530	181440

55

Communication Tradeoff Analysis

Driver	Weight	Cellular (Typ Soln)	Satellite	Acoustic	Optical	LoRa
Ease of Data Retrieval	25%	5	5	0	0	3
Communication Speed	10%	5	3	2	3	3
Surface Condition Effects	25%	2	4	2	1	2
Range	10%	4	5	1	3	4
Upfront Cost	10%	4	3	2	0	4
Ongoing Cost	20%	4	3	1	5	5
Average Grade		4	3.83	1.33	2	3.5
Weighted Average		3.3	3.39	1.03	1.59	2.87

56

Motion Tradeoff Analysis

Driver	Weight	Winch (Typical Soln)	Buoyancy Engine	Disposable Buoys	Thrust Driven	Linear Pivot Arm
Long Term Relative Cost	20%	4	4	1	3	3
Environmental Impact	20%	4	4	1	4	3
Repeatability	15%	5	4	1	3	3
Mechanical Complexity	15%	4	3	3	3	2
Biofouling / Corrosion Resistance	10%	3	3	4	3	4
Power Consumption	10%	2	3	5	1	3
Communication Integration	10%	4	4	3	4	5
Average Grade		3.71	3.57	2.57	3	3.29
Weighted Average		3.30	3.13	1.89	2.66	2.70

57

Budget ~ $5000 Total

$2453 from the SOE

~ $2500 from the RSL

Buoy Donation

MBARI Donations

Motor Donation

58

Key Findings From Interviews

- **Blue Ocean Gear**
  - Buoy Transmissibility
  - Buoy Mechanics
- **MBARI**
  - Simplicity of Design
  - Biofouling
  - Power Management
- **Vantuna Research Group**
  - Data Transmission Benchmarks
  - Environmental Conditions
  - Failures of previous attempts
- **The RSL**
  - Longevity
  - Deployability
  - Integration with existing REEF

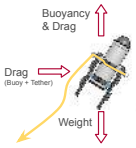
59

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Initial Powertrain Calculations

Retraction Distance	Volume	Net Force	Retraction Time	Efficiency Corrected Power (η = 0.5)	Energy Draw (F.S. = 2)
10 meters	0.027 cubic meters	244 newtons	120 seconds	41 Watts	2.7 Watt Hours

60



$$F_{\text{drag}} = \frac{1}{3}\left(\frac{1}{2}\rho v^2 C_d A\right)$$

$$T \quad \sqrt{(F_b \quad F_{\text{drag}} \quad W)^2 \quad (F_{\text{drag}})^2}$$

